

Cloudy, Hazy and Discolored Water in Freshwater Aquariums

A comprehensive research and diagnostic reference: bacterial blooms, suspended particles, algae, tannins, gases and mineral chemistry

Working research topic: "Why Is My Aquarium Cloudy? Bacterial Bloom vs Dust vs Algae vs Tannins"

Purpose and scope

This document is a research data-dump assembled as source material for a future episode of *Tank Guide Life*, a product of FishKeepingLifeCo. It is written for ingestion into a retrieval system (NotebookLM) rather than for direct publication.

It is **not** an episode script, an episode outline, or a finished consumer article. Section titles are descriptive research headings, not proposed episode titles.

The organizing premise throughout: **"cloudy water" is a symptom, not a diagnosis**. The document is built to support identification before intervention.

Document type	Research reference / evidence compilation
Intended use	NotebookLM source ingestion; podcast episode development
Prepared for	FishKeepingLifeCo — Tank Guide Life
Compiled	August 2026
Scope	Freshwater aquariums. Pond, marine and aquaculture literature is cited where it is the best available evidence, and its limitations are flagged where it is applied to home tanks.
Evidence handling	Every substantive claim carries an evidence classification. Genuine disagreements between sources are documented rather than resolved.

Executive Research Summary

What the research found, condensed. Each finding is expanded in the parts that follow.

Aquarium keepers use one word — "cloudy" — for at least eight physically distinct conditions that have different causes, different risk profiles, and different correct responses. Several of those conditions are not turbidity at all. The most consequential finding of this research is that the hobby's default response chain (see cloudiness, assume bacterial bloom, assume cycling problem, either wait or water-change) is correct often enough to feel reliable and wrong often enough to cause real harm, because it skips identification entirely.

The twelve findings that matter most

- 1 **Visible white "bacterial blooms" are almost certainly not nitrifying bacteria.** The organisms that make water milky are fast-growing heterotrophic bacteria feeding on dissolved organic carbon. Nitrifying organisms grow slowly, attach to surfaces, and gain very little energy per unit of substrate. This distinction reframes nearly every popular explanation of cloudy new-tank water.
- 2 **Water clarity and biological filtration status are only loosely coupled.** A fully cycled aquarium can bloom. An uncycled aquarium can be crystal clear with lethal ammonia. Clarity is not a cycling test and never was.
- 3 **The bloom is rarely the danger; the organic load that fuels it sometimes is.** Suspended bacteria themselves are not directly toxic in typical cases, but the decomposition driving them consumes oxygen and can coincide with ammonia. The correct question is not "is cloudiness harmful" but "what is feeding it, and what else is that condition doing to the tank."
- 4 **Color and turbidity are different physical phenomena.** Tannin-stained water can be optically clear and deeply colored at the same time. Green water is a living suspension. Microbubbles are a gas phase. Substrate dust is mineral solids. Treating them as one category is the root cause of most mismatched treatments.
- 5 **Microbubbles are a large and under-recognized share of "cloudy after a water change" reports.** Pressurized tap water carries dissolved gas that comes out of solution once it enters the tank. The signature is a milky appearance that clears from the top down within minutes to hours, with bubbles collecting on glass and plant leaves.
- 6 **Substrate dust and bacterial bloom frequently occur together in new planted tanks,** especially with active aquasoils, which release both fine particles and a large pulse of ammonium and organics. Manufacturer documentation for these products explicitly describes both effects.
- 7 **UV sterilization is genuinely excellent for one specific problem — free-floating algae — and does nothing for several others.** Its effectiveness depends on delivered dose (wattage, flow rate, contact time, lamp age, water transmittance), not on ownership. Published dose requirements for waterborne algae are meaningfully higher than for common bacteria.
- 8 **Activated carbon addresses dissolved coloration, not suspended solids.** It is a reasonable tool for tannin yellowing and a poor tool for dust, detritus, green water, or a bacterial bloom. This single distinction resolves a large fraction of "why didn't my carbon work" complaints.
- 9 **Clarifiers work by coagulation and flocculation — an established water-treatment mechanism — but they change particle size, not particle source.** They are appropriate for fine inert particulate with adequate mechanical filtration downstream, and inappropriate as a response to a decomposition problem.
- 10 **Tannins are a coloration and chemistry topic, not a water-quality failure.** Humic substances have real, measurable physiological effects in fish — protective in some contexts, stressful in others — and the aquarium marketing claims around disease prevention run well ahead of the evidence.
- 11 **The single highest-value diagnostic act is not a test kit — it is the question "what changed in the last 24 to 72 hours?"** Timing narrows the candidate list faster than appearance alone in most reported cases.

- 12 **Multiple simultaneous causes are normal, not exceptional.** New aquascapes routinely present dust, a heterotrophic bloom, wood biofilm and tannin release at once. Single-cause thinking produces treatments that partially work and are then judged as failures.

What this research does not support

The purpose here is not contrarianism. Much common hobby advice is sound. What the evidence does not support is the *universality* of that advice:

- "Cloudy water means your tank isn't cycled" — not reliable in either direction.
- "A bacterial bloom is your beneficial bacteria multiplying" — mechanistically wrong for the organisms that cause visible white water.
- "Never water-change during a bloom" and "always water-change during a bloom" — both are context-free rules applied to a context-dependent situation.
- "Blackout for three days" — a targeted algae intervention repurposed as a general cloudy-water remedy, where it accomplishes nothing.
- "Clean everything" — the response most likely to convert a cosmetic problem into a biological one.

The one-sentence version

Before trying to make an aquarium crystal clear, a keeper needs to establish what is physically suspended or dissolved in the water, what recent change introduced it, whether the livestock are showing distress, and whether the condition is self-limiting — because the correct response to dust, to gas, to algae, to dissolved organics and to decomposition are four different things and one emergency.

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FRONT MATTER

How This Document Is Organized

This reference is built for retrieval rather than for linear reading. Sections are written to be independently understandable, which means some concepts are deliberately restated where they are load-bearing. Terminology is repeated consistently rather than varied for style, because varied phrasing degrades retrieval quality.

Structural conventions

Descriptive headings	Headings state the question or the object of study, not a rhetorical hook.
Evidence labels	Highlighted statements carry a classification in brackets. See the taxonomy below.
Diagnostic tables	Comparison, timing, color, and treatment matrices are placed near the narrative they summarize and are also consolidated in the back matter.
Scenario library	Part LVIII contains 30 realistic keeper reports worked through in a consistent format: what is known, what is unknown, follow-up questions, plausible causes, safety concerns, and what not to do prematurely.
Question library	Part LXVII contains a large set of retrieval-friendly questions with direct answers, phrased the way hobbyists actually search.
Cross-references	Inline pointers use part numbers (for example, "see Part XXIII") rather than page numbers.
Citations	Author-date or organization-date form inline, resolved in the bibliography at the end. Where a widespread aquarium claim could not be traced to any reliable source, the document says so explicitly — that absence is itself a finding.

Evidence classification taxonomy

Not all statements in this document carry equal weight, and the reference is designed to make that visible rather than to flatten everything into a uniform confident voice. Ten levels are used:

Label	Meaning	Typical basis
ESTABLISHED SCIENTIFIC PRINCIPLE	Physics, chemistry or microbiology that is not in dispute and would be found in a textbook.	Textbook science; universally accepted mechanism
STRONG EVIDENCE	Well-supported by peer-reviewed research directly relevant to the claim.	Multiple studies, or one strong study with a directly applicable design
REASONABLE INFERENCE	Follows logically from established principles applied to aquarium conditions, but has not been directly tested in home aquaria.	Deduction from adjacent evidence; explicitly labelled as extrapolation
AQUACULTURE / WATER-TREATMENT PRACTICE	Standard practice in a professional adjacent field, with known differences from home aquariums noted.	RAS, hatchery, pond, drinking-water or wastewater literature
COMMON AQUARIUM PRACTICE	Widely done in the hobby with broadly consistent reported outcomes; not formally studied.	Convergent hobby experience across many independent keepers
USEFUL RULE OF THUMB	A simplification that is directionally right and practically helpful, with known exceptions.	Heuristic; explicitly acknowledged as approximate
MANUFACTURER RECOMMENDATION	Stated by a product manufacturer. Useful for understanding intended function; not independent proof of efficacy.	Product documentation, technical bulletins, package instructions

Label	Meaning	Typical basis
ANECDOTAL OBSERVATION	Reported by keepers or in community discussion; plausible but uncontrolled.	Forums, comment threads, individual reports
CONTESTED CLAIM	Credible sources actively disagree. The disagreement is documented rather than resolved.	Conflicting expert positions
UNSUPPORTED AQUARIUM FOLKLORE	Widely repeated, traceable to no reliable source, or contradicted by established mechanism.	Repetition without provenance

A note on tone for downstream content development

This document deliberately avoids replacing simple aquarium rules with new simple aquarium rules. Several pieces of standard hobby advice turn out to be well founded: bacterial blooms really are common in new tanks; waiting really is often appropriate; UV really is highly effective against green water; carbon really does remove tannin coloration; fine mechanical filtration really does capture dust. The research value is in establishing *when each of those statements applies*, not in declaring them wrong.

TERMINOLOGY AND DEFINITIONS

Part I — Defining "Cloudy Aquarium Water"

Establishing precise categories for a word hobbyists use imprecisely.

When an aquarium keeper says "my tank is cloudy," they may be describing any of at least eight physically distinct conditions. Some involve solid particles suspended in water. One involves living cells. One involves a gas phase. One involves dissolved compounds that scatter almost no light at all but absorb strongly in the blue end of the spectrum, producing color without haze. These are not variations on a theme. They are different physics with different remedies.

The vocabulary hobbyists actually use

The following words appear constantly in hobbyist descriptions. They are not standardized, and different keepers use them for different phenomena. Understanding the typical intent behind each word is useful for interpreting reports, but no descriptive word should be treated as diagnostic on its own.

Keeper's word	Usually intended to mean	Frequently actually is	Ambiguity risk
Cloudy	Reduced ability to see through the water	Anything on this list	Very high — the least specific term available
Hazy	Mild, uniform loss of clarity; objects visible but soft-edged	Early or mild bloom; fine dust; very light green water; microbubbles	High — mild green water is often called hazy, not green
Milky	Distinctly white, opaque, like diluted milk	Heterotrophic bacterial bloom; microbubbles; fine mineral or substrate dust	High — microbubbles and bacteria look nearly identical at a glance
Murky	Dirty-looking; often implies color plus particles	Detritus in suspension; disturbed substrate; heavy organic load	Moderate — usually implies visible particulate
Smoky / foggy	Diffuse, grey, appears to swirl	Fine clay or soil particulate; bloom; disturbed mulm	Moderate
Green	Obvious green tint	Suspended unicellular algae (green water)	Low when strong; mild blooms are often described as grey or hazy instead
Yellow / tea-colored / brown	Discoloration without necessarily losing clarity	Tannins and other dissolved organics; sometimes fine suspended debris	Moderate — dissolved coloration and suspended brown debris are routinely conflated
Dusty	Fine particles visible when lit	Substrate fines; filter media dust; mulm	Low to moderate
Full of particles	Discrete specks visible to the naked eye	Detritus; plant fragments; food; substrate; biofilm sloughing	Low — usually a genuinely particulate condition
Sparkly / bubbly	Glittering points that rise	Microbubbles from gas coming out of solution or an air leak	Low if observed carefully; high if only described as "cloudy"

Five categories that must be kept separate

The following five categories are the conceptual backbone of this entire document. They are physically distinct, they are measured differently, and they respond to different interventions.

1. Turbidity

Turbidity is an optical property: the degree to which suspended material causes light to be scattered and absorbed rather than transmitted in straight lines through a sample. It is a measure of how particles interfere with light, not a measure of how much material is present. The US EPA describes turbidity as a measure of water clarity determined by how much suspended material reduces the passage of light through water, with contributing materials including soil particles, algae, plankton and microbes, typically in the size range from about 0.004 mm (clay) to 1.0 mm (sand) (US EPA, Volunteer Monitoring).

[ESTABLISHED SCIENTIFIC PRINCIPLE] Turbidity is caused by light scattering from suspended particulate matter. It is an optical property, measured nephelometrically, and is distinct from the concentration of dissolved substances.

2. Coloration

Coloration is caused by dissolved compounds that absorb specific wavelengths of light. Tannin-rich water is the aquarium's clearest example: it can be almost perfectly transparent while being strongly amber. Notably, standard turbidity instruments handle color imperfectly — instruments reporting in FNU using near-infrared light sources can compensate for dissolved colored material such as humic stain, whereas white-light NTU instruments cannot (Fondriest Environmental, Fundamentals of Environmental Measurements). That measurement subtlety in professional instrumentation mirrors exactly the confusion hobbyists experience looking at a blackwater tank.

[ESTABLISHED SCIENTIFIC PRINCIPLE] Dissolved colored organic matter changes water color through absorption without necessarily increasing light scattering. Color and turbidity are independent water-quality parameters.

3. Suspended particles

Discrete solid material carried in the water column: mineral fines from substrate, clay, silt, fragments of detritus and mulm, uneaten food, plant tissue, sloughed biofilm, and dust from filter media. These particles have mass and, given adequate time and still water, most will settle. Their behavior — whether they settle, whether they are captured by fine mechanical media, whether they redistribute when flow changes — is one of the most useful diagnostic signals available to a hobbyist without instruments.

4. Microbial and algal suspensions

Living cells suspended in the water column. Two very different populations produce very different appearances: heterotrophic bacteria (white or grey milkiness) and photosynthetic microalgae (green water). Unlike inert particles, these populations reproduce, consume oxygen, and respond to the availability of their respective resources — dissolved organic carbon for the bacteria, light plus dissolved nutrients for the algae. A key consequence: they do not simply settle out, and removing them physically does not remove the conditions that produced them.

5. Dissolved compounds and dissolved gases

Two sub-cases that both fall outside "suspended particles." Dissolved organic compounds (including tannins and humic substances) produce color. Dissolved gases produce a temporary *appearance* of milkiness when they come out of solution as microscopic bubbles. Neither is filterable by mechanical media in the ordinary sense, and both are routinely misdiagnosed as bacterial.

Can water look cloudy without high turbidity?

Yes, and this is one of the most useful ideas in the document. Three distinct mechanisms produce a "cloudy" impression without a conventional suspended-solids problem:

- **Coloration without scattering.** Tannin-stained blackwater can transmit light well while absorbing strongly in the blue-green range. A keeper perceives "dirty water"; an instrument may register low turbidity and high color.

- **Gas phase, not solid phase.** Millions of microscopic bubbles scatter light very efficiently and produce an intense white appearance. They are not solids, they cannot be filtered out, and they resolve on their own as gas equilibrates.
- **Living suspensions at low density.** A modest green-water population may not read as "green" to the eye at all, appearing instead as generalized grey haze — while being unambiguously green when a sample is viewed in a white container against daylight.

Diagnostic consequence

Because at least three common conditions can look "cloudy" without being suspended solids, any diagnostic framework that begins with "how do I filter this out" starts from a false premise in a meaningful share of cases. The framework in Part II therefore begins with characterization, not with removal.

Precise definitions used throughout this document

Cloudiness	Non-specific term for any reduction in the visual clarity of aquarium water. Used in this document only as a symptom label.
Turbidity	The optical effect of suspended particles scattering and absorbing light. Measured in NTU or FNU by nephelometry.
Color (true color)	Absorption of specific wavelengths by dissolved substances, measured after filtration to remove particulates.
Apparent color	Color measured on an unfiltered sample; includes contributions from both dissolved substances and suspended particles.
Bacterial bloom	Hobby term for a rapid increase in free-floating bacteria in the water column, usually heterotrophic, producing white or grey milkiness. See Part IV for why this term is imprecise.
Heterotrophic bacteria	Bacteria that obtain carbon and energy from organic compounds. Fast-growing. The organisms typically responsible for visible white-water blooms.
Autotrophic / nitrifying bacteria	Chemolithoautotrophic organisms (ammonia oxidizers and nitrite oxidizers) that derive energy from inorganic nitrogen compounds. Slow-growing, surface-attached, and not the usual cause of visible cloudiness.
Green water	A suspension of free-floating unicellular algae (phytoplankton) dense enough to color the water column green.
Tannins	In hobby usage, the broad family of plant-derived polyphenolic and humic compounds that leach from wood, leaves and botanicals and produce yellow-to-brown coloration.
Humic substances	High-molecular-weight organic compounds formed by decomposition of plant material; the dominant fraction of dissolved organic matter in natural blackwaters.
Detritus / mulm	Fine particulate organic matter: partly decomposed waste, plant debris, food fragments and biofilm material.
Microbubbles	Very small gas bubbles suspended in the water column, produced by gas coming out of solution or entrained by equipment.
Precipitation (chemical)	The formation of an insoluble solid from dissolved ions, producing a white or chalky suspension.
Flocculation	The aggregation of destabilized fine particles into larger flocs that settle or can be captured mechanically.
Biological oxygen demand	The oxygen consumed by microorganisms decomposing organic matter in water.

DIAGNOSIS

Part II — A Visual Diagnostic Framework

The first question is not "how do I fix this" but "what does the water actually look like."

Visual appearance is where every diagnosis begins, because it is the only information a keeper has before doing anything else. It is also insufficient on its own. This part establishes what appearance can and cannot tell us, and where the overlaps are severe enough that appearance alone will mislead.

What does the water look like? — seven visual categories

The seven categories below are physically distinct. Within each, several causes are possible; between them, confusion is usually avoidable with careful observation.

1. White or milky water

Uniformly white or grey-white, opaque or translucent, no visible discrete particles. Looks like a small amount of milk has been added. Principal candidates: heterotrophic bacterial bloom, microbubbles, very fine substrate or mineral dust, and mineral precipitation. This category has the highest misdiagnosis rate in the hobby and is treated at length in Part III.

2. Green water

Green tint ranging from a faint grey-green haze to pea-soup opacity. Caused by free-floating unicellular algae. Distinguishable from surface-attached algae in that the water column itself is colored, not the glass. Discussed in Parts X through XIV.

3. Yellow, brown or tea-colored water

Amber through deep brown coloration, frequently with the water remaining optically clear — objects at the back of the tank are visible but tinted. Caused by dissolved tannins and humic substances from wood, leaves, botanicals, peat or soil-based substrates. Discussed in Parts XV through XVII. Must be distinguished from suspended brown particulate, which reduces clarity as well as adding color.

4. Grey or dusty water

Neutral grey, often with a faintly granular quality visible in a beam of light. Typical of mineral fines from sand, gravel, clay-based or soil-based substrate, and of disturbed detritus. Frequently overlaps with white/milky in mild cases.

5. Visible floating particles

Discrete specks the eye can resolve individually and track as they move. This is physically different from haze: haze is caused by particles too small to resolve. Suggests detritus, plant fragments, food, sloughed biofilm, or substrate coarse enough to see. Almost always a mechanical filtration, flow, or disturbance question rather than a microbiological one.

6. Microbubbles

A milky appearance produced by an enormous number of gas bubbles. The distinguishing features are that the cloud clears progressively from the top downward, that bubbles accumulate on glass and plant leaves, and that the appearance is often tied to equipment operation or to a recent water change. Discussed in Part XXIII.

7. Crystalline or chalky cloudiness

A distinctly white, sometimes slightly sparkling suspension, often appearing rapidly after adding a product or after mixing two products. Caused by mineral precipitation — dissolved ions combining to form an insoluble solid.

Discussed in Part XXV.

Preliminary identification table

This table is a starting point, not a verdict. Every row has exceptions, and the rightmost column exists precisely because visual identification is not definitive.

Appearance	Possible cause	Common timing	Key clues	Tests / observations	Typical response
White / milky, uniform	Heterotrophic bacterial bloom	Days 2–10 of a new setup; after overfeeding; after a die-off; after filter replacement	Develops over hours to a day; does not settle overnight; often follows an organic input	Ammonia and nitrite; recent-change interview; fish behavior	Identify and reduce the organic source; support aeration; usually self-limiting
White / milky, clears top-down within minutes to hours	Microbubbles	Immediately after a water change or filter restart	Bubbles collect on glass and leaves; appearance changes when equipment is switched off	Fill a glass and let it stand — it clears	None required; check for air leaks if persistent
White / grey, settles overnight into a film on surfaces	Substrate or mineral dust	Immediately after filling, rescaping, or adding substrate	Settles when flow stops; captured by fine floss; gritty	Flashlight beam; observe settling in a still glass	Fine mechanical filtration; patience; gentle siphoning
Chalky white, appears suddenly after dosing	Mineral precipitation	Minutes after adding buffers, remineralizers or fertilizers	Onset is nearly instantaneous and tied to a product addition	KH, GH, pH; review of what was added and in what order	Stop combining products in-tank; premix; reassess dosing
Green, faint to intense	Free-floating algae (green water)	Days to weeks; often after sunlight exposure or a long photoperiod	Sample in a white cup is unambiguously green; glass may stay relatively clean	White-cup test; photoperiod and sunlight audit; nutrient review	Light management; UV; blackout; nutrient and plant competition
Yellow to brown, water still clear	Tannins / dissolved organics	Days after adding wood, leaves or botanicals; ongoing for weeks to months	Objects remain sharply visible; color is uniform; a white background reveals amber, not haze	White-cup test; inventory of wood and botanicals	Nothing (aesthetic choice), or carbon / resin / water changes if unwanted
Brown and hazy, clarity reduced	Suspended detritus or fine soil particulate	After substrate disturbance, gravel vacuuming, or heavy waste accumulation	Particles visible in a light beam; settle partially; collect in floss	Inspect filter flow and media; check for filter bypass	Fine mechanical filtration; restore flow; reduce disturbance
Clear water with visible specks	Detritus, food, plant debris, biofilm fragments	Anytime; often after maintenance or feeding	Individual particles resolvable and trackable	Filter inspection; flow pattern observation	Mechanical capture; circulation improvement

Where appearances genuinely overlap

The table above would be dangerous if presented without this section. The following overlaps are common enough that a confident visual call is often unjustified.

Pair that looks alike	Why they overlap	What actually separates them
Bacterial bloom vs microbubbles	Both produce uniform white opacity with no resolvable particles	Time course. Microbubbles clear from the top down in minutes to hours and deposit bubbles on surfaces. A bloom persists for days and does not.
Bacterial bloom vs fine substrate dust	Both are white to grey and uniformly distributed	Settling behavior and onset. Dust appears at the moment of disturbance and settles in still water; a bloom appears hours to days after an organic input and does not settle.

Pair that looks alike	Why they overlap	What actually separates them
Mild green water vs bacterial bloom	A thin algal suspension reads as grey haze in tank lighting	The white-cup test in daylight. Algae are unmistakably green in a white container; bacteria are white or grey.
Tannins vs suspended brown debris	Both make water brown	Whether clarity is preserved. Tannin water is colored but transparent; suspended debris reduces the ability to resolve objects at depth.
Substrate dust vs mineral precipitation	Both are white and appear abruptly	The trigger. Dust follows physical disturbance; precipitation follows a chemical addition and can appear with no substrate involvement at all.
Detritus vs sloughed biofilm	Both appear as brown-grey suspended fragments	Texture and origin. Biofilm fragments are often stringy or gelatinous and follow filter or wood disturbance.
Green water vs green glass algae seen through the water	A green film on the panels tints the whole view	Wipe a small patch of glass clean and look through it, or use the white-cup test on a water sample.

[REASONABLE INFERENCE] Because at least four of these overlapping pairs are separable by simple physical observations (settling in a still glass, the white-cup test, equipment-off observation, and timing relative to a known event), a structured observation sequence should precede any product purchase in the overwhelming majority of cloudy-water cases.

What visual identification cannot do

It cannot confirm which bacterial taxa are present. Visible milkiness establishes that something is suspended, not what it is at the species level.

It cannot rule out a simultaneous chemical problem. Clear-looking water and cloudy water are both compatible with elevated ammonia.

It cannot distinguish a self-limiting condition from an escalating one in a single observation. Duration and direction of change carry information that a snapshot does not.

APPEARANCE-LED DIAGNOSIS

Part III — White or Milky Water

The single most reported appearance, and the one most reflexively misattributed.

"My water turned white" is probably the most common cloudy-water report in the freshwater hobby, and the near-universal reply it receives is "bacterial bloom." That reply is right often enough to be a reasonable prior and wrong often enough to matter. This part works through the alternatives before Part IV examines the bloom itself.

Causes of whitish or milky aquarium water

Heterotrophic bacterial bloom

A rapid proliferation of free-floating bacteria feeding on dissolved organic carbon. Characteristically develops over hours to a day or two, is uniformly distributed, does not settle out in still water, and persists for days. Typically follows an organic input: a new setup, overfeeding, a death, decaying plant material, a disturbed substrate, or a loss of established filter biomass.

Microbubbles

Gas coming out of solution or entrained by equipment. Visually near-identical to a bloom at a glance, and separable in under an hour by observation. The reference behavior: draw a glass of tank water, set it on a counter, and watch. Microbubble cloudiness clears from the top downward as bubbles rise; a bacterial suspension does not clear on that timescale. This is covered in detail in Part XXIII.

Fine substrate or mineral dust

Very fine particles from sand, gravel coatings, clay-based substrates, crushed coral or aragonite, and from broken-down aquasoil granules. Distinguishing features: onset is simultaneous with physical disturbance rather than delayed; the particles have mass and will settle in still water, though extremely fine clay fractions can remain suspended for a long time; and fine mechanical media capture them progressively.

Mineral precipitation

Dissolved ions forming an insoluble solid, most commonly involving calcium, magnesium, carbonates and phosphates. Onset is fast — often within minutes of a dosing event — and the appearance is a distinctly chalky white. Because both the products and the water were clear beforehand, keepers frequently do not connect the cause to the effect. Covered in Part XXV.

Filter disturbance and media dust

Restarting a canister filter that has been off, cleaning a filter aggressively, or adding unrinsed media can eject accumulated fine material into the water column. New activated carbon and some ceramic and clay media shed dust if not rinsed. The timing signature is unmistakable: cloudiness appears within minutes of the filter work.

Other microbial and micro-invertebrate activity

Bacteria are not the only organisms that can cloud water. Following a bacterial increase, populations of protozoa and other microscopic organisms that graze on bacteria can rise and contribute to, or extend, the cloudiness. The hobby-adjacent literature describes this as a succession from a bacterial cloud to a mixed community over a span of days (Aquarium Science, 6.2.3). This is a useful mental model for why some blooms clear on a longer timescale than expected.

[REASONABLE INFERENCE] A "bacterial bloom" that persists beyond the first several days may no longer be a purely bacterial suspension. Microbial succession from bacteria to bacterivorous protozoa is well established in aquatic microbial ecology, though it has not been systematically documented in home aquaria specifically.

Does bacterial cloudiness have a characteristic appearance?

Partially. Descriptions across hobby and retail literature converge on milky white to pale grey, with the observation that it is not green (which indicates algae) and not yellow-brown (which indicates tannins). That much is consistent and useful.

What appearance cannot do is separate bacteria from the two other white conditions — microbubbles and fine dust — because all three are white, uniform, and lack resolvable particles. The differences are in behavior over time and in what preceded the event, not in color.

[CONTESTED CLAIM] Some hobby sources state that bacterial blooms have a distinctive appearance that can be recognized on sight. The research position taken here is that color alone narrows the field to three or four candidates but does not identify among them; behavior over time is required.

Can bacterial blooms be confirmed visually?

No. Visual confirmation of a bacterial bloom in the strict sense would require microscopy or plate counts, which no ordinary hobbyist performs. What a keeper can do is exclude the alternatives:

Exclusion test	How to run it	What the result means
Standing-glass test	Fill a clear glass with tank water and leave it undisturbed on a white surface for a few hours.	Clears from the top down within minutes to an hour: microbubbles. Settles with sediment at the bottom: particulate. Stays uniformly cloudy: consistent with a microbial suspension.
White-cup color check	View the same sample against white paper in daylight.	Green tint: algae. Amber: dissolved organics. White or grey with no tint: bacteria, dust or gas.
Disturbance timeline	Establish exactly when the cloudiness began relative to physical work on the tank.	Simultaneous with disturbance: particulate. Hours to days after an organic input: consistent with a bloom.
Fine-media capture test	Add clean filter floss and check it after 12–24 hours.	Visibly loaded with sediment and the water improves: particulate. Floss looks nearly clean and the water is unchanged: the suspension is finer than the media can capture, which points toward bacteria or very fine colloids.
Equipment-off observation	Briefly switch off the filter and any air pump and observe.	Cloudiness visibly diminishes or the sparkle stops: gas-related. No change: not gas.

[COMMON AQUARIUM PRACTICE] The standing-glass observation is not a laboratory method, but it is cheap, fast, requires nothing a keeper does not already own, and reliably separates three of the four white-water candidates. It is arguably the highest-value single diagnostic step available to a hobbyist.

How reliable is "white cloudy water equals bacterial bloom"?

As a prior probability in a tank that is under about two weeks old and has recently received substrate, wood or fish, it is a reasonable first guess. As a diagnosis applied without checking, it fails in several very common situations:

- Cloudiness that appeared within minutes of a water change is far more likely to be dissolved gas.
- Cloudiness that appeared the instant sand was poured in is substrate fines.

- Cloudiness that appeared within minutes of dosing a buffer or remineralizer is chemical precipitation.
- Cloudiness that appeared immediately after restarting a canister filter is ejected debris.
- Cloudiness in a tank with direct sun exposure may be a low-density algal suspension reading as grey.

[USEFUL RULE OF THUMB] White cloudiness that appeared *at the moment* something physical happened is usually physical. White cloudiness that appeared *a day or more after* something organic happened is usually biological. This is not universal, but it correctly sorts a large majority of reports.

Preserved distinction

DUST is not a BACTERIAL BLOOM. MICROBUBBLES are not a BACTERIAL BLOOM. PRECIPITATION is not a BACTERIAL BLOOM. All four can look identical in a photograph. They differ in onset timing, settling behavior, response to mechanical filtration, and what preceded them — and they require four different responses.

MICROBIOLOGY

Part IV — What a "Bacterial Bloom" Actually Is

The most important scientific distinction in this entire document: heterotrophic bacteria are not nitrifying bacteria.

The hobby uses "bacterial bloom" to describe a visible white suspension in the water column. The term is imprecise in two ways. First, blooms of bacteria can occur without visible cloudiness, and cloudiness can occur without bacteria. Second, and far more consequentially, the popular explanation of what the bacteria are is mechanistically wrong in a way that produces bad advice.

Two fundamentally different bacterial lifestyles

Aquarium water contains, among many other organisms, two functionally distinct groups of bacteria that behave in almost opposite ways.

Property	Heterotrophic bacteria	Nitrifying (chemolithoautotrophic) organisms
Energy and carbon source	Organic compounds — carbohydrates, proteins, fats dissolved in the water	Inorganic nitrogen — ammonia or nitrite — with carbon fixed from dissolved CO ₂
Energy yield per unit substrate	High. Organic matter is energy-dense.	Very low. Ammonia and nitrite oxidation yield little energy, and much of it is spent fixing carbon.
Growth rate	Fast. Doubling times of roughly 20–30 minutes are typical for many heterotrophs under favorable conditions.	Slow. Reported doubling times for ammonia-oxidizing bacteria such as <i>Nitrosomonas europaea</i> range from about 7.4 to 24 hours depending on conditions (Hommes et al. 2003, cited in Gupta et al. 2020).
Preferred habitat	Will grow suspended in the water column when dissolved organics are abundant; also form biofilms	Overwhelmingly surface-attached biofilm on filter media, substrate and hardscape
Visible as cloudiness?	Yes — can reach densities high enough to scatter light strongly	Not realistically. Population densities and growth rates are far too low to produce visible water-column turbidity.
Role in the aquarium	Mineralize organic waste. Beneficial when established in the filter; a nuisance when suspended.	Convert ammonia to nitrite and nitrite to nitrate. The functional basis of "cycling."

[ESTABLISHED SCIENTIFIC PRINCIPLE] Heterotrophic bacteria obtain carbon and energy from organic compounds and grow rapidly when dissolved organic carbon is abundant. Nitrifying organisms are chemolithoautotrophs with characteristically slow growth rates and a strong preference for attached growth. These are different organisms with different ecology.

Is "a bacterial bloom is your beneficial bacteria multiplying" accurate?

Not as usually stated. The energy argument is decisive. Organic compounds provide orders of magnitude more energy per unit mass than ammonia or nitrite, which is why a small pulse of dissolved organic carbon can support an explosive heterotrophic population in the water column while the same tank's nitrifying population remains invisible (Aquarium Science, 6.2.3). Free-floating growth requires enough energy to support reproduction without the advantages of a fixed position in a flow of substrate-bearing water; nitrifiers do not have that energy budget.

[STRONG EVIDENCE] The organisms responsible for visible white-water blooms in aquaria are heterotrophic bacteria feeding on dissolved organic compounds, not nitrifying bacteria. Multiple independent hobby-science and retail-technical sources converge on this, and it follows directly from established microbial energetics.

The steelman: where the popular claim has partial merit

The claim is not entirely without a mechanistic basis, and honest treatment requires saying so. In wastewater treatment, suspended bacterial flocs form in the water column through processes including quorum sensing, and nitrifying organisms can and do live within suspended flocs in activated-sludge systems before attaching as biofilm. A reader raised exactly this objection on the Aquarium Science article and the author acknowledged the nuance while maintaining the core distinction.

Two caveats limit how far this can be carried into aquarium practice. Activated-sludge systems operate at nutrient loadings and biomass concentrations utterly unlike a home aquarium. And even in those systems, nitrifiers are a minority of the floc biomass rather than its cause. The floc-to-biofilm mechanism is, however, a genuinely useful explanation for a phenomenon keepers observe: why a persistently cloudy tank sometimes clears almost overnight once the filter matures.

[CONTESTED CLAIM] Whether nitrifying organisms make any meaningful contribution to visible aquarium blooms is contested at the margins. The mainstream position — that they do not cause the visible cloudiness — is well supported. The wastewater-derived floc argument is mechanistically real but has not been demonstrated to be quantitatively relevant in home aquaria.

Suspended bacteria versus attached biofilms

The distinction between planktonic (free-floating) and sessile (attached) bacterial life is the reason the same organism can be described as a problem in one place and an asset in another. Aquarium biofilters are a fixed-film bioreactor: carrier materials such as ceramic beads and sponges support biofilm growth, with flow controlling biofilm thickness and ensuring oxygen and nutrient transfer into the film (Bagchi et al. 2025, home-aquarium biofilter succession study).

A practical consequence follows. Organic matter in an aquarium will be processed by microorganisms somewhere. The question is where. If a system has abundant, mature, well-oxygenated biofilm surface area, most of that processing happens in the filter, out of sight. If it does not, a larger share happens in the water column, in view. This reframing — that the bloom is a symptom of insufficient established biofilm relative to organic load — is the most actionable model available.

[REASONABLE INFERENCE] Visible water-column blooms are best understood as an imbalance between the rate of dissolved organic carbon input and the capacity of established surface-attached microbial communities to consume it. This model explains why blooms are common in new systems, after filter media replacement, and after sudden organic inputs to otherwise stable tanks.

Microbial succession: why a bloom is not always the same thing on day 8

Bacteria are prey. A dense bacterial population in the water column is a food source for bacterivorous protozoa and other microorganisms, and those populations can rise over a span of days. The Aquarium Science treatment describes the sequence as: dissolved organic compounds released by food and waste, followed by a heterotrophic bacterial increase, followed over days by an increase in minute organisms feeding on those bacteria, which can extend the cloudiness. The collective hobby term for the resulting mixed community is "infusoria."

[REASONABLE INFERENCE] Prolonged cloudiness may reflect a successional community rather than a static bacterial population. This is consistent with established aquatic microbial ecology (the microbial loop) but has not been characterized specifically in home aquaria.

Are the bacteria in a bloom dangerous organisms?

The overwhelming majority of the time, no. Heterotrophic bacteria in aquarium water are predominantly environmental decomposers rather than fish pathogens. The Aquarium Science treatment notes that these bacteria *can* rarely be pathogenic but that the vast majority of the time they are not. That framing — possible but uncommon — is the honest one.

Where risk rises meaningfully is in the extreme organic-overload scenario: a large quantity of food dumped into a tank, or a substantial die-off. The concern there is not the cloudiness itself but the magnitude of decomposition, the associated oxygen demand, and the possibility of opportunistic bacteria overwhelming stressed fish.

[CONTESTED CLAIM] The degree to which bloom bacteria pose direct pathogenic risk to fish is not well quantified in home-aquarium literature. Sources agree that the risk is low in typical cases and higher in gross organic overload, but no reliable quantitative data was located.

MICROBIOLOGY

Part V — Why Bacterial Blooms Happen

Evaluating the standard list of triggers against the underlying mechanism.

Every trigger in the list below can be evaluated against a single question: does it increase dissolved organic carbon in the water column, or does it decrease the established microbial capacity to consume it? Triggers that do neither are unlikely to be real causes, and several commonly cited ones fall into that category.

Proposed trigger	Mechanism	Evidence assessment
Newly established aquarium	No established biofilm anywhere in the system; organics from substrate, wood and first feedings have no established consumers	STRONG EVIDENCE — the most consistently reported trigger across all sources
New nutrient-rich or active substrate	Releases both fine particulate and a pulse of organics and ammonium	STRONG EVIDENCE — explicitly documented in manufacturer literature (ADA, Aqua Soil Amazonia)
New driftwood or botanicals	Leach sugars, tannins and other soluble organics that directly feed heterotrophs	STRONG EVIDENCE — also the direct cause of surface biofilm on new wood (Part XXVII)
Overfeeding	Uneaten food is a large, concentrated organic input	STRONG EVIDENCE — mechanistically direct and universally reported
A dead fish, snail or other animal	Decomposing tissue releases a large, sustained organic pulse	STRONG EVIDENCE
Decaying plant material / plant melt	Same mechanism at lower intensity; substantial in newly planted tanks where emerged growth melts	STRONG EVIDENCE
Increased stocking	Raises waste production, potentially beyond current biofilm capacity	REASONABLE INFERENCE — mechanistically sound; magnitude depends on how large the increase is relative to existing capacity
Replacing filter media or cartridges	Removes the established biofilm that was consuming the organic load	STRONG EVIDENCE — cited as a leading cause; particularly associated with cartridge-based hang-on-back filters
Aggressive filter cleaning	Same mechanism, partial rather than total	STRONG EVIDENCE for aggressive cleaning; effect of light rinsing is much smaller
Complete filter replacement	The new filter has no established community and must develop one	STRONG EVIDENCE
Filter failure or power outage	Flow stops; biofilm loses oxygen and substrate delivery; material may be released on restart	REASONABLE INFERENCE — mechanistically expected; commonly reported
Disturbed substrate / major cleaning	Releases trapped organics and detritus into the water column	STRONG EVIDENCE for the particulate component; REASONABLE INFERENCE for a subsequent bacterial response
Antibiotic or antimicrobial medication	Suppresses established bacterial populations, including in the filter	REASONABLE INFERENCE — mechanistically clear; magnitude varies enormously by compound and dose

Proposed trigger	Mechanism	Evidence assessment
Inadequate aeration	Limits aerobic processing capacity in the biofilm	COMMON AQUARIUM PRACTICE — widely asserted; mechanistically plausible; not independently quantified for home tanks
Coarse substrate trapping food	Food falls below the reach of fish and decomposes in place	REASONABLE INFERENCE — mechanistically sound and frequently reported
Adding bottled bacterial products	Introduces a large bacterial inoculum plus a carrier medium	ANECDOTAL OBSERVATION — commonly reported as producing temporary cloudiness; product formulations vary widely and no controlled data was located
Previously dry wood, rock or decorations returning to water	Residual organics and dried biofilm rehydrate and become available	ANECDOTAL OBSERVATION — plausible; poorly documented

The unifying model

One sentence that covers almost every trigger

A bacterial bloom occurs when the supply of dissolved organic carbon in the water column temporarily exceeds the capacity of established, surface-attached microbial communities to consume it — either because supply rose sharply, or because that established capacity was removed.

This model has real diagnostic power, because it converts "why is my tank cloudy" into two answerable questions: **what recently added organic material to this system**, and **what recently removed established biological capacity from it**. In practice, most reported cases resolve to one or both.

Triggers that are frequently cited but hold up poorly

- **"Water changes cause bacterial blooms."** Water changes remove dissolved organics rather than adding them. Post-water-change cloudiness is far more often dissolved gas (Part XXIII) or disturbed substrate. There is a separate and more defensible argument that very large water changes during cycling slow the establishment of the filter community by removing substrate for growth, but that is a different claim from "causes a bloom."
- **"Hard water causes cloudiness."** Dissolved minerals do not scatter light. Precipitated minerals do. See Part XXVI.
- **"Too much light causes bacterial blooms."** Light drives photosynthetic algae, not heterotrophic bacteria. Confusing these produces the very common error of blacking out a tank that has a bacterial problem.
- **"High nitrate causes bacterial blooms."** Nitrate is not an energy source for heterotrophs in this context. High nitrate may co-occur with high organic load because both reflect accumulated waste, but the causal arrow runs through the organics.

[UNSUPPORTED AQUARIUM FOLKLORE] The claim that light exposure or nitrate concentration drives heterotrophic bacterial blooms is not supported by any mechanism and appears to arise from generalizing algae control advice to a bacterial problem.

MICROBIOLOGY

Part VI — Bacterial Blooms and the Nitrogen Cycle

Related in some circumstances. Not interchangeable in any of them.

This part deliberately does not become a cycling guide. Its purpose is narrower and more useful: to establish exactly how much information about biological filtration status can be read from water clarity. The answer is: much less than the hobby assumes.

Six questions that separate clarity from cycling status

Can a bacterial bloom happen during cycling?

Yes, and this is the most common context. A new aquarium has organic input from substrate, wood, initial feeding or an ammonia source, and has no established biofilm anywhere in the system. Both conditions for a bloom are satisfied simultaneously. This co-occurrence is precisely why the two phenomena get conflated.

Can a bacterial bloom happen in an already cycled aquarium?

Yes. A cycled aquarium has an established nitrifying population sized to its ammonia load. That says nothing about whether it can absorb a sudden large input of dissolved organic carbon. Overfeeding, a death, plant die-off or filter media replacement can produce a bloom in a tank that has been stable for years. The nitrifying population and the heterotrophic processing capacity are related but not identical.

[STRONG EVIDENCE] Established aquariums can and do experience visible bacterial blooms. Blooms are not diagnostic of an incomplete or crashed nitrogen cycle.

Does bacterial cloudiness mean ammonia or nitrite must be elevated?

No. The two have different drivers. Cloudiness is driven by dissolved organic carbon availability; ammonia and nitrite concentrations are driven by the balance between nitrogen input and nitrifying capacity. A tank can have abundant organic carbon and adequate nitrification simultaneously.

Can an aquarium have cloudy bacterial water while ammonia and nitrite remain zero?

Yes, and this is a frequently reported and frequently distressing situation for keepers, who reasonably conclude that their test kit must be broken or their tank must be sick. It is neither.

The mechanism: heterotrophic bacteria consume organic carbon and, in doing so, incorporate nitrogen into their own biomass rather than releasing it as free ammonia. Meanwhile an established or partially established nitrifying population processes what ammonia does appear. The result is a visibly cloudy tank with a clean test panel.

[REASONABLE INFERENCE] Cloudy water with zero ammonia and zero nitrite is a coherent and expected state, not a contradiction. It indicates that the system is processing nitrogen adequately while carrying more suspended organic-consuming biomass than the established biofilm is absorbing.

Can an aquarium have dangerous ammonia without cloudy water?

Yes, and this is the more dangerous asymmetry. Ammonia is a dissolved gas in equilibrium with ammonium ion. It has no color and no light-scattering properties at aquarium concentrations. A tank can be optically perfect and lethal. This is the core of Part XXXVI.

[ESTABLISHED SCIENTIFIC PRINCIPLE] Ammonia and nitrite are colorless dissolved species at aquarium concentrations and produce no visible change in water appearance. Water clarity carries no information about their concentration.

Does clear water mean an aquarium is cycled?

No. Clear water means nothing is scattering light. Newly filled tanks with washed inert gravel and no organic input are frequently crystal clear and completely uncycled. Conversely, some fully mature planted aquariums pass through cloudy phases after rescaping.

The relationship, stated precisely

Tank state	Clarity	Biological filtration status	Correct interpretation
New setup, day 3, milky	Cloudy	Almost certainly incomplete	The two co-occur but the cloudiness is not the evidence — test results are
New setup, day 3, crystal clear	Clear	Almost certainly incomplete	Clarity provides false reassurance. Test.
Established tank, milky after overfeeding	Cloudy	Likely intact	Organic overload, not a cycling failure
Established tank, milky after cartridge replacement	Cloudy	Possibly compromised	Both the bloom and a potential ammonia risk share one cause: removed biofilm
Established tank, clear, ammonia 0.5 ppm	Clear	Compromised	A genuine emergency with no visual signal at all
Planted tank, tea-colored, clear	Colored but clear	Unrelated to color	Coloration is a dissolved-organics phenomenon and carries no cycling information

Preserved distinction

WATER CLARITY and BIOLOGICAL FILTRATION STATUS are related only through shared upstream causes — principally organic load and the presence or absence of established biofilm. They are not proxies for one another in either direction. A test kit answers the cycling question. Nothing about the appearance of the water does.

RISK ASSESSMENT

Part VII — Are Bacterial Blooms Dangerous?

Separating the cloudiness from the condition that produced it.

This is where the welfare stakes live. The question "is a bacterial bloom harmful" is badly formed, because it conflates a visual phenomenon with the underlying situation. A more useful formulation: **does the condition that is producing this cloudiness pose a risk to the animals, independent of how the water looks?**

Direct risk from the bacteria themselves

Low in typical cases. Suspended heterotrophic bacteria in an aquarium are predominantly environmental decomposers. The commonly repeated hobby position that blooms are "almost always harmless, temporary, and a normal part of establishing a new tank" is a fair summary of the typical case, though it should not be extended to atypical ones.

[COMMON AQUARIUM PRACTICE] The consensus position across hobby, retail and hobby-science sources is that visible bacterial blooms in new aquariums are usually not directly harmful. This is a broad convergence of reported experience rather than a controlled finding.

Indirect risks, which are the real ones

Oxygen consumption and biological oxygen demand

This is the most substantiated indirect risk. A large population of aerobic heterotrophic bacteria consumes oxygen, and so does the ongoing decomposition of the organic material feeding them. In limnology and wastewater engineering, biological oxygen demand is the standard measure of exactly this: oxygen consumed by microorganisms breaking down organic matter. A dense bloom in a warm, poorly aerated, heavily stocked tank is a plausible route to depressed dissolved oxygen.

[ESTABLISHED SCIENTIFIC PRINCIPLE] Aerobic microbial decomposition of organic matter consumes dissolved oxygen. In enclosed systems with limited gas exchange, high organic loading can depress dissolved oxygen concentrations.

Two aggravating factors deserve explicit mention. Warm water holds less dissolved oxygen than cool water, so a tropical tank has less headroom to begin with. And oxygen demand does not pause overnight, whereas in planted tanks photosynthetic oxygen production does — making the pre-dawn hours the highest-risk window.

Underlying ammonia

A bloom triggered by a death, a food dump, or filter media removal may be accompanied by elevated ammonia from the same event. Here the cloudiness is a visible signal of an invisible hazard — useful information, but the hazard is the ammonia, not the haze.

Ongoing decomposition of a large organic mass

The extreme case is a substantial quantity of food or a dead animal in the tank. Aquarium Science describes this scenario as requiring urgent large-scale water changes rather than patience, on the basis that the objective is to remove the food and dissolved organics from the system as quickly as possible.

[CONTESTED CLAIM] The recommendation for very large (90–100 percent) emergency water changes in gross organic overload situations is advocated by some experienced sources and regarded as excessive by others who prioritize parameter stability. The disagreement is about the relative risk of a large parameter swing versus continued exposure. Both camps agree the organic material itself must be physically removed.

When "just wait it out" is reasonable

- The tank is new, lightly stocked or unstocked, and the cloudiness developed without an identifiable acute event such as a death.
- Ammonia and nitrite are at or near zero, or the tank is fishless.
- All livestock are behaving normally — feeding, normal respiration, using the full water column.
- There is no foul odor.
- The cloudiness is stable or improving rather than intensifying day over day.
- Aeration and surface agitation are adequate.

When waiting is not appropriate

Red flags that override "wait it out"

Fish gasping at the surface, or gathering at the outflow where oxygen is highest.

Rapid or labored gill movement across multiple animals.

Any detectable ammonia or nitrite in a stocked tank.

One or more animals already dead, or a dead animal that has not been located.

A strong foul, rotten or sulfurous odor.

A known large organic input — a food dump, a dead fish left in place, a large mass of decaying plant matter.

Loss of filter flow, or a filter that has been off for an extended period.

Cloudiness that is intensifying rather than stabilizing after several days.

The logic of this list is worth making explicit: none of these red flags are about the cloudiness. Every one of them is about the underlying condition or about livestock welfare. That is the correct place to put the attention.

Separating the cloudiness from the underlying condition

Situation	Is the cloudiness itself harmful?	Is the underlying condition harmful?	What deserves attention
New fishless tank, day 4, milky	No	No — this is expected succession	Nothing urgent. Continue the cycling process.
New stocked tank, day 4, milky, ammonia 0	Not directly	Watch closely — nitrification is likely incomplete	Testing frequency, feeding restraint, aeration
Established tank, milky after a food dump	Not directly	Yes — large organic load and oxygen demand	Physically remove the food; increase aeration; test ammonia
Established tank, milky, fish gasping	Possibly contributing via oxygen demand	Yes	Aeration immediately; test; find and remove the organic source
Tank milky after replacing all filter media	Not directly	Yes — nitrifying capacity has been removed	Ammonia and nitrite testing; treat as a partial recycle
Tank milky after a dead fish was found and removed	No	Diminishing — the source is gone	Aeration; check for a second casualty; monitor parameters

TIME COURSE

Part VIII — How Long Do Bacterial Blooms Last?

Why no honest universal number exists, and what actually controls duration.

Reported durations in hobby literature range from under 24 hours to several weeks. That range is not sloppiness — it reflects genuine variance driven by factors that differ enormously between tanks. Any specific number offered as universal should be treated as a guess.

[ANECDOTAL OBSERVATION] Commonly reported durations cluster in the range of a few days to two or three weeks, with new-setup blooms often described as resolving within one to two weeks. No controlled study establishing a duration distribution for home aquaria was located during this research.

Factors that plausibly influence duration

Factor	Direction of effect	Reasoning	Evidence level
Amount of organic material available	More organics, longer bloom	The population persists as long as its energy source does	ESTABLISHED PRINCIPLE
Whether the organic source is ongoing or a one-off	Ongoing sources extend indefinitely	A decaying animal or continuous overfeeding keeps supplying substrate	ESTABLISHED PRINCIPLE
Maturity and surface area of the filter biofilm	More mature biofilm, faster resolution	Attached communities outcompete and consume suspended populations	REASONABLE INFERENCE
Total biofilm surface area in the system	More surface area, faster resolution	Greater attached capacity relative to load	REASONABLE INFERENCE
Dissolved oxygen and circulation	Better aeration, faster processing	Aerobic decomposition is oxygen-limited when oxygen is scarce	REASONABLE INFERENCE
Temperature	Warmer, faster microbial turnover but also lower oxygen	Metabolic rates rise with temperature; oxygen solubility falls	ESTABLISHED PRINCIPLE
Grazing by protozoa and micro-invertebrates	Grazing shortens the bacterial phase but may extend visible cloudiness	Predators consume bacteria but are themselves suspended and visible	REASONABLE INFERENCE
Mechanical filtration fineness	Minimal direct effect on bacteria	Individual bacteria are generally below the capture threshold of ordinary aquarium media (Part LXXI)	ESTABLISHED PRINCIPLE
UV sterilization	Dramatically shortens the visible phase	Inactivates suspended organisms passing through the unit	STRONG EVIDENCE for the visual effect; does not address the cause
Water changes	Variable — see Part IX	Dilutes both the organisms and their substrate; effect depends on the source of the organics	CONTESTED
Lighting and photoperiod	No meaningful effect	Heterotrophic bacteria are not photosynthetic	ESTABLISHED PRINCIPLE

Evaluating three common duration claims

"Bacterial blooms always disappear in two to three days."

Not supported as an "always." Some do; many reported cases run substantially longer, particularly in new tanks with active substrates or with an ongoing organic source. The claim is best understood as a description of a favorable case presented as a rule.

[CONTESTED CLAIM] A two-to-three-day resolution is within the reported range but is not typical enough to state as a rule. Advising a keeper to expect it sets up unnecessary alarm when day four arrives.

"Never do a water change during a bacterial bloom."

This rule has a real rationale in one specific context and is misapplied outside it. The rationale: during fishless cycling, removing ammonia and organics slows the establishment of the microbial community that will eventually resolve the bloom, so water changes prolong it. Aquarium Science states this position directly, arguing that changing water removes the food and ammonia necessary for the filter to build up biofilm.

The misapplication: extending that reasoning to a stocked tank with a dead animal in it, or to a tank with detectable ammonia, or to a tank where fish are showing respiratory distress. In those situations the priority is not optimizing the microbial timeline, it is livestock welfare.

[CONTESTED CLAIM] Whether water changes prolong bacterial blooms is genuinely disputed. The mechanism proposed (removal of substrate needed for biofilm establishment) is coherent for fishless cycling. Its applicability to stocked tanks with welfare concerns is rejected by most welfare-oriented sources. The disagreement is real and should not be papered over.

"Water changes make bacterial blooms worse."

This is a stronger claim than the previous one and is less defensible. A water change removes both suspended organisms and dissolved organics; there is no mechanism by which it adds organic carbon. The most likely origin of this belief is misattribution: post-water-change cloudiness from dissolved gas or disturbed substrate is interpreted as a worsening bloom. That is a different phenomenon entirely (Parts XXIII and XXIV).

[UNSUPPORTED AQUARIUM FOLKLORE] No mechanism supports water changes increasing heterotrophic bacterial populations. The observation that generates this belief is almost certainly microbubble or substrate cloudiness misidentified as a bloom.

INTERVENTION ANALYSIS

Part IX — Should You Water-Change a Bacterial Bloom?

Four situations, four different answers. This is why a universal rule fails.

The reason this question generates such fierce disagreement in the hobby is that both camps are arguing from real cases. The "never change" camp is thinking about a cycling tank. The "always change" camp is thinking about a tank with a dead fish in it. Both are right about their own scenario and wrong to generalize.

What a water change actually does to each variable

Variable	Effect of a partial water change	Notes
Suspended bacteria	Reduced proportionally to volume removed	A 50 percent change removes roughly half the suspended population. They reproduce quickly, so the visual improvement is often short-lived if the substrate remains.
Dissolved organic carbon	Reduced proportionally	This is the meaningful effect — it removes the fuel, not just the fire.
Ammonia and nitrite	Reduced proportionally	Directly relevant to livestock safety when either is elevated.
Dissolved oxygen	Usually improved	New water is typically closer to saturation, and the process itself agitates the surface.
Established filter biofilm	Essentially unaffected	The biofilm is attached; changing water does not remove it. This is a separate question from cleaning the filter.
Rate of biofilm establishment (in a cycling tank)	Potentially slowed	Removing substrate reduces the resource driving community growth. This is the legitimate core of the "don't change water" argument.
Particulate matter	Reduced, plus additional removal if siphoned from the substrate	For dust-driven cloudiness this is often the most effective single action.

Four situations compared

Situation	Water change?	Rationale	Cautions
Mild cloudiness, ammonia and nitrite zero, fish behaving normally, new tank	Optional; a modest change is low-risk but may not shorten the bloom, and in a cycling tank may lengthen it	Nothing is currently threatening the animals. Time and biofilm establishment are doing the work.	Avoid changing water purely because the tank looks unappealing; that is the impulse that produces repeated large changes with no improvement.
Cloudiness with detectable ammonia or nitrite in a stocked tank	Yes — this is the priority action	The nitrogenous compounds are the hazard. Dilution reduces exposure immediately.	Match temperature; dechlorinate; do not simultaneously clean the filter, which would reduce the capacity that is already insufficient.
Cloudiness with identifiable dead or decaying organic material	Yes — combined with physically removing the material	The organic mass is an ongoing source of both oxygen demand and ammonia. Dilution alone does not remove it.	Physical removal matters more than the water change. Search thoroughly — a missing animal is a common hidden cause.
Cloudiness with signs of oxygen stress — gasping, surface gathering, rapid gill movement	Yes, and increase aeration first	Aeration acts within minutes; a water change also brings in oxygenated water and dilutes oxygen-demanding material.	Aeration is the faster intervention. Do it before, not after, preparing water.

The decision rule that replaces both universal rules

Do not ask "should I water-change a bacterial bloom." Ask "is anything in this tank currently threatening the animals?" If yes, a water change is one of several tools for reducing that threat and the cloudiness is incidental. If no, a water change is an aesthetic choice with a possible cost to the microbial timeline, and patience is defensible.

What water changes will not do

- They will not remove dissolved tannins meaningfully in one pass — dilution is proportional, and wood keeps releasing more.
- They will not eliminate a green-water population, which regrows from the remaining cells under the same light and nutrient conditions.
- They will not fix inadequate mechanical filtration, though they will temporarily mask it.
- They will not address microbubbles, which resolve on their own and may briefly worsen from the refill.
- They will not, on their own, remove a decaying animal.

ALGAE

Part X — Green Water

A living suspension of free-floating microalgae — biologically unrelated to the algae growing on the glass.

Green water is a phytoplankton bloom occurring inside a glass box. The organisms are unicellular photosynthetic algae suspended in the water column, reproducing in response to light and dissolved nutrients. This makes it categorically different from every other cause of cloudiness in this document: it is not debris, not gas, not a decomposition product, and not a chemical precipitate. It is a population.

Which organisms are involved

Freshwater green-water communities are typically dominated by small unicellular green algae, with *Chlorella* and *Scenedesmus* the most commonly named genera in aquaculture and hobby culture contexts, alongside the flagellate *Euglena*. Cultured green water sold commercially for feeding *Daphnia* and *Moina* is described as consisting primarily of single-celled species including *Chlorella*, *Scenedesmus*, *Nannochloropsis* and *Euglena* (Hydralife). Notably, commercial aquarium-derived cultures are often sold explicitly as community cultures rather than isolated strains, reflecting a stable functional phytoplankton community rather than one species.

[STRONG EVIDENCE] Green water in freshwater systems is caused by suspended unicellular photosynthetic algae, commonly including *Chlorella*, *Scenedesmus* and *Euglena*. It is a community rather than a single organism in most real-world cases.

Cell sizes are relevant to treatment. Commercial phytoplankton cultures describe cells in the range of roughly 2 to 15 micrometres depending on species. That size range sits above individual bacteria and below most detritus, which has direct consequences for what filtration can and cannot capture (Part LXXI).

What drives a green-water population

Three resources must all be available: light of adequate intensity and duration, dissolved inorganic nutrients (principally nitrogen and phosphorus, with micronutrients including iron), and dissolved carbon dioxide. Water temperature and the absence of effective competition or grazing modulate the outcome.

Resource	Role	Typical aquarium source	Practical lever
Light	Drives photosynthesis; usually the limiting factor keepers can actually control	Aquarium fixture, room lighting, and above all direct or indirect sunlight	Photoperiod reduction; intensity reduction; blocking sunlight
Nitrogen	Structural and metabolic requirement	Fish waste, decomposition, fertilizers, tap water	Feeding discipline; maintenance; plant competition
Phosphorus	Structural and metabolic requirement	Fish food, waste, fertilizers, some tap water	Feeding discipline; phosphate media if genuinely limiting
Micronutrients (esp. iron)	Enzyme cofactors; often supplied deliberately in planted tanks	Fertilizer dosing; substrate leaching	Dosing review
Dissolved CO ₂	Carbon source	Respiration, atmosphere, injection systems	Rarely a practical lever
Absence of competition	Fast-growing plants take up the same nutrients and shade the water column	Under-planted or newly planted tanks lack this	Floating plants; increased fast-growing plant mass

Resource	Role	Typical aquarium source	Practical lever
Absence of grazing	In outdoor and pond systems, zooplankton crop algae; in most aquariums this control is absent	Filtered, fish-stocked aquaria have few surviving grazers	Rarely a practical lever indoors

Green water versus surface-attached algae

This distinction is essential because it determines which treatments can possibly work. Free-floating algae pass through the filter and through any UV unit in the flow path. Attached algae never leave their surface.

Algae type	Where it lives	Appearance	Does it cloud the water column?	Responds to UV?
Green water (phytoplankton)	Suspended in the water column	Green tint to the water itself	Yes — this is the definition	Yes — highly effective
Green spot algae	Hard-attached to glass and slow-growing leaves	Small hard green dots	No	No
Green dust / film algae	Loosely attached to glass	Uniform green film on panels	No — but tints the view through the glass	Partially, if it releases motile spores into the water
Hair / thread algae	Attached to plants, hardscape, equipment	Long green filaments	No	No
Black beard algae	Attached to leaf edges, wood, equipment	Dark tufts	No	No
Diatoms ("brown algae")	Attached film on all surfaces	Brown dusty coating	No — see Part XXIX	No
Cyanobacteria	Mats on substrate, glass, plants	Slimy sheets, often blue-green, often with an earthy smell	Not typically in aquaria — see Part XXVIII	Only the planktonic forms, which are uncommon in aquaria

Preserved distinction

GREEN WATER is not SURFACE ALGAE. They are different organisms occupying different niches, and almost no intervention works on both. A UV sterilizer is close to a complete solution for one and completely irrelevant to the other. An algaecide, an algae scraper, and an algae-eating animal are aimed at the other and do essentially nothing for the first.

Aquarium age and green water

Green water is reported both in very new aquariums — where nutrient levels are unstable and plant uptake has not established — and in mature aquariums that undergo a change in light exposure, such as seasonal sun angle shifts or a new lighting fixture. Unlike heterotrophic bacterial blooms, green water is not particularly associated with tank age on its own. Light exposure is the stronger predictor.

ALGAE

Part XI — How to Identify Green Water

Simple observations that reliably separate an algal suspension from a bacterial one.

Green water at high density is unmistakable. Green water at low density is one of the most commonly misidentified conditions in the hobby, because a thin algal suspension viewed through aquarium lighting, against a dark background, reads as generalized grey haze rather than as green.

Does green water always look green?

No. Three factors suppress the perception of green in situ. First, density: at low cell counts the tint is subtle. Second, viewing conditions: dark substrate and dark backgrounds absorb the light that would otherwise reveal the color, and many aquarium LED fixtures shift color perception. Third, path length: a short viewing path through a nano tank shows far less color than a long path through a large one.

[REASONABLE INFERENCE] Mild green-water blooms are readily mistaken for bacterial haze because perceived color depends on cell density, path length, background and illumination spectrum. Removing the sample from those conditions is the obvious correction.

The white-cup test

This is the single most useful field test in this entire document, and it costs nothing.

- 1 Fill a clear, colorless glass or white cup with aquarium water.
- 2 Carry it away from the aquarium and away from aquarium lighting.
- 3 View it in natural daylight — near a window is sufficient — against a sheet of white paper.
- 4 Compare it, if possible, with a second glass of tap or dechlorinated source water held alongside it.

What the sample looks like	Interpretation
Distinctly green, even faintly	Suspended algae (green water)
White or grey, no color cast	Bacterial suspension, fine dust, or microbubbles
Amber, yellow or brown but transparent	Dissolved organics / tannins
Clear with visible specks	Particulate matter — detritus or substrate
Clear with fine rising bubbles that vanish within an hour	Microbubbles
Looks identical to the tap water sample	The problem may be in the viewing conditions, on the glass, or in lighting — see Part LVII

[COMMON AQUARIUM PRACTICE] The white-cup or white-background test is widely used and is mechanistically sound: it removes the confounding effects of tank lighting, background color and viewing path length. Its limitation is that a very dilute suspension may still not show color in a small sample volume.

Other useful observations

The flashlight beam

Shining a narrow, bright beam through the water in a darkened room reveals suspended particulate by scattering — the same effect that makes a light beam visible in a dusty room. This helps establish that *something* is suspended and gives a sense of particle size and movement, but it does not distinguish algae from bacteria from dust. It is a presence test, not an identification test.

White background behind the aquarium

Temporarily placing a sheet of white paper or card behind the tank, with room lighting rather than aquarium lighting, makes color casts far more visible than a dark background does. Useful as a quick check before drawing a sample.

The glass-wipe check

Wipe a small patch of the front glass clean and look through that patch. If the view clarifies dramatically, the problem was a film on the glass, not in the water. This distinguishes green film algae from green water — a distinction that matters because they respond to completely different treatments.

Depth-of-view estimation

A rough field analogue of a Secchi measurement: note whether the back glass, the hardscape at the back, or only the front third of the tank is visible, and record it. Repeating the observation daily converts a static impression into a trend, which is far more informative. A condition that is improving needs different handling from one that is intensifying.

[USEFUL RULE OF THUMB] Recording whether the back wall of the aquarium is visible, each day at the same time and under the same lighting, is a practical low-effort proxy for tracking turbidity trend without instruments.

Green water versus bacterial bloom — direct comparison

Feature	Green water	Bacterial bloom
Color in a white cup in daylight	Green	White or grey
Typical trigger	Light exposure plus available nutrients	Dissolved organic carbon input, or loss of established biofilm
Onset speed	Days to weeks, often gradual	Hours to a day or two, often abrupt
Relationship to sunlight	Strong — direct sun is a classic trigger	None
Effect of blackout	Substantial — removes the energy source	None — the organisms are not photosynthetic
Effect of UV	Highly effective	Effective at clearing, does not address the cause
Effect of fine mechanical filtration	Limited — cells are small and reproduce continuously	Minimal — individual bacteria are below typical media capture size
Effect of reducing feeding	Indirect and slow	Direct and meaningful
Day/night oxygen pattern	Pronounced — produces oxygen by day, consumes it by night	Consumes oxygen continuously
Self-resolving?	Only if the driving conditions change	Often, as biofilm establishes and organics are consumed

ALGAE

Part XII — Why Green Water Happens

The interaction between light and nutrients, not either alone.

Reducing green water to "too many nutrients" is the single most common analytical error in hobby algae discussion. Heavily fertilized high-tech planted aquariums routinely run at nutrient concentrations that would be considered extreme by fish-only standards and remain free of green water for years. Meanwhile, tanks with low measured nitrate develop green water when moved next to a sunny window. Nutrients are necessary; they are rarely the variable that changed.

Triggers, assessed

Trigger	Mechanism	Assessment
Direct sunlight on the aquarium	Sunlight delivers intensity far beyond typical aquarium fixtures and varies seasonally	STRONG EVIDENCE — the most consistently reported single trigger
Excessively long photoperiod	Extends the daily energy budget available to algae	COMMON AQUARIUM PRACTICE — widely observed; reducing photoperiod is a standard and generally effective first response
Excessive light intensity	Especially where plant growth cannot use the available light	COMMON AQUARIUM PRACTICE
New aquarium instability	Nutrients are released before plant uptake and microbial processing establish	REASONABLE INFERENCE — consistent with reported patterns
Overfeeding	Adds nitrogen and phosphorus to the water column	REASONABLE INFERENCE — contributes but rarely sufficient alone
Fertilization	Adds the same nutrients deliberately	REASONABLE INFERENCE — in well-planted tanks this rarely causes green water; in sparsely planted tanks with strong light it can contribute
Disturbed substrate	Releases stored nutrients into the water column	REASONABLE INFERENCE
Outdoor tubs and ponds	Full-spectrum unfiltered sunlight, often with high nutrient loading	STRONG EVIDENCE — green water is near-universal in outdoor culture and is frequently cultivated deliberately
Seasonal changes in sun angle	A window that delivered no direct sun in winter may deliver hours of it in spring	REASONABLE INFERENCE — explains sudden green water in tanks that have been stable for months
Loss of plant mass	Removes the competitor for the same dissolved nutrients	REASONABLE INFERENCE — consistent with the frequent report of green water following a heavy prune or a plant die-off

The light-nutrient interaction

A more accurate model treats green water as the outcome of an imbalance rather than of a single excess. Algae bloom when light energy exceeds what the existing photosynthetic biomass (higher plants) can convert, and dissolved nutrients are simultaneously available in the water column rather than being sequestered in plant tissue. Either factor alone is usually insufficient:

- **High light, low water-column nutrients, dense fast-growing plants:** plants outcompete algae for nutrients and shade the column. Green water is uncommon.

- **High nutrients, low light:** algae lack the energy to build the population. Green water is uncommon.
- **High light, high water-column nutrients, sparse plants:** the classic green-water condition.
- **Direct sunlight, any nutrient level:** sunlight is intense enough that even modest nutrient availability supports a bloom. This is why the sunlight case is reported so consistently.

[REASONABLE INFERENCE] Green water is better modeled as a light-to-plant-uptake imbalance than as a nutrient excess. This model explains the otherwise puzzling observation that heavily fertilized planted tanks are often less prone to green water than lightly fertilized sparsely planted ones.

Why this matters for treatment

If green water is a nutrient problem, the intervention is to strip nutrients — which in a planted tank harms the plants that are the main long-term defense. If it is a light-versus-uptake imbalance, the interventions are to reduce light, increase plant mass, or physically remove the algae population. The second framing produces better advice and is better supported.

ALGAE

Part XIII — Is Green Water Harmful?

Deliberately cultivated in aquaculture; treated as a crisis in the display hobby. Both positions have a basis.

Green water occupies an unusual position: it is simultaneously a nuisance that display keepers work hard to eliminate and a valuable resource that breeders and aquaculturists deliberately produce. Any honest treatment has to explain both.

Green water as a cultivated resource

Green water has a long history of intentional use in aquaculture. It is described as improving water quality in rearing systems while sheltering cultured organisms from direct sunlight and thereby reducing stress, with documented survival benefits in larval culture — in one cited example, adding live microalgae to cobia rearing water raised larval survival from 8.2 percent to 23.3 percent (Kwan et al., *Fish and Shellfish Immunology Reports*).

In the hobby, green water is a standard food for *Daphnia* and *Moina*, and through them an indirect food chain for fry. The American Killifish Association describes green water as an excellent food for *Daphnia* and for zooplankton such as rotifers and larger protozoa, making it useful for fry rearing, and notes that this runs contrary to the widely accepted rule of keeping fry in clean water while nevertheless being demonstrated to work by many aquarists.

[STRONG EVIDENCE] Green water is deliberately cultured in aquaculture and by hobbyist breeders as a live-food substrate and a rearing medium, with documented benefits for larval survival in several species. It is not inherently toxic.

One important nuance for downstream content: fry generally do not eat the algae directly. *Chlorella* cells are described as too large for very small fry to digest, and green water functions mainly by supporting the zooplankton and microorganisms that fry actually consume. Presenting green water as "fry food" without that intermediate step overstates the mechanism.

Real risks associated with green water

Day and night oxygen cycling

This is the most substantiated risk. A dense algal population photosynthesizes during the light period, producing oxygen, and respire continuously, consuming it. The result is a diel oxygen swing: highest in late afternoon, lowest just before lights come on. In a dense bloom in a warm, heavily stocked tank, the pre-dawn minimum is the risk window.

[ESTABLISHED SCIENTIFIC PRINCIPLE] Photosynthetic organisms produce oxygen in light and consume it continuously through respiration, generating diel dissolved-oxygen cycles in productive waters. This is standard limnology and applies directly to dense aquarium blooms.

The inverse risk also exists and is less discussed: intense photosynthesis in a very dense bloom can drive dissolved oxygen above saturation. Veterinary sources note that heavy algal blooms have been implicated in gas supersaturation by producing more oxygen than can diffuse out of the water, and that pond oxygen levels are commonly kept below about 125 percent for this reason (WikiVet; Toledo Zoo fish health Q and A). This is primarily a pond and outdoor-culture concern rather than a typical indoor aquarium one.

pH variation

Photosynthesis consumes dissolved carbon dioxide, which raises pH; respiration releases it, which lowers pH. In a dense bloom with low carbonate buffering, the resulting daily pH swing can be substantial. In well-buffered water it

is muted. This is a real mechanism, most relevant in soft-water and outdoor systems.

[ESTABLISHED SCIENTIFIC PRINCIPLE] Photosynthetic CO₂ uptake raises pH and respiratory CO₂ release lowers it. The magnitude of the resulting diel pH swing is inversely related to carbonate alkalinity.

Loss of visibility

Underrated as a welfare consideration rather than a chemical one. If a keeper cannot see the animals, they cannot observe feeding behavior, detect disease, notice a missing animal, or find a casualty before it decomposes. A dense bloom degrades the keeper's ability to provide care even when the water chemistry is fine.

[REASONABLE INFERENCE] The most defensible welfare argument against dense green water in a display aquarium is not toxicity but loss of observation. Undetected mortality and undetected disease are genuine risks when livestock cannot be seen.

Why "green water equals toxic aquarium" is an oversimplification

- The organisms are the same class of primary producers that support natural aquatic food webs and are deliberately cultured for that purpose.
- Fish are routinely and successfully raised in green water in aquaculture and by hobbyist breeders.
- The identified risks are indirect — oxygen cycling, pH swing, loss of visibility — and are density-dependent and system-dependent rather than intrinsic.
- Green water does not generally coincide with elevated ammonia; a functioning algal population actively takes up inorganic nitrogen.

The balanced position

Green water in a moderately stocked, adequately aerated indoor aquarium is primarily an aesthetic and observational problem rather than a toxicity problem. Green water in a dense bloom, in a warm, heavily stocked, poorly aerated system, or in soft water, carries real oxygen and pH risks that deserve attention. Both statements are true and neither is a general rule.

Considerations for specific groups

Group	Considerations in green water
Adult fish, moderate stocking	Generally tolerant. Monitor for pre-dawn respiratory signs in dense blooms.
Fry	Often benefits from the associated microfauna; this is a recognized rearing method. Density and oxygen still matter.
Shrimp	Generally tolerant; grazing invertebrates benefit from the associated biofilm and microorganisms. Sensitive to sudden large water-parameter changes made in an attempt to clear the water — the treatment poses more risk than the algae.
Snails	No specific concern identified.
Labyrinth fish (bettas, gouramis)	Can breathe atmospheric air, which provides a degree of insulation from dissolved-oxygen depression. This should not be treated as a licence to allow low oxygen — the rest of the community has no such option.
Heavily stocked tanks	Highest risk group for the nighttime oxygen minimum. Supplemental aeration is a reasonable precaution.

[ANECDOTAL OBSERVATION] Claims of species-specific sensitivity to green water beyond the general oxygen and pH mechanisms were not substantiated by the sources reviewed. This document declines to assert species-specific harm without evidence.

ALGAE

Part XIV — Treating Green Water

Including a detailed examination of UV sterilization, the one intervention that is close to definitive.

Green water is unusual among the conditions in this document in that a highly effective, well-understood physical treatment exists. That treatment is UV, and understanding why it works so well here and so poorly elsewhere clarifies a great deal about treatment matching generally.

Intervention comparison

Intervention	Mechanism	Effectiveness for green water	Limitations
Reduce photoperiod	Reduces daily energy available to algae	Moderate; often slows a bloom but rarely eliminates an established one alone	Ineffective if sunlight is the light source; slow
Eliminate direct sunlight	Removes the dominant energy input	High where sunlight is the driver — frequently the single most effective non-equipment action	Requires physically moving the tank or blocking the window
Complete blackout (several days, tank fully covered)	Removes light entirely; algal cells die off	Often high for green water specifically	Stressful to plants; must be total, including ambient room light; does nothing about the conditions that caused it, so recurrence is common; irrelevant to every non-algal cause of cloudiness
UV sterilizer	UV-C radiation damages nucleic acids in cells passing through the unit	Very high when correctly sized and plumbed	Depends entirely on delivered dose; does nothing to attached algae or to the conditions driving the bloom
Water changes	Dilutes the population and dissolved nutrients	Low to moderate alone — the remaining cells regrow under the same conditions	Useful as an adjunct; not a solution
Fine mechanical filtration / polishing	Physical capture	Low to moderate — cells around 2–15 micrometres are at or below the practical capture range of ordinary fine media	Media clogs quickly; population reproduces continuously
Diatomaceous earth filtration	Very fine physical capture down to a few micrometres	High as a rapid clearing method	Specialist equipment; clogs rapidly in a dense bloom; does not prevent recurrence
Floating plants	Shade the water column and compete for dissolved nutrients	Moderate to high as a preventive and long-term measure	Slow to act on an established bloom; changes the aesthetic; requires ongoing harvesting to actually export nutrients
Increased fast-growing plant mass	Nutrient competition	Moderate as prevention	Requires the plants to be growing and, critically, to be pruned and removed — uptake without export only stores nutrients
Algaecides	Chemical toxicity to algal cells	Variable; can work but kills the population all at once	A mass die-off releases the algal biomass as dissolved organics, which can drive a subsequent bacterial bloom and an oxygen sag; risk to plants and to invertebrates; treats the symptom only
Daphnia introduced as grazers	Filter-feeding on the algal population	Effective in fishless containers; impractical in stocked tanks	Fish eat the Daphnia
Flocculant / clarifier	Aggregates cells into larger flocs for mechanical capture	Can clear the water temporarily	Requires adequate mechanical filtration downstream or the flocs simply settle and later redistribute; does not stop reproduction

[CONTESTED CLAIM] The effectiveness of a three-day blackout for green water is widely reported but variable in practice. Failures are commonly attributable to incomplete light exclusion or to unchanged underlying conditions. It is a reasonable attempt, not a reliable protocol.

UV sterilization examined in detail

UV works by exposing water flowing past a UV-C lamp to radiation that damages the nucleic acids of organisms in the water, preventing reproduction. The critical concept is **dose**: the product of radiation intensity and exposure time. Aquaculture engineering sources are explicit that UV units cannot be compared by wattage alone, and that the full amount of UV energy required must reach the organism after the energy leaves the lamp, passes through the quartz sleeve, accounts for lamp aging, and passes through any turbidity and color in the water (Integrated Aqua Systems technical brief).

The five variables that determine whether a UV unit works

Lamp wattage	Determines the radiation output available. A necessary but far from sufficient specification.
Flow rate through the unit	Determines contact time. The same lamp delivers a high dose at low flow and an inadequate dose at high flow. This is the variable keepers most often get wrong, because faster flow feels like more water treated.
Contact / dwell time	The duration each parcel of water spends inside the irradiated chamber. Determined by chamber geometry and flow rate together.
Lamp age	UV-C output declines substantially over operating life while the lamp continues to emit visible light. A lamp that still glows may be delivering a fraction of its rated output. Replacement intervals are typically specified in months of operation, not in whether the lamp has failed.
Water clarity and transmittance	Suspended particles and dissolved color absorb and scatter UV before it reaches target organisms. This has been demonstrated experimentally: reducing turbidity from 23.32 to 10.93 NTU increased log reduction of <i>E. coli</i> by more than 2.15 log under otherwise identical UV treatment (Chhetri et al., <i>Food Science and Nutrition</i>).

[STRONG EVIDENCE] UV disinfection efficacy is a function of delivered dose, which depends on lamp output, flow rate, contact time, lamp age and water transmittance. Turbidity measurably reduces UV effectiveness. This is established water-treatment engineering, demonstrated experimentally.

Dose requirements: algae need more than bacteria

This is a specific, practically important finding. Published aquaculture guidance gives a required UV exposure of approximately 15,000 microwatt-seconds per square centimetre for common bacteria and approximately 22,000 for waterborne algae (Integrated Aqua Systems). Broader aquaculture guidance places typical required doses across target organisms in the range of roughly 30 to 180 millijoules per square centimetre (Aquatic Equipment and Design). For context, research indicates minimal risk of photoreactivation — organisms repairing UV damage — at applied doses of at least 15 mJ/cm² (UltraAqua).

[AQUACULTURE / WATER-TREATMENT PRACTICE] Waterborne algae require a higher UV dose than common bacteria for inactivation — roughly 22,000 versus 15,000 microwatt-seconds per square centimetre in one widely cited aquaculture reference. Units sized for bacterial control may be undersized for algal clearing, and units sized for parasite control are typically larger again.

A note on unit conventions, because they cause confusion: 1,000 microwatt-seconds per square centimetre equals 1 millijoule per square centimetre. Figures quoted in microwatt-seconds therefore look far larger than the same dose quoted in millijoules.

Why UV works so well for green water and not for attached algae

UV only treats water that passes through the unit. Free-floating algae circulate continuously with the water and therefore pass through repeatedly, receiving cumulative exposure. Attached algae never enter the flow path at all — green spot algae on the glass, hair algae on a rock, and black beard algae on a leaf edge are never irradiated. The same reasoning explains why UV does nothing for dust, detritus, dissolved tannins or microbubbles: two of those are not organisms, and the other two are not affected by damage to nucleic acids.

[ESTABLISHED SCIENTIFIC PRINCIPLE] UV-C acts only on organisms that physically pass through the irradiated chamber, and only by damaging nucleic acids. It has no effect on non-living suspended matter, on dissolved compounds, on gases, or on organisms attached to surfaces outside the flow path.

Separating water clarification from disease control

A UV unit sized and plumbed to clear green water is not automatically providing meaningful pathogen control, and vice versa. Different target organisms require different doses — some viruses require doses more than an order of magnitude higher than bacteria. Clearing water and controlling disease are two different specifications that happen to use the same device.

A realistic treatment sequence for green water

- 1 Confirm it is actually green water using the white-cup test, and confirm it is in the water column rather than on the glass.
- 2 Identify and eliminate direct sunlight if present. This is the highest-yield single action in sunlight-driven cases.
- 3 Reduce photoperiod and, if applicable, light intensity to a level the plants can actually use.
- 4 Address obvious nutrient inputs — review feeding, review fertilizer dosing, remove decaying material — without stripping a planted tank of the nutrients its plants require.
- 5 Increase competitive plant mass, particularly fast-growing stems and floating plants, and export biomass by pruning rather than letting it decay in place.
- 6 If a rapid result is needed, deploy UV at an appropriate flow rate, or a blackout, or diatom filtration — recognizing that all three treat the population, not the conditions.
- 7 Maintain aeration throughout, particularly if a large algal die-off is expected.

[REASONABLE INFERENCE] Sequencing matters: physically removing the population without changing the light and nutrient conditions predicts recurrence. Changing the conditions without removing the population predicts a slow resolution. Doing both predicts the best outcome, and the evidence base for each component individually supports that combination.

DISSOLVED ORGANICS

Part XV — Tannins and Tea-Colored Water

Coloration without turbidity: the clearest example in the hobby of "looks cloudy" being the wrong description.

Yellow-to-brown aquarium water caused by dissolved plant compounds is the one condition in this document that frequently is not a problem at all. It is also the condition most likely to be misdescribed by the keeper reporting it, because "cloudy" and "colored" are used interchangeably in casual speech and they are physically different.

What "tannins" means in aquarium usage

In hobby usage, "tannins" is a loose umbrella term for the family of water-soluble plant-derived organic compounds that leach out of submerged wood and plant material and tint water yellow to brown. Chemically this includes true tannins — polyphenolic compounds — but the coloration in an aquarium comes from a broader mixture that includes humic and fulvic acids.

Humic substances are described in the water-treatment literature as complex high-molecular-weight organic compounds of aromatic-aliphatic character containing carbon, oxygen, hydrogen and nitrogen, with relative molecular weights ranging from a few hundred to tens of thousands, formed through the decomposition of organic matter and constituting a common component of natural waters (Kucerova et al., *Engineering Proceedings*). Their charge derives principally from carboxyl and phenolic functional groups, which can ionize and donate protons — which is the mechanistic basis for their acidifying tendency (Chavez-Mejia et al., *Polymers*).

[ESTABLISHED SCIENTIFIC PRINCIPLE] Humic substances are high-molecular-weight organic compounds produced by the decomposition of plant material. They carry ionizable carboxyl and phenolic groups, are the dominant fraction of dissolved organic carbon in many natural waters, and produce yellow-to-brown coloration.

Sources of tannins and humic substances in aquariums

Source	Typical intensity	Duration of release	Notes
Driftwood (Malaysian, mopani, bogwood)	High initially	Weeks to many months; large or dense pieces can release for a year or more	The most common source and the most common reason for surprise
Spider wood, manzanita, cholla	Low to moderate	Weeks to a few months	Generally lighter-releasing woods; often chosen for this reason
Indian almond (catappa) leaves	High per unit mass	Days to weeks per leaf; replaced continuously in blackwater setups	Deliberately used for coloration and chemistry
Other leaf litter (oak, magnolia, guava)	Moderate	Weeks; decomposes into the system	Also a food and biofilm substrate for invertebrates
Seed pods and botanicals (alder cones, casuarina, etc.)	Moderate to high	Weeks	Frequently the deliberate approach for blackwater aesthetics
Peat	High	Ongoing while in use	Historically used for softening and acidifying; sustainability concerns exist
Soil-based and active substrates	Low to moderate	Predominantly in the first weeks	ADA documentation explicitly notes humic acid seeping into the water and turning it yellow, especially in the initial setup period

Source	Typical intensity	Duration of release	Notes
Decaying plant material	Low	Continuous at low level in planted systems	Contributes to the gradual yellowing of long-established tanks

Colored but clear versus genuinely turbid

This distinction is the practical core of the entire part. Two aquariums can both be brown and be in completely different physical states.

Property	Tannin coloration	Suspended brown particulate
Physical mechanism	Dissolved compounds absorbing light	Solid particles scattering light
Can you see the back glass clearly?	Yes — tinted but sharp	No — objects become indistinct with depth
Behavior in a still glass overnight	Unchanged — color remains uniform	Sediment collects at the bottom; the water above clears
Response to fine mechanical filtration	None	Progressive removal
Response to activated carbon	Substantial reduction in color	Essentially none
Response to a water change	Proportional dilution, then re-darkening as the source continues releasing	Proportional removal, particularly if combined with siphoning
Appearance in a beam of light	Beam passes through with little scattering	Beam is clearly visible as scattered light
Effect on turbidity measurement	Low turbidity, high color — and note that white-light NTU instruments do not compensate for dissolved color while near-infrared FNU instruments do	High turbidity

[ESTABLISHED SCIENTIFIC PRINCIPLE] Dissolved coloration and suspended-particle turbidity are independent optical phenomena. Tannin-stained water can be strongly colored and optically clear simultaneously.

Preserved distinction

TANNIN COLORATION is not necessarily TURBIDITY. A keeper who describes a blackwater tank as "cloudy" is usually describing color. The single clarifying question is: "can you still see the back of the tank clearly, just tinted?" If yes, this is a coloration question and most of the cloudy-water diagnostic tree does not apply.

DISSOLVED ORGANICS

Part XVI — Are Tannins Harmful?

Real physiological effects, real evidence, and marketing claims that run well ahead of both.

Tannins and humic substances are among the few aquarium topics where genuine peer-reviewed fish physiology exists, and where the hobby marketing built on top of it substantially exceeds what that research supports. Both facts need stating.

What the physiological research actually shows

Blackwater habitats and ion regulation

Amazonian blackwaters are among the most naturally acidic, dissolved-organic-carbon-rich and ion-poor waters on Earth, and the fish that inhabit them face substantial ionoregulatory challenges. Research has demonstrated that dissolved organic carbon from the upper Rio Negro protects zebrafish against ionoregulatory disturbances caused by low pH exposure (Duarte, Smith, Val and Wood, *Scientific Reports*, 2016). This is a real, measured protective effect — not a marketing claim.

[STRONG EVIDENCE] Dissolved organic carbon and humic substances can protect fish against ionoregulatory disturbance under low-pH conditions. This is documented in controlled physiological studies.

The double-edged finding

The picture is not uniformly positive, and the most useful single paper on this is titled to reflect that. Holland and colleagues, working with eastern rainbowfish (*Melanotaenia splendida splendida*), found that respiratory stress increased with increasing acidity but decreased significantly in treatments containing humic substances at 10 and 20 mg/L compared with those without, while separately noting increased morbidity attributed to possible ion loss under prolonged humic-substance exposure in a stressful low-pH environment (Holland et al., *Environmental Science and Pollution Research*, 2013). The paper is explicitly framed as describing a double-edged effect.

[STRONG EVIDENCE] Humic substances have context-dependent effects on fish: protective against respiratory stress at low pH in some conditions, and potentially contributing to ion loss and morbidity under prolonged exposure in others. The direction of effect depends on concentration, water chemistry and species.

Broader physiological effects

A proteomics study in Japanese medaka summarizes the state of the field usefully: a number of studies describe both stress responses — including heat-shock protein expression, modulation of biotransformation enzymes, oxidative stress and hormone-like effects — and positive effects including increased vigor, growth, lifespan and stress resistance, with the overall conclusion that humic substances interfere with various metabolic pathways in freshwater organisms (Morozov et al., *International Journal of Molecular Sciences*). That study specifically found suppression of innate immunity and coagulation proteins under humic acid exposure.

[CONTESTED CLAIM] Humic substances are biologically active in fish, but the literature reports both beneficial and adverse effects depending on species, concentration and water chemistry. The blanket hobby framing of tannins as simply beneficial is not supported; neither is a blanket framing of them as harmful.

Evaluating the four most common aquarium claims

Claim	What the evidence shows	Evidence assessment
"Tannins prevent disease"	No aquarium-scale controlled evidence establishing disease prevention was located. Physiological studies show effects on stress and, in at least one case, suppression of innate immunity proteins — which cuts against a simple prophylactic story. The plausible indirect mechanism (reduced stress improving disease resistance) is not the same as prevention.	UNSUPPORTED for the strong form; the reduced-stress pathway is a REASONABLE INFERENCE at best
"Tannins are antibacterial"	Antimicrobial activity of plant polyphenols is well documented in laboratory settings against specific organisms at specific concentrations. Whether the concentrations achieved by adding leaves to an aquarium produce meaningful antibacterial effect in that system is a separate question that the laboratory work does not answer.	CONTESTED — laboratory findings are real; extrapolation to aquarium outcomes is not established
"Tannins are antifungal"	Same situation as the antibacterial claim. Additionally, aquarium "fungus" is often water mold (oomycete) rather than true fungus, which complicates extrapolation from antifungal laboratory work.	CONTESTED — mechanism plausible, aquarium-scale efficacy unestablished
"Tannins boost immunity"	Directly contradicted in at least one proteomics study, which found suppression of innate immunity proteins under humic acid exposure in medaka. Other work reports increased stress resistance. The literature does not point in one direction.	CONTESTED, with evidence on both sides

The honest framing for downstream content

The defensible statement is: humic substances are biologically active, and blackwater conditions are the natural habitat of many popular aquarium species, which is a good reason to consider replicating them. The indefensible statement is that adding leaves to a tank prevents disease. Laboratory antimicrobial findings should be described as laboratory findings, not as home-aquarium outcomes.

Effects on pH and carbonate hardness

The mechanism is real: humic and fulvic acids can donate protons from their carboxyl and phenolic groups, which tends to acidify water. Whether that produces a measurable pH change in a given aquarium depends almost entirely on the carbonate buffering capacity of that water.

Water type	Expected pH effect of tannins	Why
High KH tap water (moderately to very hard)	Little to no measurable change from typical botanical loads	Carbonate alkalinity neutralizes the added acidity. The buffer must be depleted before pH moves.
Low KH water	Measurable downward drift	Less buffering capacity to absorb the acidity
RO or RO-remineralized water with minimal KH	Substantial and potentially rapid change	Almost no buffering capacity; this is the configuration used deliberately in serious blackwater setups

[ESTABLISHED SCIENTIFIC PRINCIPLE] Whether an added weak acid changes pH depends on the carbonate buffering capacity of the water. Tannins do not "automatically lower pH" — they exert an acidifying influence that is opposed by alkalinity.

An important corollary that is frequently missed: KH is a consumable reserve, not a permanent lock. A tank with moderate KH may show no pH movement for weeks and then begin drifting once that reserve is depleted by ongoing acid production from both botanicals and nitrification. This is why "my tannins never affected pH" and "my pH crashed from botanicals" can both be accurate reports from different keepers, and even from the same keeper at different times.

[REASONABLE INFERENCE] Delayed pH movement in botanical-heavy tanks is consistent with gradual depletion of carbonate alkalinity. Monitoring KH, not just pH, gives earlier warning than monitoring pH alone.

Other reported effects

Fish coloration	Many species display more intense coloration in tinted water with dark substrate. The plausible mechanism is behavioral and perceptual — reduced light penetration and greater perceived security — rather than a direct chemical effect on pigment. ANECDOTAL to REASONABLE INFERENCE.
Stress and shyness	Reduced light penetration is reported to lower stress in shy and nocturnal species. Consistent with the aquaculture finding that microalgae shading reduces stress in rearing systems. REASONABLE INFERENCE.
Algae suppression	Reduced light transmission through tinted water plausibly reduces energy available to algae. Mechanistically sound; not quantified for aquariums. REASONABLE INFERENCE.
Ammonia and nitrite toxicity	A research question has been raised in the literature as to whether dissolved aquatic humic substances can reduce the toxicity of ammonia and nitrite in recirculating aquaculture systems (Meinelt et al.). This is an active research area, not a settled result, and should not be presented as a reason to relax nitrogen management.
Metal bioavailability	Dissolved organic carbon is well established to moderate the bioavailability and toxicity of metals such as copper to fish (McGeer et al. 2002). STRONG EVIDENCE, though of limited direct relevance to cloudy-water diagnosis.

A caution about adding botanicals too quickly

One practical risk is well recognized and worth carrying into any content: adding a large quantity of botanical material at once introduces a large mass of decomposable organic matter. That is precisely the condition described in Part V as a bacterial bloom trigger, and it carries the associated oxygen demand. The common hobby guidance to start with a small quantity and observe over a week is a sensible application of the same reasoning used everywhere else in this document.

[REASONABLE INFERENCE] Heavy botanical additions are an organic loading event and should be expected to produce the same category of response as any other large organic input: possible bacterial bloom, biofilm growth, and increased oxygen demand.

DISSOLVED ORGANICS

Part XVII — Removing Tannins (If You Want To)

An aesthetic decision in most cases, not a water-quality intervention.

The framing of this part matters. Tannin coloration in an otherwise healthy aquarium is a preference question. A keeper who likes it should not be told to remove it, and a keeper who dislikes it should not be told they are damaging their tank by removing it. Both positions are legitimate.

Removal methods compared

Method	Mechanism	Effectiveness	Practical notes
Water changes	Dilution	Proportional and temporary. A 50 percent change removes roughly half the dissolved color, which then partially returns as the source continues releasing.	Cheap and safe. Effective as ongoing management rather than a fix. Frequency matters more than volume.
Activated carbon	Adsorption of dissolved organic molecules onto a high-surface-area porous medium	Effective for coloration. Pilot-plant work on granular activated carbon reported over 70 percent removal of humic substances, though only about 50 percent reduction in measured water color under the technology used (Kucerova et al.).	Capacity is finite; exhausted carbon stops working and must be replaced. Replacement intervals of weeks are typical. Also adsorbs some medications.
Synthetic adsorbent resins (Purigen-type)	Adsorption of dissolved organic nitrogenous waste and organics onto a macroporous polymer	Generally effective for coloration and organics; regenerable	Manufacturer positioning is as an organic-adsorbing resin. Note that at least one independent hobby-science test reported it ineffective specifically for clearing bacterial-bloom cloudiness — a different application from removing dissolved color.
Time	The source depletes	Eventually effective for wood; ineffective while botanicals continue to be replaced	Duration is highly variable. Dense hardwoods can release visible color for many months.
Boiling wood before use	Extracts soluble compounds; also saturates the wood so it sinks	Partial. Reduces the initial release; does not eliminate ongoing release from large pieces	Practical only for pieces that fit in a pot. Boiling has a second effect worth noting: at least one hobbyist report argues that boiling provides a fresh sterile surface that biofilm then recolonizes.
Extended soaking of wood	Same extraction mechanism, slower and gentler	Partial; works on any size piece given a container	Water must be changed repeatedly or it saturates. Weeks may be required for large pieces.
Ozone / advanced oxidation	Chemical oxidation of colored organic molecules	Highly effective in professional water treatment	Not a practical or advisable home-aquarium intervention. Mentioned for completeness only.

[STRONG EVIDENCE] Activated carbon adsorbs humic substances from water, with demonstrated removal efficiencies well above 50 percent in pilot-scale drinking-water treatment. Adsorption efficiency is affected by pH, initial concentration, adsorbent type and contact time.

[MANUFACTURER RECOMMENDATION] ADA documentation for Aqua Soil Amazonia advises that keepers uncomfortable with humic-acid coloration can remove excess organic matter through frequent water changes, or use carbon media to adsorb dissolved humic acid. This is a manufacturer statement about its own product and is consistent with the independent adsorption literature.

How long does driftwood keep releasing color?

No reliable general answer exists, and any specific number offered is a guess. The variables that matter are wood species and density, piece size and surface-to-volume ratio, how long the wood was seasoned before use, whether it was pre-soaked or boiled, water chemistry, and how much water volume it is releasing into. Reported durations range from a few weeks for small, light, pre-soaked pieces to well over a year for large dense pieces.

[ANECDOTAL OBSERVATION] Reported tannin-release durations for driftwood span weeks to more than a year. No systematic study of release kinetics for aquarium driftwood was located. Keepers should be given a range and the variables, not a number.

Aesthetic preference versus water-quality problem

Question to ask	If the answer is...	Then this is...
Can you still see the back of the tank clearly?	Yes, just tinted	A coloration situation, not a clarity problem
Are ammonia and nitrite at zero?	Yes	Not a biological filtration problem
Are the fish behaving normally — feeding, normal respiration, using the whole tank?	Yes	Not an acute welfare problem
Has pH or KH changed substantially and unexpectedly?	No	Not a chemistry problem
Do you dislike how it looks?	Yes	A legitimate aesthetic reason to remove it, and no justification needed beyond that
Do you like how it looks?	Yes	A legitimate reason to leave it, and no intervention is required

Do not assume tannins need removal

A significant share of "how do I fix my brown water" questions come from keepers who have been told, incorrectly, that brown water means a dirty or unhealthy aquarium. The most useful response is often to establish that the water is chemically fine, explain what the color is, and let the keeper decide. Many decide they like it once they understand it.

PHYSICAL CAUSES

Part XVIII — Substrate Dust

The most visually alarming and least biologically significant cause of cloudiness in a new tank.

Substrate-derived cloudiness has a signature that makes it one of the easiest conditions to identify correctly: it appears at the exact moment of physical disturbance, not hours or days later. Keepers who understand that single fact avoid a large share of misdiagnosis.

Cloudiness by substrate type

Substrate	Dust profile	Rinse?	Notes
Play sand / pool filter sand	Very high initially. Fine silt fraction can remain suspended for hours to days.	Yes — thoroughly, and expect it to take longer than you think	Pool filter sand is graded and generally rinses cleaner than play sand. Rinse in small batches until the runoff is clear.
Aquarium-branded sand	Moderate to high; varies widely by brand	Yes, unless the manufacturer says otherwise	Coated or dyed products may have a different profile; follow the packaging.
Standard aquarium gravel	Low to moderate — mainly manufacturing dust and coating residue	Yes	Rinses quickly. Rarely a prolonged problem.
Aquasoil / active substrates	Variable. Granules can crumble to powder in transit or if allowed to dry out.	No — see Part XIX	Manufacturer instructions generally prohibit rinsing. Fill technique matters more than rinsing here.
Soil-based substrates (dirted tanks)	High, and the fine clay fraction is the most persistent of any substrate type	Typically capped rather than rinsed	Very fine clay colloids can remain suspended for extended periods because settling velocity scales with particle size.
Clay-based substrates (fired clay pellets, laterite)	Moderate; some products shed continuously if disturbed	Usually yes, gently	Aggressive rinsing can break down softer products.
Crushed coral / aragonite	Moderate to high — fine carbonate dust	Yes	The dust is chemically active as well as visually annoying: it is a carbonate and will contribute to hardness and alkalinity.
Inert mineral substrates (quartz, basalt, river rock)	Low to moderate	Yes	Generally straightforward.

Should substrate be rinsed?

Usually yes, with one important category of exception. The general principle: inert mineral substrates benefit from rinsing, while nutrient-rich active substrates generally should not be rinsed because rinsing removes the nutrients they exist to provide and can physically break down the granules.

[MANUFACTURER RECOMMENDATION] Manufacturer instructions are the primary authority for any given product and should be followed over general advice. For active planted-tank substrates, the near-universal manufacturer position is not to rinse.

[UNSUPPORTED AQUARIUM FOLKLORE] The blanket claim that "all aquarium substrate should be rinsed" is wrong as stated, because it would damage the performance of an entire category of products. The correct version is "most inert substrates should be rinsed; check the instructions for active substrates."

Why rinsed sand can still cloud the water

A frequent and frustrating report: the keeper rinsed thoroughly and the tank is still opaque. Several mechanisms explain this:

- **The finest fraction is very hard to rinse out.** Rinsing removes particles fast enough to be carried away by the runoff. The finest silt-sized fraction remains behind and re-suspends when water hits it.
- **Filling technique matters as much as rinsing.** Pouring water directly onto the substrate excavates it and lifts fines throughout the water column. Pouring onto a plate, a bowl, or a plastic bag laid over the substrate prevents this almost entirely.
- **Very fine particles settle very slowly.** Settling velocity depends strongly on particle size; clay-sized particles can remain in suspension for a long time even in still water.
- **Circulation keeps them up.** A filter return aimed across the substrate will keep re-suspending material that would otherwise settle.
- **The cloud may not be substrate at all.** If the tank has been running for a day or more, the source may have shifted to a bacterial response to organics released during setup.

What actually clears substrate dust

Approach	Effectiveness	Notes
Fine mechanical filtration (filter floss, polishing pads)	High — this is the correct tool for this problem	Expect the media to load quickly and require replacement or rinsing every few days while the tank clears.
Time and still water	Moderate to high for coarser fractions; slow for clay	Turning off circulation for a period allows settling, after which the settled material can be siphoned.
Water changes with substrate siphoning	High — physically removes the material	Most effective of the low-tech options because it exports rather than redistributes.
Redirecting flow away from the substrate	Moderate, preventive	Stops the cycle of re-suspension.
Clarifier / flocculant	Can be effective — this is the application these products are actually designed for	Requires adequate mechanical filtration to capture the resulting flocs, otherwise they settle and are re-suspended later. See Part XLI.
Activated carbon	None	Carbon adsorbs dissolved compounds. Suspended mineral particles are neither dissolved nor adsorbable.
UV sterilizer	None	UV damages nucleic acids. Sand has none.
Bacterial additives	None	No mechanism connects bacterial supplementation to mineral particle removal.

[ESTABLISHED SCIENTIFIC PRINCIPLE] Mineral particulate is removed by physical means — sedimentation, mechanical capture, or aggregation followed by capture. It is not affected by adsorption media, UV, or biological additives, because none of those act on inert suspended solids.

PHYSICAL CAUSES

Part XIX — Aquasoil and Active Substrates

Where dust and bacterial bloom reliably occur together.

Active planted-tank substrates deserve separate treatment because they do three things simultaneously that other substrates do not: they can release fine particulate, they release a substantial pulse of ammonium, and they release dissolved organics including humic substances. All three have visible consequences, and keepers routinely attribute all of them to one cause.

What manufacturer documentation actually says

ADA's own technical material for Aqua Soil Amazonia is unusually explicit and is worth treating as a primary source, with the standard caveat that manufacturer documentation describes intended behavior rather than providing independent proof.

- **On ammonium release:** Amazonia is described as containing rich nitrogen, with a large amount of ammonium leaching into the water immediately after it is laid in the aquarium, and the guidance is not to add fish or shrimp during the initial high-ammonium stage — with a stated recommendation to wait two to three weeks after planting.
- **On dust:** the granules are softly baked soil, and if a bag is left open and the granules dry excessively, or if granules crumble into powder during transportation, the water can become cloudy when filled.
- **On filling technique:** the documentation advises removing cloudy water and refilling slowly, repeating if necessary, and specifically to turn on the filter only after the water becomes clear — because running the filter in cloudy water suspends fine soil particles and this can cause white turbidity due to overgrowth of bacteria feeding on organic compounds contained in those particles.
- **On humic coloration:** humic acid is described as seeping into the water and turning it yellow, more likely during the initial setup period, with the statement that humic acid is a natural substance that will not disturb the health of fish and plants, and that frequent water changes or carbon can reduce it if the keeper finds it objectionable.
- **On bloom-type cloudiness:** the documentation notes that at the initial setup stage, with immature bacterial filtration, the water may turn cloudy because of excessive organic matter dissolved into the water, that microbes fed by those organics are likely floating in the water, and that a UV sterilizer is effective in such a case.

[MANUFACTURER RECOMMENDATION] ADA documentation explicitly describes three distinct visual effects from the same product — particulate cloudiness, humic yellowing, and microbial cloudiness from released organics — and gives different guidance for each. This is a manufacturer's account of its own product and is notably more nuanced than typical hobby discussion.

"Cloudy after adding aquasoil" — particles, bloom, or both?

The answer is frequently both, in sequence. The two phases are separable by timing and by behavior.

Phase	Timing	Appearance	Behavior	Response
Phase 1: particulate	Immediately on filling; may recur with any disturbance	Brown to grey, sometimes described as muddy	Settles in still water; captured by fine mechanical media; visible sediment after settling	Careful filling technique; drain and refill; fine filtration; patience
Phase 2: microbial	Typically days 2–10 after setup	White to grey, uniform	Does not settle; not captured by ordinary mechanical media	Usually self-limiting as biofilm establishes; UV if a fast result is needed

Phase	Timing	Appearance	Behavior	Response
Overlapping: humic coloration	First weeks, gradually declining	Yellow to amber tint	Neither settles nor is mechanically captured; responds to carbon and water changes	Optional — aesthetic decision

[REASONABLE INFERENCE] A new aquasoil setup that is cloudy on day one and still cloudy on day six has probably experienced two different phenomena, not one continuous one. Treating it as a single condition leads keepers to conclude that interventions "did not work" when they addressed a phase that had already passed.

Substrate categories compared

Category	Nutrient content	Ammonium release	Dust risk	Rinse	Longevity
Baked-soil aquasoils (ADA Amazonia, Tropica, similar)	High	Substantial in the first weeks	Moderate; higher if granules have dried or been crushed	No	Depletes over time; eventually needs root tab supplementation
Lower-ammonium aquasoils (various "shrimp-safe" formulations)	Moderate to high	Lower by design	Moderate	No	Similar depletion pattern
Clay-based enriched gravels (fired clay, laterite blends)	Moderate	Minimal	Moderate	Usually yes, gently	Long-lasting structurally; nutrient content depletes
Mineralized topsoil / dirted tanks	High	Variable and can be substantial	High — persistent clay fraction	Capped, not rinsed	Very long-lasting
Inert sand and gravel	None	None	Variable by product	Yes	Indefinite; all nutrition must be supplied by root tabs or water column

Why this is not a product review

Product-level comparisons date quickly and vary by production batch. The durable insight is categorical: substrates that supply nutrients release those nutrients into the water column at first, which has visible consequences; substrates that supply nothing do not, and require the nutrients to be added another way.

TIMING-BASED DIAGNOSIS

Part XX — Cloudiness After Maintenance

The "what changed in the last 24 to 72 hours" framework.

Timing is the most underused diagnostic axis in the hobby. Appearance narrows the field; timing frequently closes it. Building the habit of asking what changed, and when, before asking what the water looks like would improve the average quality of cloudy-water advice more than any other single change.

The framework

WHAT CHANGED IN THE LAST 24 TO 72 HOURS?

Physical: did anything get moved, stirred, vacuumed, planted, uprooted, added or removed?

Equipment: was a filter cleaned, restarted, replaced, or did it stop? Was media changed? Was an air pump, powerhead or heater added or removed?

Chemical: was anything dosed — conditioner, fertilizer, buffer, remineralizer, medication, bacterial supplement, algaecide, clarifier?

Biological: were animals added? Did any die? Did plants melt? Was there a spawning event?

Nutritional: was feeding increased, was a new food used, did someone else feed the tank?

Environmental: did light exposure change? Did the room temperature change? Was there a power outage? Did the seasons shift the sun angle?

Source water: did the utility flush the mains, was there a supply interruption, has anything changed about the water coming out of the tap?

Maintenance actions and their characteristic cloudiness

Action	Onset	Appearance	Most likely cause	Typical duration
Gravel vacuuming	Immediate	Brown-grey haze with visible particles	Disturbed detritus and substrate fines	Hours
Stirring or disturbing sand	Immediate	Grey-white haze	Substrate fines; possibly trapped organics	Hours to a day
Moving rocks or hardscape	Immediate	Localized then general haze	Detritus trapped beneath and around the hardscape	Hours
Uprooting or replanting plants	Immediate	Brown haze, sometimes with a soil component	Substrate disturbance; released mulm; in aquasoil tanks, granule breakdown	Hours to a day; longer in soil-based tanks
Cleaning a filter	Immediate to a few hours	Brown particulate, or later white haze	Released debris immediately; possible bacterial response if cleaning was aggressive	Hours; days if biofilm was substantially removed
Restarting a canister filter after it has been off	Immediate	Brown, sometimes with an odor	Anaerobic material and accumulated debris ejected from the canister	Hours; the odor is a warning sign worth investigating
Replacing filter media or a cartridge	Hours to days	White, uniform	Loss of established biofilm capacity	Days to weeks
Cleaning decorations	Immediate	Brown haze	Dislodged biofilm and detritus	Hours
Moving an aquarium	Immediate	Heavy, often severe	Total substrate disturbance plus disrupted filtration	Days

Action	Onset	Appearance	Most likely cause	Typical duration
Large water change	Minutes to an hour	White, uniform, clears top-down	Dissolved gas coming out of solution	Minutes to hours
Adding substrate to an established tank	Immediate	Heavy grey or brown	Substrate fines	Hours to days
Rescaping	Immediate, then again days later	Two phases: particulate, then possibly white	Disturbance first; then a possible bacterial response to released organics	Days

[USEFUL RULE OF THUMB] Cloudiness that began within minutes of physical work is almost always physical. Cloudiness that began a day or more after biological disruption (media replacement, a death, overfeeding) is usually microbial. Cloudiness that began within minutes of a chemical addition is usually chemical. These three timing signatures cover most reported cases.

Why timing beats appearance

Consider two identical photographs of a white, uniformly cloudy 20-gallon aquarium. In one, the keeper did a water change forty minutes ago. In the other, the keeper replaced the filter cartridge four days ago. The photographs are indistinguishable; the conditions are entirely different; the correct responses are "do nothing" and "test for ammonia immediately." No amount of looking at the water resolves this. One question does.

PHYSICAL CAUSES

Part XXI — Fine Detritus and Organic Particles

Water that is chemically fine and physically full of things.

An aquarium can have a perfect test panel and still look dirty, because test kits measure dissolved chemistry and eyes see suspended solids. These are separate properties, and confusing them produces the reasonable but incorrect conclusion that visible particles mean bad water quality.

What is actually suspended

Fish waste	Fecal material breaks down into progressively finer particles that remain in suspension until captured or settled.
Uneaten food	Both intact fragments and the fine particulate produced as food softens and disintegrates.
Plant debris	Fragments of decaying leaves, root hairs, and material released during pruning.
Mulm	The general term for accumulated fine organic sediment: partially decomposed material that collects in low-flow areas and in the substrate.
Biofilm fragments	Sloughed material from surfaces, filter media and wood. Often stringy or gelatinous rather than granular.
Microorganism biomass	Dead and living microbial material, and the remains of microfauna.

Why clear chemistry and visible debris coexist

Nitrifying bacteria convert ammonia to nitrite to nitrate. That process handles the dissolved nitrogenous products of decomposition. It does not remove particles. A well-cycled aquarium with inadequate mechanical filtration will read zero ammonia, zero nitrite and low nitrate while carrying visible suspended debris, because nothing in the nitrogen cycle physically removes solids from the water column.

[ESTABLISHED SCIENTIFIC PRINCIPLE] Biological filtration transforms dissolved nitrogen compounds. Mechanical filtration removes suspended solids. These are different processes addressing different fractions of the waste stream, and adequacy in one does not imply adequacy in the other.

Causes of persistent suspended detritus

Cause	How to recognize it	What addresses it
Inadequate mechanical filtration	Filter contains only coarse sponge or biological media; no fine stage present	Add a fine polishing stage downstream of the coarse stage
Filter bypass	Water is visibly taking a path around rather than through the media; flow is high but capture is poor	Reseat media, close gaps, check gaskets and media baskets
Clogged media reducing flow	Return flow is visibly weaker than it was	Rinse or replace the mechanical stage — the coarse stage, not the biological one
Media too coarse for the particle size	The filter is clean, flow is good, and particles pass through anyway	Finer media — see Part XXXVIII for particle-size limits
Poor circulation creating dead zones	Debris accumulates in specific corners and never reaches the intake	Reposition returns; add circulation; direct flow toward accumulation zones

Cause	How to recognize it	What addresses it
High waste production	Heavy stocking, large messy eaters, or generous feeding	Feeding discipline; more frequent maintenance; larger mechanical capacity
Ongoing substrate disturbance	Flow aimed at the substrate; active digging species	Redirect flow; accept a degree of it with digging species
Recent disturbance still settling	A known event occurred within the last day or two	Time and mechanical capture

An important reframing

Suspended detritus is a filtration and flow question, not a water-quality question in the chemical sense. It is worth addressing for observation and aesthetics, and because accumulated organics eventually contribute to oxygen demand — but a tank full of visible mulm with a clean test panel is not an emergency, and treating it as one leads to the over-cleaning described in Part XLVI.

EQUIPMENT

Part XXII — Filter-Related Cloudiness

When the filter is the cause rather than the solution.

The reflexive response to cloudy water is to do something to the filter. That is sometimes correct and sometimes the action that caused the problem. This part sorts the cases.

Filter situation	Type of cloudiness produced	Mechanism	Correct response
Filter undersized for the bioload	Persistent haze; slowly accumulating debris	Insufficient mechanical capture and insufficient biofilm surface area relative to waste production	Increase filtration capacity or reduce load. Adding media to an already-full filter does not add capacity if flow is already restricted.
Clogged mechanical media	Gradually worsening haze and visible particles	Flow falls, so less water is processed per unit time and capture rate drops	Rinse or replace the mechanical stage. Leave the biological stage alone.
Filter bypass	Visible particles despite apparently strong flow	Water takes the path of least resistance around the media	Reseat media, close gaps, check that baskets and gaskets are correctly positioned
Improper media order	Rapid clogging of fine media; poor overall capture	Fine media placed before coarse media blinds immediately	Coarse first, then fine, then biological, then chemical if used
Brand-new filter	White haze developing over days	No established biofilm; organics are processed in the water column instead	Time; seed with media from an established filter; monitor ammonia if stocked
Recently and aggressively cleaned filter	White haze developing over hours to days	Established biofilm was removed along with the debris	Time; avoid repeating; clean only the mechanical stage in future, and rinse it in tank water rather than under a tap
Replaced filter cartridge	White haze over days; possible ammonia	Cartridge replacement discards the entire biofilm along with the mechanical medium	Where possible, replace only part of the media at a time, or convert cartridge filters to reusable foam so mechanical and biological stages are separable
Insufficient fine mechanical stage	Fine particles never captured	Media pore size is larger than the particles	Add filter floss or a polishing pad after the coarse stage
Filter ejecting trapped debris	Sudden brown cloud	Accumulated material dislodged by flow changes, restart, or handling	Usually self-limiting; clean the mechanical stage
Canister restarted after being off	Sudden brown cloud, sometimes with a foul odor	Material inside the canister went anaerobic while flow was stopped and is ejected on restart	If the filter has been off for more than a short period, clean it before restarting rather than pushing its contents into the tank
Air leak on the intake side of a canister	Persistent fine bubbles and a milky appearance	Air drawn in and chopped into microbubbles by the impeller	Check O-rings, hose connections and the intake for cracks. See Part XXIII.

[COMMON AQUARIUM PRACTICE] The convention of arranging media coarse-to-fine in the direction of flow, and of cleaning mechanical media frequently while disturbing biological media rarely, is standard practice across the hobby and is consistent with filtration engineering principles.

The connection worth making explicit

A filter performs at least three separable jobs: capturing particles, hosting the biofilm that processes dissolved waste, and moving water. Cloudy-water problems can arise from a failure in any of the three, and the responses are different. "Clean the filter" is not a response to a diagnosis; it is a response to an absence of one.

PHYSICAL CAUSES

Part XXIII — Microbubbles

Not particles, not organisms, not a water-quality issue — and one of the most common misdiagnoses in the hobby.

Microbubble cloudiness deserves its own part because beginners reliably mistake it for a bacterial bloom, and because it is the one cause of white water that requires no action at all and resolves without help. Recognizing it saves keepers from unnecessary water changes, unnecessary product purchases, and unnecessary worry.

The mechanism

Gas solubility in water depends on pressure and temperature. Municipal tap water is delivered under pressure — commonly cited figures for household supply are in the range of 30 to 80 psi — and at that pressure the water holds more dissolved gas than it can hold at atmospheric pressure. When that water enters an aquarium and the pressure drops to atmospheric, the excess gas comes out of solution as vast numbers of microscopic bubbles. Each bubble scatters light, and collectively they produce an intense white appearance (Aquarium Science, 6.2.3).

[ESTABLISHED SCIENTIFIC PRINCIPLE] Gas solubility in water increases with pressure and decreases with temperature. Water saturated at elevated pressure will release excess dissolved gas when depressurized, forming bubbles. This is the same physics that produces bubbles in a freshly opened carbonated drink.

Sources of microbubbles in aquariums

Source	Typical pattern	Persistence
Pressurized tap water added during a water change	Immediate milkiness following the refill, clearing progressively	Minutes to a few hours — usually gone by the next morning
Water heated rapidly, or cold water warming in the tank	Bubbles form as warming water holds less gas	Hours
Air leak on the suction side of a canister filter	Continuous fine bubble stream from the outlet; persistent milkiness	Continuous until the leak is found and fixed
Hang-on-back filter with the outflow entraining air	Bubbles generated at the point where the return breaks the surface	Continuous; often changes with water level
Air stones and air-driven filters	Obvious bubble column; fine stones produce very small bubbles	Continuous while running
CO2 injection with a fine diffuser	Fine bubble mist through the tank during injection hours	Follows the injection schedule
Powerheads or pumps drawing air	Intermittent bursts of fine bubbles	Until corrected
Very high surface agitation	Air entrained and driven downward	Continuous while the flow pattern persists

How to recognize microbubbles

Clue	Why it works
The cloud clears from the top downward	Bubbles rise. A suspension of solids or organisms does not clear directionally.

Clue	Why it works
Bubbles collect on the glass, on plant leaves and on hardscape	They adhere to surfaces before coalescing and rising. Nothing else in the cloudy-water list does this.
The appearance changes when the filter or air pump is switched off	If a piece of equipment is producing the bubbles, stopping it stops production.
A sample in a glass clears within minutes to an hour	The definitive test. A bacterial suspension will not.
It appeared within minutes of adding water	The timing signature of depressurization
A flashlight beam shows glittering rising points rather than drifting specks	Bubbles rise consistently; particles drift with the current
The water is clear when you look into the tank from above but milky from the side	Path length through the bubble cloud is shorter looking down

[COMMON AQUARIUM PRACTICE] The observation that post-water-change cloudiness appears within an hour and disappears overnight, and is caused by dissolved air forming microbubbles, is consistently reported and is mechanistically well explained. It is described as harmless but confusing — which is exactly right.

When microbubbles are worth investigating

Transient microbubbles after a water change require no action. Persistent microbubbles do, for two reasons. First, a continuous fine bubble stream from a canister filter usually indicates an air leak on the suction side, which is a mechanical problem that can worsen. Second, in extreme cases, genuine gas supersaturation is a recognized fish health hazard.

Gas bubble disease — the serious end of this topic

Gas bubble disease occurs when the total pressure of dissolved gases in water exceeds ambient atmospheric pressure and gases come out of solution within the fish's tissues and blood vessels, forming emboli. Veterinary sources describe it as analogous to decompression sickness in divers (WikiVet; PetMD; Toledo Zoo).

Cited aquarium causes include leaks in pumps or valve systems that draw air under pressure, and sudden temperature changes. Heavy algal blooms have also been implicated in ponds through photosynthetic oxygen production exceeding what can diffuse out.

Signs include visible bubbles in the eyes, fins, skin or gills, and abnormal buoyancy. Fish near the surface develop more severe signs faster because of the lower hydrostatic pressure.

Perspective matters: this is uncommon in normal home aquarium practice and is associated with equipment faults, not with the ordinary milky appearance after a routine water change. Persistent, continuous microbubble production is the situation that warrants investigating the equipment.

[STRONG EVIDENCE] Gas bubble disease from total dissolved gas supersaturation is a documented condition in fish, with experimental work establishing tolerance thresholds in salmonids and external signs appearing above roughly 130 percent total dissolved gas in one silver carp study. Equipment air leaks are a recognized aquarium-scale cause.

Preserved distinction

MICROBUBBLES are not a BACTERIAL BLOOM. They are a gas phase, not a biological population. They cannot be filtered out, they are unaffected by UV, carbon, clarifiers or water changes, and they resolve on their own. Any advice that treats them as a bacterial problem will appear to fail, and any advice that waits will appear to succeed — for the wrong reason in both cases.

TIMING-BASED DIAGNOSIS

Part XXIV — Cloudiness After a Water Change

A specific timing signature with a short list of explanations.

This is one of the most frequently searched cloudy-water questions, and it has an unusually tractable answer, because the timing narrows the candidate list sharply. This part addresses visual cloudiness only — the separate and important topic of livestock problems following water changes is outside its scope.

Cause	Onset	Appearance	How to confirm	Action
Microbubbles from dissolved gas	Within minutes; often peaks around an hour	Uniform white; clears top-down	Glass sample clears within the hour; bubbles on glass and leaves	None. It resolves overnight.
Disturbed substrate from the fill	Immediate	Grey or brown, with visible particles	Settles in still water; captured by floss	Fill onto a plate or bowl next time; fine filtration
Disturbed detritus from vacuuming	Immediate	Brown haze with particles	Same as above	Mechanical capture; usually resolves within hours
Temperature-driven gas release	Within an hour	Uniform white	Associated with a noticeable temperature difference between old and new water	None; match temperature more closely in future
Mineral precipitation from mixing water chemistries	Minutes	Chalky white, sometimes with a slight sparkle	Occurs when a buffer, remineralizer or hard tap water meets a different chemistry; no settling of organic character	Review what is being added and in what order; premix outside the tank
Source-water turbidity	Immediate and proportional to the volume added	Matches the appearance of the tap water itself	The comparison test in Part LXIX — a glass of tap water beside a glass of tank water	Pre-filter, settle, or age the change water
A delayed bacterial response	A day or more later, not immediately	White, uniform, persistent	The delay itself is the clue. This is not caused by the water change; it coincides with whatever else happened during maintenance.	See Parts IV–IX

[USEFUL RULE OF THUMB] Cloudiness that appears within an hour of a water change and clears by morning is almost always dissolved gas. Cloudiness that appears a day or more later is a separate event that happens to follow the water change chronologically. Distinguishing them prevents the widespread belief that water changes cause blooms.

The order-of-operations problem

Maintenance sessions typically bundle several actions: vacuuming, water removal, filter rinsing, refilling, dosing. If cloudiness follows, the keeper has no way to attribute it to a specific action because everything happened at once. This is the single most common reason cloudy-water troubleshooting stalls, and it has a simple fix: separate maintenance actions in time when troubleshooting, so that cause and effect remain distinguishable. This principle recurs in Part LXXV.

CHEMISTRY

Part XXV — Mineral Precipitation and Chemical Cloudiness

How two clear liquids produce white water.

This is the cause keepers are least likely to identify on their own, because it violates an intuition: nothing visible was added, and yet the water turned white. The explanation is straightforward chemistry.

The mechanism, in beginner-friendly terms

Water can hold a certain quantity of dissolved ions — charged particles such as calcium, magnesium, carbonate and phosphate. Some combinations of these ions are highly soluble and stay dissolved and invisible. Others are poorly soluble: when those particular partners meet at sufficient concentration, they combine into a solid that will not stay dissolved and appears as a fine white suspension.

A useful analogy: dissolving salt in water gives a clear solution; dissolving too much gives crystals at the bottom. Precipitation is the same principle, except that the limit is reached not by adding more of one substance but by introducing a partner ion that forms a poorly soluble compound with what is already there.

[ESTABLISHED SCIENTIFIC PRINCIPLE] When the product of the concentrations of two ions exceeds the solubility product for the compound they form, that compound precipitates out of solution as a solid. Calcium carbonate and calcium phosphate are the most commonly relevant examples in aquarium water.

Relevant interactions in aquariums

Interaction	Result	Typical aquarium context
Calcium plus carbonate at elevated pH	Calcium carbonate precipitation — white, chalky	Adding a carbonate buffer to already-hard water; raising pH sharply in hard water; localized precipitation around a dosing point
Calcium plus phosphate	Calcium phosphate precipitation	Phosphate-containing fertilizers dosed into hard, high-pH water
Magnesium plus carbonate or hydroxide at high pH	Magnesium compounds precipitating	Aggressive pH-raising products in hard water
Remineralizer components added together in concentrated form	Local precipitation before dilution occurs	Adding two remineralizing products to the same small volume of RO water without allowing each to dissolve fully
Buffer plus acid-buffer products used simultaneously	Reaction products and precipitation	Using a "pH up" and a "pH down" product together, or layering multiple chemistry-adjusting products
Iron-containing fertilizer plus phosphate	Iron phosphate precipitation; also loss of the iron from availability	Dosing iron and phosphate fertilizers simultaneously into the same spot
CO2 injection stopping in hard water	pH rises, reducing carbonate solubility, potentially producing haze	End of the CO2 photoperiod in very hard water

Recognizing chemical cloudiness

- **Onset is fast** — minutes, sometimes seconds, following an addition.
- **It is tied to a dosing event**, not to physical disturbance and not to an organic input.
- **The appearance is distinctly chalky**, sometimes with a faint sparkle in a light beam.

- **It often occurs in hard water or in water where hardness is being adjusted.**
- **It may recur reproducibly** every time the same product is dosed the same way — which is the strongest confirmation available.

Preventing it

- 1 Dose one product at a time, with time between them, rather than adding several to the same spot at once.
- 2 Premix remineralizers and buffers in a separate container of water and allow them to dissolve fully before adding.
- 3 Add products to a high-flow area so they dilute rapidly rather than reaching high local concentration.
- 4 Avoid using pH-adjusting products in already-well-buffered water, where they frequently accomplish little except driving precipitation.
- 5 Check whether two products are chemically compatible before combining them.

[REASONABLE INFERENCE] The characteristic recurrence of chemical cloudiness with a specific dosing action makes it one of the more confirmable causes: if the keeper can reproduce it deliberately, the diagnosis is established.

CHEMISTRY

Part XXVI — Hard Water and Cloudiness

Dissolved minerals do not scatter light. Precipitated ones do.

The claim "your water is cloudy because it is hard" is heard often enough to need direct treatment. As stated, it is wrong, and the reason it is wrong clarifies the whole category.

Dissolved versus precipitated

Property	Dissolved minerals	Precipitated minerals
Physical state	Ions in solution	Solid particles suspended in water
Visible?	No — individual ions are far smaller than the wavelength of visible light	Yes — particles large enough to scatter light
Measured by	GH, KH, TDS, conductivity	Turbidity
Typical cause	The source water's mineral content	A chemical change that exceeded a solubility limit
Removed by	RO, dilution, ion exchange	Mechanical filtration, settling, or reversing the chemistry change
Aquarium significance	Determines suitability for species; affects buffering	A visual symptom that usually indicates a dosing or chemistry problem

[ESTABLISHED SCIENTIFIC PRINCIPLE] Dissolved ions do not scatter visible light appreciably. Water with very high GH and KH can be optically indistinguishable from soft water. Hardness alone does not cause cloudiness.

Where hard water genuinely does contribute

Hard water is not irrelevant — it is a precondition rather than a cause. High concentrations of calcium and carbonate mean that a smaller chemical push is required to exceed a solubility limit. So hard water raises the probability of precipitation events without itself causing cloudiness.

- Precipitation is more likely when a pH-raising product is added, because carbonate solubility falls as pH rises.
- Carbonate-based substrates such as crushed coral and aragonite contribute both physical dust and dissolving carbonate.
- Evaporation concentrates dissolved minerals; topping off with hard tap water rather than with RO or distilled water steadily raises concentrations over time.
- Scale deposits on heaters and equipment in hard water are the same chemistry expressed on a surface rather than in suspension.

[REASONABLE INFERENCE] Hard water should be understood as a factor that increases susceptibility to precipitation-driven cloudiness, not as a direct cause of cloudy water. The distinction matters because the remedy differs: adjusting a dosing practice versus changing the water source.

Preserved distinction

DISSOLVED MINERALS are not PRECIPITATED MINERALS. A GH or KH test result says nothing about whether the water will be clear. If a keeper reports cloudy water and hard water, the hardness is context, and the question is what changed to push something out of solution.

ADJACENT PAIN POINT

Part XXVII — Driftwood Biofilm and "Fungus"

Not cloudy water, frequently reported alongside it, and almost always harmless.

The white slimy growth that appears on new driftwood is not cloudy water, but it arrives in the same conversations, arises from the same underlying process, and alarms the same keepers. It is included because it is a genuinely common new-aquarium pain point and because the mechanism illuminates the bacterial-bloom discussion.

What it is

New submerged wood releases soluble organic compounds — sugars, tannins and other molecules — as it waterlogs. Microorganisms colonize that food source rapidly, forming a visible biofilm on the wood surface. The organisms involved are described variously as heterotrophic bacteria, saprophytic fungi, or a mixture; the consistent point across sources is that they are decomposers feeding on dead organic material rather than pathogens attacking living tissue.

[COMMON AQUARIUM PRACTICE] The white or grey slimy coating on new driftwood is consistently described across hobby sources as a saprophytic biofilm feeding on organics leaching from the wood. Reports converge strongly on it being harmless and self-limiting.

Answering the recurring questions

Is this cloudy water?

No. It is a surface growth. It becomes relevant to the cloudy-water discussion in two ways: the same dissolved organics that feed it also feed suspended bacteria in the water column, so the two frequently appear together; and pieces of it can detach.

Can pieces detach and become suspended?

Yes. Biofilm fragments dislodged by flow, by grazing animals, or by a keeper scrubbing the wood will appear as stringy or gelatinous suspended material, distinguishable from mineral dust by its texture and from a bacterial bloom by being resolvable as discrete fragments rather than uniform haze.

Is it harmful?

Sources converge on no, for fish, shrimp and plants. The qualifications offered are mechanical rather than toxic: heavy growth could clog a filter intake or smother delicate moss attached to the wood. A large mass of any decomposing organic material contributes to oxygen demand, which is the general principle rather than a specific property of wood biofilm.

Do shrimp and snails eat it?

Yes, consistently reported. Shrimp and snails graze it actively, and some breeders deliberately introduce new wood to provide supplemental grazing for shrimplets. Otocinclus, plecos and various other grazers are also reported to consume it.

Does it resolve on its own?

Yes, generally within a few weeks, as the readily soluble organics in the wood surface are depleted and grazing animals and competing microbial communities establish. Reported timeframes cluster around two to four weeks.

Should I scrub it off?

It can be removed manually if the appearance is objectionable, but this does not address the cause. The wood continues releasing organics and the film regrows. Scrubbing also puts the removed material into suspension, briefly worsening water clarity. One hobbyist report specifically argues against boiling a colonized piece, on the reasoning that boiling provides a clean surface for recolonization.

How do I tell biofilm from a fish disease?

Location. Biofilm grows on wood, hardscape and substrate — on dead organic material. Cotton-like growths on a fish are a different situation entirely and require a different response. Growth on wood is not a signal that fish are at risk.

How do I tell biofilm from wood rot?

Structure. Surface growth that can be wiped or siphoned away is biofilm. Wood that has become soft, is structurally breaking down, or smells foul is a different problem and may mean the piece is unsuitable.

Why this belongs in a cloudy-water document

The wood biofilm and the water-column bacterial bloom in a new tank are the same process expressed in two places. Dissolved organic carbon enters the system; microorganisms consume it; where they establish depends on available surface. Understanding that connection makes both phenomena predictable rather than mysterious — and explains why they so often appear in the same week.

MISIDENTIFICATION PREVENTION

Part XXVIII — Cyanobacteria

Almost always a mat, almost never the reason the water column is cloudy.

Cyanobacteria are photosynthetic bacteria, not algae, despite the persistent common name "blue-green algae." In freshwater aquariums they overwhelmingly grow as attached mats on substrate, glass, plants and hardscape rather than as a suspended population. This matters because "my tank is green" produces cyanobacteria advice that is irrelevant to a green-water problem, and vice versa.

Benthic versus planktonic growth

The environmental literature distinguishes cyanobacteria growing as attached mats in benthic habitats from those forming planktonic blooms in the water column, and notes that benthic growths present different assessment challenges precisely because they are patchy and are not visible from the surface in the way planktonic blooms are (ITRC, Harmful Cyanobacterial Blooms guidance). Both forms exist in nature. In home freshwater aquariums, the benthic mat form is what keepers encounter.

[STRONG EVIDENCE] Cyanobacteria occur in both attached benthic and suspended planktonic forms. In freshwater aquariums, the attached mat form dominates reported cases; planktonic cyanobacterial blooms of the kind seen in eutrophic lakes are not a typical aquarium presentation.

Distinguishing cyanobacteria from green water and from diatoms

Feature	Cyanobacteria	Green water	Diatoms
Location	Attached mats on surfaces	Suspended in the water column	Attached film on surfaces
Does the water column look colored?	No	Yes — that is the definition	No
Color	Blue-green, dark green, brown, black or reddish	Green	Brown, mustard or rusty
Texture	Slimy; peels off in sheets or clumps	Not applicable — it is suspended cells	Gritty; wipes off rather than peeling
Smell	Characteristically earthy or musty	Usually none distinctive	None distinctive
Primary drivers	Low flow plus available nutrients; light	Light plus available nutrients	Silicates plus light plus nutrients in an immature system
Responds to UV?	No — it is not in the flow path	Yes — highly effective	No
Typical timeline	Persistent until conditions change	Persistent until light or population is addressed	Usually self-limiting as the tank matures

[COMMON AQUARIUM PRACTICE] The reported practical distinctions — cyanobacteria peels off in sheets and smells earthy, diatoms feel gritty and wipe away — are consistently described across hobby sources and are useful field discriminators.

When cyanobacteria could contribute to water-column appearance

- Mats can detach and float. Environmental guidance specifically notes that benthic cyanobacterial mats can detach from the bottom, float to the surface and accumulate. In an aquarium this presents as visible floating clumps rather than as generalized cloudiness.
- Disturbing a mat during maintenance releases fragments into the water column, producing temporary particulate cloudiness.
- A large die-off following treatment adds organic matter to the system, which can drive a subsequent heterotrophic bacterial bloom — a genuine but indirect route from cyanobacteria to cloudy water.

Do not classify every green aquarium as cyanobacteria

The word "green" covers green water (suspended algae), green film algae on the glass, green spot algae, hair algae, and cyanobacteria in its green presentations. These are five different problems with five different responses. The first question is always: is the color in the water, or on a surface?

MISIDENTIFICATION PREVENTION

Part XXIX — Diatoms

A surface film, not a water-column condition.

Diatoms are the near-universal new-tank "brown algae." They are relevant to this document mainly for exclusion: they are frequently reported in the same breath as cloudy water, and they are not a cause of it.

What diatoms are

Diatoms are single-celled algae with silica cell walls, appearing as a dusty brown layer on glass, rocks, substrate and leaves. In an aquarium they grow as an attached film. The gritty texture keepers report comes from the silica structures the cells build.

[ESTABLISHED SCIENTIFIC PRINCIPLE] Diatoms are silica-walled unicellular algae. In aquarium conditions they predominantly form attached biofilms on surfaces rather than dense suspended populations.

Do diatoms cause water-column cloudiness?

Not typically. The standard aquarium presentation is a surface coating. A keeper reporting both a brown film on surfaces and cloudy water most likely has two separate things happening — which is entirely normal in a new tank, where a heterotrophic bacterial bloom and a diatom film commonly occur within the same few weeks.

One retail source describes the sequence explicitly: bacterial blooms cause the white cloudiness seen in the first weeks of a new aquarium, after which a thin brownish film — the diatom film — coats substrate and surfaces (That Pet Place). That ordering matches the widely reported new-tank experience.

[REASONABLE INFERENCE] Diatom films and heterotrophic bacterial blooms commonly co-occur in new aquariums as sequential stages of establishment. Their co-occurrence does not mean the diatoms are causing the cloudiness.

Drivers and timeline

- Silicates, light and dissolved nutrients are the standard listed requirements. Tap water is the main silicate source, and municipal levels vary substantially by region.
- Diatom blooms are strongly associated with immature systems, and most sources report resolution as the tank matures — commonly quoted windows are four to eight weeks, though this is a reported range rather than a measured distribution.
- The mechanism for resolution is competition: as plants, bacteria and grazing microfauna establish, diatoms lose their competitive advantage. This is consistent with the observation that mature planted tanks with high light and high nutrients nonetheless keep diatoms in check.

[ANECDOTAL OBSERVATION] The commonly cited four-to-eight-week resolution window for new-tank diatom films is a reported range across hobby and retail sources rather than a measured finding. Duration varies with silicate availability, plant mass, grazers and light.

The misdiagnosis this section prevents

A keeper with brown film on the glass and clear water does not have "brown cloudy water" and does not need any of the water-column interventions in this document. A keeper with brown water and clean glass is looking at tannins or suspended debris, not diatoms. The distinguishing question is where the brown is.

CONTEXT

Part XXX — Tank Age as a Diagnostic Clue

Shifts the probabilities. Never decides the answer.

How long an aquarium has been running changes which causes are likely, because it changes how much established biofilm exists, how much organic material has accumulated, and what has recently been done to the system. It is context, not diagnosis.

Tank age	Most likely causes	Less likely	Key questions
Day 1–3	Substrate dust; microbubbles; hardscape residue; filter media dust; the very beginning of an organic-driven bloom	Green water (insufficient time in most cases); tannins at full intensity; detritus accumulation	What substrate, and was it rinsed? How was the tank filled? Is there wood? Was the filter running during filling?
Week 1–4	Heterotrophic bacterial bloom; ongoing substrate fines; tannin release; wood biofilm; early diatom film	Detritus from accumulated waste; established-tank causes	Is it stocked? What do ammonia and nitrite read? Has anything died? What is the feeding schedule?
Months 1–6	Green water if light exposure changed; detritus and inadequate mechanical filtration; response to maintenance events; continued tannin release	Pure substrate dust (unless disturbed); new-setup bloom	What changed recently? Has light exposure changed? What is the filter maintenance history?
Long established (many months to years)	A specific recent event: a death, overfeeding, filter media replacement, aggressive cleaning, power outage, new livestock, new food, medication, substrate disturbance, source-water change	Spontaneous causes — a stable system rarely changes without an input	What is different this week compared to last week? Anything, however minor?

[REASONABLE INFERENCE] Tank age functions as a prior probability rather than as evidence. A four-day-old tank with white water is probably experiencing setup-related phenomena; a four-year-old tank with white water has almost certainly experienced a specific triggering event. Neither conclusion should be accepted without checking.

The trap to avoid

Tank age is frequently used as a conversation-ender: "it is a new tank, that is just a bacterial bloom." That reply is often right and is a poor habit, because the same appearance in the same new tank could be dust, gas, or precipitation — and because in a stocked new tank the question of whether ammonia is present is genuinely urgent and gets skipped.

CONTEXT

Part XXXI — Cloudy Water in New Aquariums

"I just set up my tank and now it is cloudy" — the single most common report in the hobby.

New-tank cloudiness usually has more than one cause running concurrently, which is why single-cause answers so often produce partial results that keepers interpret as failure.

The candidate list for a new setup

Cause	Timing signature	Confirming observation	Response
Substrate fines	Present from the moment of filling	Settles in a still glass; captured by floss; sediment visible after settling	Fine mechanical filtration; drain and refill; careful fill technique
Microbubbles	Minutes after filling or after a top-up	Clears top-down; bubbles on glass and leaves	None
Hardscape residue	Present from filling; may recur when rocks are moved	Associated with specific unwashed rocks or wood	Remove, rinse and replace the pieces
Filter media dust	Minutes after the filter starts	Began exactly when the filter was switched on	Rinse media before use; run floss for a day
Heterotrophic bacterial bloom	Days 2–10	Delayed onset; does not settle; not captured by floss	Usually time; reduce inputs; maintain aeration
Tannin release	Days, building gradually	Amber color with clarity preserved	Optional — aesthetic
Source-water turbidity	Present from filling, matching the tap	A glass of tap water looks the same as a glass of tank water	Settle or pre-filter the fill water
Aquasoil ammonium and organics	Days 1–21	Product-specific; combined with a measurable ammonia reading	Follow the manufacturer protocol; do not stock until it clears

Decision framework for a new tank

- 1 **Is anything alive in the tank?** If no, this is a patience-and-observation situation, not a welfare situation, and there is time to work through the diagnostics carefully.
- 2 **If yes, are the animals behaving normally?** Distress overrides everything else in this list. Go to Part LX.
- 3 **If stocked, what do ammonia and nitrite read?** In a new stocked tank this is not optional. The cloudiness may be irrelevant; the nitrogen readings are not.
- 4 **When exactly did the cloudiness start, relative to filling?** At the moment of filling points to physical causes; two or more days later points to biological ones.
- 5 **What does the white-cup test show?** Color identifies the category.
- 6 **Does a still sample settle, clear from the top, or stay uniform?** This separates particulate, gas and microbial.
- 7 **Does fine mechanical media load up within a day?** If yes, the cloudiness is particulate and being removed. If the media stays clean, the suspension is finer than the media or is not particulate at all.
- 8 **Is there wood, active substrate, or unrinsed material that could explain an organic input?**
- 9 **Only then: decide whether intervention is required at all.**

Why "multiple causes simultaneously" is the default expectation for new tanks

A typical new planted setup introduces, on day one: a substrate that may shed fines and release organics and ammonium, wood that leaches sugars and tannins, pressurized tap water carrying dissolved gas, plants that will partially melt, a filter with no established biofilm, and possibly unrinsed media. Every one of those is on the cloudy-water cause list. Expecting one answer is the mistake.

CONTEXT

Part XXXII — Cloudy Water in Established Aquariums

A stable system that changes has been changed. Find the input.

When an aquarium that has been clear for months or years suddenly clouds, the diagnostic logic inverts. In a new tank, the question is which of many simultaneous setup effects is responsible. In an established tank, the question is what single thing is different — because a system at equilibrium does not leave equilibrium without a push.

The established-tank investigation checklist

Possible change	Type of cloudiness	How to check
A dead animal that has not been found	White, developing over a day or two; possibly an odor	Census the livestock. Check behind hardscape, under substrate edges, inside decorations and in the filter intake and canister.
Overfeeding, including by someone else in the household	White, developing over a day or two	Ask directly. Uneaten food visible in the substrate is confirmatory.
Filter malfunction or reduced flow	Gradual haze and increasing visible particles	Compare current return flow to normal. Inspect impeller and media.
Filter media replaced or cartridge changed	White, over days; possible ammonia	Maintenance history. Test ammonia and nitrite.
Aggressive filter or tank cleaning	White, over days	Maintenance history. How much was cleaned, and with what?
Plant die-off or a heavy prune left to decay	White, over days; sometimes brown particulate	Inspect for melting or decaying plant tissue.
New livestock added	White, over days	Increased bioload beyond current capacity; also possible medication or bag-water introduction.
New food introduced	White or particulate	Different foods disintegrate differently and have different acceptance rates.
Medication dosed	White, over days	Antibacterial compounds can suppress the established microbial community.
Power outage	Brown on restart, then possibly white	Filter contents may have gone anaerobic; biofilm may have been damaged.
Substrate disturbed	Brown particulate, immediate	Recent planting, rescaping, digging fish, or vacuuming.
Lighting changed or photoperiod extended	Green, over days to weeks	New fixture, new timer setting, or a room lighting change.
Sunlight exposure changed	Green, over days to weeks	Seasonal sun angle, curtains left open, furniture moved, a tree pruned outside.
Fertilizer regime changed	Green, or chalky white if precipitation	Dosing history.
Source water changed	Variable; may match the tap	Compare a glass of tap water to a glass of tank water. Contact the utility if a main was flushed or treatment changed.
Equipment added that entrains air	White with fine bubbles	New powerhead, air pump, or a filter reseated imperfectly.

[USEFUL RULE OF THUMB] In an established aquarium, sudden cloudiness with no identifiable recent change most often means the change has not been identified yet — commonly a death that has not been discovered or a feeding that was not observed. Widening the search is usually more productive than concluding the cause is spontaneous.

Different questions for a different situation

In a new tank, "is this normal?" is a reasonable opening question and the answer is often yes. In a long-established tank, "is this normal?" is the wrong question, because it was normal until something changed. The right question is "what is different?"

DIAGNOSIS

Part XXXIII — Smell as a Diagnostic Clue

Genuinely informative, frequently overinterpreted, and worth a safety note.

Odor carries real diagnostic information because different microbial processes produce different volatile compounds. It is also the least reliable sense to formalize, because perception varies enormously between people and descriptions are subjective.

A safety note first

This document does not encourage deliberately inhaling directly from an aquarium, a filter canister, or a disturbed substrate. Anaerobic pockets in substrate and in stagnant filters can release hydrogen sulfide, which is hazardous at sufficient concentration. Normal ambient awareness of how a tank smells during routine maintenance is entirely different from putting one's face into a canister and inhaling deliberately. The advice is to notice, not to sample.

Odor	Commonly associated with	What it suggests	Reliability
No noticeable smell, or a faint clean earthy-fresh scent	A healthy, well-maintained aquarium	Nothing requiring action	Moderate — a healthy tank can still have a mild smell
Earthy or musty	Cyanobacteria; also certain soil substrates	Look for mats on surfaces rather than a water-column problem	Moderate — frequently reported as a discriminator for cyanobacteria
Swampy or stagnant	Accumulated organic matter; poor circulation; dead zones	Detritus accumulation and possibly low oxygen in parts of the system	Moderate
Rotten-egg / sulfurous	Hydrogen sulfide from anaerobic decomposition	A significant anaerobic pocket, commonly in deep substrate or a filter that has been off. Treat as a warning sign.	Reasonably specific — the compound is distinctive
Strong decaying-organic / fishy-rotten	A dead animal, or a large mass of decaying material	Search actively for a casualty or a food mass	High as a prompt to search; not proof
Ammonia-like or sharp	Sometimes reported at very high ammonia concentrations	Test immediately	Low — do not rely on smell to detect ammonia; ammonia is generally not detectable by smell at concentrations that are already dangerous to fish
Chemical or perfumed	Contamination from cleaning products, aerosols, or hands	Investigate what has been near or in the tank	Low but worth noting

[ANECDOTAL OBSERVATION] Odor-based aquarium diagnosis rests on convergent hobby reporting rather than on measurement. It is useful for generating hypotheses — particularly "there may be something dead in here" — and should not be used to rule anything out.

Limitations of smell as a diagnostic tool

- Olfactory sensitivity varies enormously between individuals and declines with sustained exposure — a keeper standing over their own tank daily adapts to its smell.
- Descriptions are not standardized. "Earthy," "musty," "swampy" and "fishy" mean different things to different people.

- The most dangerous condition in this entire document — elevated ammonia — is not reliably detectable by smell at aquarium concentrations.
- A tank can smell entirely normal while carrying a serious problem, and can smell slightly off while being perfectly healthy.

[REASONABLE INFERENCE] Smell should be treated as a hypothesis generator, never as a substitute for testing. Its single highest-value use is prompting a thorough search for decaying material that has not yet been found.

DIAGNOSIS

Part XXXIV — Water Tests and What They Actually Tell You

Essential for safety. Nearly useless for identifying the source of cloudiness.

This part makes an uncomfortable point clearly: no ordinary aquarium test kit measures anything that identifies why water is cloudy. Test kits measure dissolved chemistry. Cloudiness is, in most cases, suspended matter or a gas phase. The tests still matter enormously — for the animals — but they answer a different question than the one the keeper is asking.

Test	What it measures	What it tells you about cloudiness	What it tells you about safety
Ammonia	Total ammonia nitrogen (free ammonia plus ammonium)	Almost nothing. Blooms occur at zero ammonia and dangerous ammonia occurs in clear water.	Critical. The single most important test in any stocked tank showing any abnormality.
Nitrite	Nitrite nitrogen	Almost nothing directly. Elevated nitrite indicates incomplete nitrification, which may share a cause with the cloudiness.	Critical. Directly toxic to fish.
Nitrate	Nitrate nitrogen	Very little. High nitrate indicates accumulated waste, which correlates loosely with organic load, but the correlation is weak enough to be unreliable.	Moderate. Chronic elevation matters for long-term health.
pH	Hydrogen ion activity	Relevant in two narrow cases: a rising pH can precipitate carbonates, and large diel swings suggest a dense photosynthetic population.	Important, primarily through its effect on ammonia toxicity — the proportion of free ammonia rises sharply with pH.
KH (carbonate hardness)	Carbonate and bicarbonate alkalinity	Relevant to precipitation risk and to whether tannins will move pH.	Important as a stability reserve. Low and falling KH is an early warning that pH may become unstable.
GH (general hardness)	Calcium and magnesium concentration	Context for precipitation risk only. Does not indicate cloudiness.	Important for species suitability and for invertebrate moulting.
Phosphate	Orthophosphate concentration	Some relevance to algae questions and to calcium phosphate precipitation risk.	Low direct toxicity concern at aquarium concentrations.
Dissolved oxygen	Dissolved O ₂ concentration	Highly relevant to the consequences of a bloom, though not to its identity. Rarely measured by hobbyists.	Critical, and underused. If measurable, this is the test that matters most during a heavy bloom.
TDS / conductivity	Total dissolved solids, inferred from conductivity	Minimal. It measures dissolved material, and most cloudiness is suspended.	Useful for tracking source-water consistency and for invertebrate keeping.
Chlorine / chloramine	Residual disinfectant	None for cloudiness.	Critical after any water change, and after any utility supply event.
Silicate	Dissolved silica	Relevant only to diatoms, which are a surface film rather than a water-column condition.	No direct toxicity concern.

Two statements that must both be held simultaneously

The symmetry that resolves most confusion

A normal test panel does NOT identify the source of cloudiness. Zero ammonia, zero nitrite and moderate nitrate are entirely compatible with a heavy bacterial bloom, dense green water, heavy substrate dust, or severe microbubbles.

Cloudy water does NOT automatically indicate poor ammonia or nitrite control. The two conditions have different drivers and coincide only when they share an upstream cause.

[ESTABLISHED SCIENTIFIC PRINCIPLE] Colorimetric aquarium test kits measure dissolved chemical species. They have no sensitivity to suspended particulate, to microbial cell density, or to dissolved gas beyond their specific analytes. A clean panel and a cloudy tank are not contradictory results.

So why test at all during a cloudy-water event?

Because the purpose of testing is not to diagnose the cloudiness. It is to determine whether the animals are currently in danger, which is a separate and more urgent question. A keeper who tests during a bloom is not looking for the cause of the haze — they are checking whether the condition that produced the haze has also produced something toxic.

[REASONABLE INFERENCE] The correct framing for testing during a cloudy-water event is triage, not diagnosis. Test to establish safety, then use physical observation to establish cause.

CONCEPTUAL

Part XXXV — Is Cloudy Water Always a Water-Quality Problem?

No. And the cases where it is not are more numerous than the hobby assumes.

Answering this directly: cloudiness ranges from completely inconsequential to a visible signal of a genuine emergency, and the appearance alone does not tell you where on that range you are.

Condition	Is water quality affected?	Is livestock welfare affected?	Assessment
Tannin coloration	Chemistry is altered — dissolved organics present, possible pH influence in low-KH water. This is not degradation.	Generally no; potentially positive for blackwater species	Aesthetic and chemistry topic, not a quality failure
Substrate dust	Essentially no chemical effect for inert substrates	Minimal in typical concentrations; extremely heavy prolonged suspended solids could irritate gills, though this is not a commonly reported aquarium outcome	Cosmetic; resolves with filtration and time
Microbubbles (transient)	No	No	Entirely benign; requires no action
Microbubbles (persistent, from an equipment fault)	Potentially — gas supersaturation is a real hazard at sufficient levels	Yes at sufficient supersaturation	The equipment fault is the problem, not the appearance
Green water	Not directly degraded; dissolved nutrients are being taken up by the algae	Indirect: diel oxygen and pH cycling; loss of observation	Worth addressing, but not equivalent to poisoned water
Mild bacterial bloom in a new, lightly stocked or unstocked tank	Indicates available organics; not itself degradation	Usually no	Normal succession
Bacterial bloom from a large organic input (death, food dump)	Yes — oxygen demand, likely ammonia	Yes, potentially severely	The cloudiness is a visible symptom of a genuine problem
Suspended detritus	Not directly; accumulated organics eventually contribute to oxygen demand	Generally no at moderate levels	Filtration and flow question
Mineral precipitation	Indicates a chemistry event; the precipitate itself is usually inert	Generally no, but the underlying chemistry swing may matter	Investigate the dosing practice

The conclusion this table supports

Cloudiness is a signal with a very wide range of meanings, from "nothing is wrong" through "something is inconvenient" to "act immediately." What determines which one you are looking at is not how cloudy it is but what is causing it and what else is happening in the tank — particularly to the animals.

[REASONABLE INFERENCE] Because the relationship between cloudiness and harm is mediated entirely by cause, teaching keepers to identify cause is more valuable than teaching them any specific remedy. This is the core editorial conclusion of this research.

CONCEPTUAL

Part XXXVI — Clear Water Does Not Always Mean Healthy Water

The mirror-image misconception, and the more dangerous one.

If cloudy water can be harmless, clear water can be lethal. This part is deliberately brief; its purpose is to establish one principle, not to expand into a general water-quality guide.

Dangerous conditions that are optically invisible

Condition	Visible?	Why it is invisible	How it is detected
Ammonia	No	A dissolved species with no light-scattering or absorbing properties at aquarium concentrations	Test kit
Nitrite	No	Same	Test kit
High nitrate	No	Same	Test kit
Chlorine and chloramine	No	Dissolved gases and compounds	Test kit; or reliably prevented by conditioning all incoming water
Low dissolved oxygen	No	A dissolved gas	DO meter or test; or inferred from fish behavior
Gas supersaturation	Not directly	A dissolved gas condition	Total dissolved gas measurement; or inferred from persistent microbubbles and clinical signs
Heavy metals (e.g. copper from plumbing)	No	Dissolved ions	Specific test kits; often only investigated after unexplained invertebrate losses
Pathogens	No	Microscopic and at low density	Observation of clinical signs in the animals
pH outside a species' tolerance	No	A chemical property	Test kit
Temperature problems	No	Not an optical property	Thermometer

[ESTABLISHED SCIENTIFIC PRINCIPLE] All of the principal chemical hazards in freshwater aquaria — ammonia, nitrite, nitrate, chlorine, chloramine, dissolved metals — are dissolved species that produce no visible change in water appearance at relevant concentrations. Optical clarity carries no information about any of them.

The principle, stated once and plainly

CLARITY IS NOT A WATER-QUALITY TEST. It is one observation among many. A crystal-clear aquarium may be perfectly healthy or may be killing its occupants, and looking at it will not distinguish those cases. This is the reason the entire diagnostic framework in this document begins with livestock behavior and testing rather than with appearance.

A brief note on the aesthetic pursuit of crystal-clear water

Wanting a visually pristine aquarium is a legitimate goal and is achievable through generous mechanical filtration, adequate biological capacity, appropriate stocking and consistent maintenance. Nothing in this document argues against it. The argument is narrower: clarity should be pursued as an aesthetic outcome and as a byproduct of good husbandry, not treated as evidence that husbandry is good.

INTERVENTIONS

Part XXXVII — Water Changes Across All Causes

The same action producing five different outcomes, depending entirely on what is in the water.

A water change is not a treatment. It is a dilution and a physical export. Whether that helps depends entirely on whether the thing causing the problem is dilutable, exportable, or neither.

Cause	What a water change does	Net effect	Better or complementary actions
Heterotrophic bacterial bloom	Removes a proportion of suspended cells and a proportion of the dissolved organics feeding them	Variable. Short-term visual improvement is common; recurrence is likely if the organic source remains. In a cycling tank it may lengthen the process.	Identify and remove the organic source; maintain aeration; allow biofilm to establish
Green water	Removes a proportion of the algal population and dissolved nutrients	Low. The remaining population regrows under unchanged light and nutrient conditions.	Address the light source; UV; blackout; plant competition
Substrate dust	Physically exports suspended particles, particularly when combined with siphoning	High. This is one of the most effective actions available for this cause.	Fine mechanical filtration; avoid re-disturbing
Tannins	Dilutes dissolved color proportionally	Moderate and temporary. Wood continues releasing, so color partially returns.	Carbon or adsorbent resin if removal is desired; or acceptance
Suspended detritus	Exports the particles present	Moderate to high, especially with substrate siphoning. Does not fix the underlying filtration or flow inadequacy.	Improve mechanical filtration and circulation
Mineral precipitation	Dilutes the ions involved and removes the precipitate	Moderate, and can be counterproductive if the incoming water is itself the source of the ion imbalance	Change the dosing practice — this is the actual fix
Microbubbles	Nothing useful; a fresh fill of pressurized water may briefly add more	None or slightly negative in the short term	Wait; investigate equipment if persistent
Underlying ammonia or nitrite	Directly reduces exposure	High — this is the priority action regardless of what the water looks like	Address the cause of the nitrogen; feed less; verify filtration
Decaying organic material present	Reduces dissolved products but does not remove the material	Partial. Necessary but insufficient.	Physically remove the material — this is the actual fix

Four possible outcomes of the same action

HELP: when the problem is dilutable or exportable — dust, detritus, ammonia, nitrite.

LITTLE EFFECT: when the problem regenerates or is not in the water — green water, microbubbles, an unaddressed organic source.

MASK A SYMPTOM TEMPORARILY: when it improves appearance without touching the cause — the most common failure mode, because the visible improvement is read as a solution.

BE IMPORTANT FOR AN UNDERLYING PROBLEM: when nitrogenous compounds or a heavy organic load are present, in which case the cloudiness is beside the point and the water change is doing real work.

[REASONABLE INFERENCE] Neither "always water change" nor "never water change" survives contact with the range of causes. The useful question is what the water change is expected to remove and whether that thing is actually the problem.

INTERVENTIONS

Part XXXVIII — Mechanical Filtration

Which forms of cloudiness can physically be trapped, and which cannot.

Mechanical filtration works by physical exclusion: particles larger than the effective pore size are retained, smaller ones pass through. That single principle predicts almost everything about when adding filter floss will work.

Media types and their roles

Medium	Approximate role	Best used for	Limitations
Coarse sponge (10–20 ppi)	Bulk particle capture; large surface for biofilm	First stage; protecting finer media downstream	Passes fine particles freely
Fine sponge (30–45 ppi)	Intermediate capture	Second stage; good compromise between capture and flow	Clogs faster than coarse; still passes very fine material
Filter floss / polyester wool	Fine depth filtration — the workhorse for polishing	Fine dust, fine detritus, and clearing after disturbance	Loads quickly and must be replaced or rinsed every few days when working hard; restricts flow when packed tightly
Polishing pads (pinkie pads, micron pads)	Fine surface filtration	Final polishing stage	Clogs rapidly in a heavy load; can cause bypass if it blinds completely
Diatomaceous earth filter	Very fine filtration, into the low-micrometre range	Rapid clearing of very fine particulate and of algal cells	Specialist equipment; clogs quickly; typically run intermittently rather than continuously
Ceramic and sintered biological media	Biofilm habitat, not mechanical capture	Biological filtration	Contributes essentially nothing to clarity; frequently misunderstood on this point

What can and cannot be mechanically captured

Suspended material	Approximate size scale	Mechanically capturable?	Practical outcome
Sand and coarse silt	Tens to hundreds of micrometres	Yes, readily	Captured by almost any medium; also settles
Fine silt and detritus fragments	A few to tens of micrometres	Yes, with fine media	Filter floss and polishing pads are effective
Clay-sized particles	Below a few micrometres	Partially, with very fine media	Difficult; DE filtration or flocculation plus capture is the reliable route
Algal cells (green water)	Roughly 2–15 micrometres depending on species	Marginally, with very fine media; readily with DE filtration	Ordinary media give partial results at best, and the population reproduces continuously
Individual bacteria	Roughly 0.5–2 micrometres	Not with ordinary aquarium media	This is why filter floss does not clear a bacterial bloom
Bacterial flocs and aggregates	Variable, can be much larger	Partially	Explains why some blooms respond somewhat to fine filtration and others do not

Suspended material	Approximate size scale	Mechanically capturable?	Practical outcome
Dissolved tannins and humic substances	Molecular — far below any filtration threshold	No	Requires adsorption, not filtration
Dissolved gases / microbubbles	Not applicable — a gas phase	No	Resolves on its own

[ESTABLISHED SCIENTIFIC PRINCIPLE] Mechanical filtration removes particles above the effective pore size of the medium. Individual bacteria, at roughly 0.5 to 2 micrometres, are below the practical capture range of standard aquarium mechanical media, which is the mechanistic reason fine filtration does not clear a bacterial bloom.

Practical guidance

- Arrange media coarse-to-fine in the direction of flow. Fine media placed first blinds immediately and causes bypass.
- Expect fine media to require frequent attention when it is doing real work. Floss that stays white after two days is not capturing anything.
- Do not pack floss so tightly that flow is substantially restricted — reduced flow means less water processed and can worsen the outcome.
- Treat the mechanical stage and the biological stage as separate maintenance items. Cleaning the first frequently is good practice; cleaning the second frequently is the mistake described in Part XLIV.
- If floss loads heavily and the water is still cloudy, there are probably two things in suspension: one being captured and one not.

[COMMON AQUARIUM PRACTICE] Polishing filtration with floss or fine pads is widely used and effective for particulate clarity. It requires maintenance on a scale of days rather than weeks when working hard, and its flow restriction is a real trade-off rather than a theoretical one.

INTERVENTIONS

Part XXXIX — Activated Carbon and Adsorbent Media

Excellent for dissolved coloration. Useless for suspended anything.

Activated carbon is one of the most misunderstood products in the hobby, because it is marketed as a general water-quality improver and is mechanistically a specific tool for a specific class of molecule.

How it works

Activated carbon is an extremely porous material with a very large internal surface area. Dissolved organic molecules adhere to that surface through adsorption. Two consequences follow immediately: it acts only on dissolved substances, and its capacity is finite — once the available surface is occupied, adsorption stops.

Pore structure matters more than most keepers realize. Water-treatment literature uses different indicator numbers for different molecular sizes, with the tannic acid number specifically indicating a carbon's capacity for large, high-molecular-weight natural organic molecules, and the phenol number indicating capacity for smaller polar organics. Different carbons genuinely differ in their suitability for removing tannin-type coloration.

[ESTABLISHED SCIENTIFIC PRINCIPLE] Activated carbon removes dissolved organic molecules by adsorption onto internal pore surfaces. Capacity is finite and pore size distribution determines which molecular sizes are removed effectively.

What carbon can and cannot address

Target	Effective?	Explanation
Tannins and yellowing	Yes	The primary legitimate use for clarity purposes. Pilot-scale drinking-water work reported over 70 percent removal of humic substances by granular activated carbon, though only around 50 percent reduction in measured color (Kucerova et al.).
Dissolved organic compounds generally	Yes, partially	Reduces the pool of dissolved organics, which indirectly reduces the substrate available to heterotrophic bacteria.
Medications and some chemical residues	Yes	Which is why carbon is removed during medication courses.
Odors from dissolved compounds	Yes, partially	Same adsorption mechanism.
Bacterial bloom (the suspended cells)	No	Bacteria are suspended particles, not dissolved molecules. There is no adsorption mechanism for them, and they are far too large to enter the pore structure.
Green water	No	Same reasoning — suspended cells.
Substrate dust	No	Suspended mineral particles.
Detritus	No	Suspended particulate.
Microbubbles	No	A gas phase.
Ammonia	Not meaningfully	Standard activated carbon is a poor ammonia adsorbent; zeolite is the medium used for that purpose.

[UNSUPPORTED AQUARIUM FOLKLORE] The claim that activated carbon clears cloudy water generally is not supported. It clears one specific category — dissolved organic coloration — and has no mechanism to act on the suspended particles and cells that cause most cloudiness.

Practical considerations

- Carbon exhausts. Replacement intervals in the range of a few weeks are typical, depending on load. Exhausted carbon is not harmful but does nothing.
- New carbon can shed fine dust. Rinsing before use avoids adding a particulate problem while solving a dissolved one.
- Carbon removes medications, which is desirable after treatment and undesirable during it.
- Carbon will remove tannins whether or not the keeper wants them removed — a blackwater keeper running carbon is working against themselves.
- It occupies filter space that could hold mechanical or biological media. That trade-off should be deliberate.

Synthetic adsorbent resins

Macroporous synthetic polymer resins marketed for organic removal are positioned as higher-capacity, regenerable alternatives to carbon for dissolved organic nitrogenous waste and organics. For dissolved coloration, they are a reasonable option.

[CONTESTED CLAIM] At least one independent hobby-science source reports testing a leading adsorbent resin specifically for clearing bacterial-bloom cloudiness and finding it ineffective (Aquarium Science, 7.4.6). This is not a contradiction of the product's intended function — removing dissolved organics is a different task from removing suspended bacteria — but it is directly relevant to keepers who buy it as a cloudy-water remedy.

The distinction that resolves the confusion

An adsorbent removes dissolved molecules. A bacterial bloom is suspended cells. Reducing dissolved organics may slow a bloom by removing its food, which is a real but indirect and slow effect. It will not clear the water today, and a product judged on that expectation will be judged as having failed.

INTERVENTIONS

Part XL — UV Sterilizers

Consolidated reference. The core dose discussion appears in Part XIV.

Part XIV covers UV mechanism and dose in the context of green water, where UV is most clearly indicated. This part consolidates the cross-cause picture and the setup variables in a form suited to retrieval.

Effectiveness by target

Target	Effective?	Notes
Green-water algae	Yes — highly	The organisms circulate through the unit repeatedly and receive cumulative exposure. Published aquaculture guidance puts required exposure for waterborne algae around 22,000 microwatt-seconds per square centimetre, higher than for common bacteria.
Suspended bacteria	Yes, for clearing	Will visibly clear a bloom, often quickly. Does not remove the dissolved organics driving it, so the underlying condition persists.
Free-swimming parasite stages	Partially, dose-dependent	A different and generally higher dose specification. Owning a clarifying-grade unit does not imply parasite control.
Viruses	Only at high doses	Some are highly UV-resistant — one aquaculture reference cites IPNV as requiring at least 246 mJ/cm ² .
Attached algae on glass, plants or hardscape	No	Never passes through the unit.
Tannins and dissolved organics	No	UV damages nucleic acids. Dissolved organic molecules have none. Additionally, colored water absorbs UV and reduces the unit's effectiveness against actual targets.
Substrate dust and detritus	No	Not living. Also reduces UV transmittance.
Microbubbles	No	A gas phase.
Ammonia and nitrite	No	Dissolved inorganic species.

The setup variables, restated

Variable	Effect if wrong	Practical guidance
Flow rate too high	Insufficient contact time; inadequate delivered dose	Slower flow through the unit generally means higher dose per pass. Manufacturer flow specifications for "clarification" and for "sterilization" typically differ substantially, and the sterilization figure is the slower one.
Undersized wattage	Insufficient intensity	Wattage alone is not a specification; it must be paired with flow.
Aged lamp	Output declines while the lamp still appears to work	Replace on the manufacturer's schedule, not on failure. A lamp that still emits visible light may be delivering a small fraction of its rated UV-C output.
Dirty or scaled quartz sleeve	Blocks UV before it reaches the water	Clean the sleeve as part of routine maintenance.

Variable	Effect if wrong	Practical guidance
Turbid or colored water	Absorbs and scatters UV before it reaches organisms	Demonstrated experimentally: reducing turbidity from 23.32 to 10.93 NTU improved log reduction of <i>E. coli</i> by more than 2.15 log at the same UV treatment (Chhetri et al.).
Placement in the flow path	Treats water that has not been mechanically cleared first	Positioning after mechanical filtration improves transmittance and therefore delivered dose.

[AQUACULTURE / WATER-TREATMENT PRACTICE] UV disinfection is standard practice in recirculating aquaculture and in drinking-water treatment, with well-established dose-response relationships. The engineering principle that unit performance cannot be inferred from wattage alone is explicit in aquaculture technical literature.

Why owning a UV unit does not guarantee results

A keeper reporting that their UV sterilizer "is not working" on green water is most often describing one of four situations: flow through the unit is too fast for the required dose, the lamp is past its service life, the sleeve is fouled, or the unit is undersized for the tank. A fifth possibility deserves checking first: the green is on the glass rather than in the water, in which case no UV unit could ever help.

A note on the "band-aid" critique

One independent hobby-science source takes the position that while UV will clear cloudy water, often overnight, the underlying problem remains and UV is therefore a band-aid rather than a solution for cloudy water — while still recommending UV as a worthwhile addition to any aquarium (Aquarium Science, 6.2.3). That framing is well taken for bacterial cloudiness, where the dissolved organics remain untouched. It is less applicable to green water, where physically removing the population, combined with light management, genuinely can resolve the situation.

[REASONABLE INFERENCE] UV is closer to a solution for green water and closer to a cosmetic intervention for bacterial cloudiness, because in the first case removing the population plus changing the light regime addresses the system, and in the second the food source is left in place.

INTERVENTIONS

Part XLI — Aquarium Water Clarifiers

A real mechanism from water-treatment engineering, applied to a narrower set of problems than the marketing implies.

Clarifiers are the product category most directly aimed at the symptom this document is about, which makes evaluating them squarely relevant. The underlying chemistry is legitimate and well established. The question is what it can and cannot accomplish.

Mechanism: coagulation and flocculation

Fine suspended particles in water typically carry a surface electrical charge that causes them to repel one another, keeping them dispersed. Coagulation neutralizes that charge, allowing particles to approach. Flocculation then bridges destabilized particles into larger aggregates — flocs — that settle rapidly or can be captured by mechanical media that would otherwise pass them.

[ESTABLISHED SCIENTIFIC PRINCIPLE] Coagulation and flocculation are standard unit processes in drinking-water and wastewater treatment. A coagulant destabilizes colloidal particles and a flocculant aggregates them into settleable or filterable flocs.

Common coagulant and flocculant chemistries in water treatment include aluminium sulfate, iron salts, synthetic polymers such as polyacrylamide, and biopolymers such as chitosan. Performance data from the treatment literature is instructive: in one aquaculture wastewater study, chitosan achieved 87.67 percent turbidity removal at a 20 mg/L dose after 30 minutes settling time (Ahmad et al., *Polymers*), while a review comparing biopolymers with conventional coagulants noted chitosan removing about 68.9 percent turbidity against more than 95 percent for aluminium sulfate and iron chloride in the compared studies.

Manufacturer positioning

[MANUFACTURER RECOMMENDATION] API describes its ACCU-CLEAR product as a colorless inorganic polymer that rapidly eliminates all forms of cloudy water caused by dirt, silt, dust and other small floating debris, working by causing tiny suspended particles to clump together. Note the scope in that claim: dirt, silt, dust and floating debris — that is, particulate. Aqueon's clarifier instructions advise waiting at least 24 hours after use and then performing a partial water change to remove settled particles.

That second manufacturer instruction is important and frequently ignored. It acknowledges directly that the product does not remove material from the system — it relocates it. Something still has to physically take it out.

What clarifiers can and cannot help

Cause	Clarifier effective?	Reasoning
Substrate dust and fine mineral particulate	Yes	This is the intended application. Aggregation converts uncapturable fines into capturable flocs.
Fine detritus	Yes, generally	Same mechanism, provided mechanical capture follows.
Post-maintenance particulate haze	Yes	Explicitly cited in manufacturer material as an application.
Green water	Partially	Can aggregate algal cells for capture, but the population continues reproducing and the driving conditions are unchanged.

Cause	Clarifier effective?	Reasoning
Bacterial bloom	Partially at best	May aggregate some suspended material and improve appearance. Does nothing about dissolved organics, and the population regrows.
Tannins	No	Dissolved molecules are not colloidal particles. Coagulation of very high-molecular-weight humic acids is possible in water treatment but is not what aquarium clarifiers are dosed to do, and low-molecular-weight fulvic acids are removed poorly even in that context.
Microbubbles	No	A gas phase; there is nothing to flocculate.
Mineral precipitation	Cosmetically	Aggregates the precipitate but does not address the chemistry that produced it.

Risks and limitations

- **Filter clogging.** The product's whole purpose is to make particles larger and more capturable, which means the mechanical media will load faster. In a heavily loaded tank this can restrict flow significantly.
- **Overdosing.** More is not better in coagulation chemistry — excess coagulant can restabilize particles rather than aggregating them, a well-known effect in water treatment. Following the dose instruction is not merely caution.
- **Repeated use masking a cause.** A keeper dosing weekly to control a problem that recurs weekly has a filtration, feeding or flow problem being managed cosmetically.
- **Settled material returning.** If flocs settle and are not removed, later disturbance re-suspends them.
- **Gill considerations.** Concern about aggregated material and fish gills is commonly raised in hobby discussion. No aquarium-specific evidence quantifying this risk was located, and this document declines to assert it as established. The defensible statement is that heavy suspended-solids loads are known to affect gill function in fish generally, which is a reason for caution rather than a documented clarifier harm.

[ANECDOTAL OBSERVATION] Claims of direct gill harm from aquarium clarifier products at label doses were not substantiated by any source located in this research. The general principle that high suspended-solids concentrations affect fish gills is well established in fisheries science but is not the same claim.

The honest summary

Clarifiers do what they say for the problem they are designed for: fine inert particulate, with adequate mechanical filtration downstream and a follow-up water change to export the settled material. They are not a general cloudy-water remedy, and using them without identifying the cause is how keepers end up dosing repeatedly with no lasting improvement.

INTERVENTIONS

Part XLII — "Quick Fix" Products Evaluated

Four questions asked of every product category.

For each product category: what problem does it actually address, what does it not address, when is it useful, and when is it treating appearance only?

Product	Actually addresses	Does NOT address	Genuinely useful when	Treating appearance only when
Water clarifiers (floculants)	Fine inert particulate suspension	Dissolved compounds; gas; the causes of any of it	Post-maintenance dust that fine filtration alone is not clearing fast enough	Dosed repeatedly for a recurring problem, or used on a bloom
Bacterial additives	Introducing nitrifying and heterotrophic organisms to a new or disrupted system	Suspended particulate; dissolved coloration; gas; excess organic input	Establishing or re-establishing a filter, particularly after media replacement	Dosed at an established tank's bloom in the belief it clears water
Algaecides	Killing algal cells	The light and nutrient conditions that produced them; any non-algal cloudiness	Rarely — and only after algae is confirmed and other options are unavailable	Used on cloudiness that has not been confirmed as algal, which is common
UV sterilizers	Suspended living organisms passing through the unit	Anything attached, dissolved, gaseous or inert	Green water; rapid clearing of a microbial suspension	Used as a permanent substitute for addressing organic load
Activated carbon	Dissolved organic molecules including tannins; medications	Anything suspended	Removing unwanted coloration; post-medication cleanup	Added in response to particulate cloudiness
Synthetic adsorbent resins	Dissolved organic waste and coloration	Suspended cells and particles	Ongoing organic management; coloration removal	Bought as a bacterial-bloom remedy
Phosphate removers	Dissolved phosphate	Cloudiness of any kind directly	A specific, tested phosphate problem in an algae context	Used speculatively for cloudy water
Ammonia-binding conditioners	Temporarily reducing free ammonia toxicity	Cloudiness; the cause of the ammonia	Bridging a genuine ammonia emergency while the real fix is applied	Used routinely instead of addressing filtration or stocking
"Sludge remover" bacterial products	Claimed digestion of accumulated organics	Verified in aquarium conditions — independent evidence is scarce	Uncertain	Substituted for physical removal of detritus, which is reliable

[CONTESTED CLAIM] The efficacy of bottled bacterial products varies substantially by formulation. A study cited in recent home-aquarium biofilter research found that only one of five tested quick-start nitrification products significantly reduced aquarium ammonia concentrations. This is a finding about ammonia reduction, not about clarity, but it is directly relevant to how much confidence to place in the category.

The question to teach

Before buying anything: what, physically, is in my water, and does this product act on that? A product that adsorbs dissolved molecules cannot remove suspended cells. A product that aggregates particles cannot remove dissolved color. A device that damages nucleic acids cannot affect sand. Almost every disappointed product review in this category is a mismatch between mechanism and problem.

INTERVENTIONS

Part XLIII — Should You Turn Off the Light?

A precise intervention for one cause, and a null action for every other.

Lighting reduction and blackouts are among the most frequently recommended responses to cloudy water, and they only act on organisms that require light. That is a very short list.

Cause	Effect of reducing light or blacking out	Why
Green water	Substantial — potentially decisive	Removes the energy source for photosynthetic organisms
Attached algae	Moderate, slower	Same mechanism, but attached forms often have stored reserves and recover
Cyanobacteria	Moderate — blackout is a recognized approach	Photosynthetic, though many species are also capable of using very low light
Heterotrophic bacterial bloom	None	These organisms derive energy from organic carbon, not light
Substrate dust	None	Mineral particles do not respond to light
Detritus	None	Same
Tannins	None — possibly a very slow reduction in photodegradation	Dissolved organics are not light-dependent in any practically relevant way
Microbubbles	None	A gas phase
Mineral precipitation	None	A chemistry event

[ESTABLISHED SCIENTIFIC PRINCIPLE] Light reduction affects photosynthetic organisms and nothing else. Applying it to non-photosynthetic causes of cloudiness has no mechanism of action.

Practical notes on blackouts

- A blackout must be total. Ambient room light through the glass is often enough to sustain a bloom, which is a common reason blackouts are reported as failing.
- Plants tolerate a few days of darkness but are stressed by it; some species melt. Plant losses add organic material to the tank, which can produce a bacterial bloom after the algae clears.
- Feeding should be reduced during a blackout, since the fish cannot see well and uneaten food adds to the organic load at exactly the wrong moment.
- Aeration should be maintained or increased. Photosynthetic oxygen production stops entirely during a blackout while respiration continues.
- A blackout removes the population but not the conditions. Without a change to light exposure or plant competition, recurrence is the expected outcome.

[COMMON AQUARIUM PRACTICE] Blackouts of roughly three to five days are widely used for green water with mixed reported success. Failures are commonly attributable to incomplete light exclusion or to unchanged underlying conditions rather than to the method being unsound.

The error this part exists to prevent

A keeper with a milky new-tank bacterial bloom who is told to black out the tank for three days will emerge with the same cloudy water, stressed plants, and a lost week. The intervention was not wrong in general — it was matched to the wrong cause.

INTERVENTIONS

Part XLIV — Should You Clean the Filter?

"My water is cloudy, so I cleaned my filter" — sometimes the fix, sometimes the cause.

This is a genuinely two-sided question, and the answer turns on which part of the filter is being cleaned and how thoroughly.

Scenario	Outcome	Reasoning
Mechanical media is visibly clogged and flow has dropped	Cleaning helps	Restoring flow restores the rate at which water is processed and particles captured. This is the case where the instinct is correct.
Mechanical media is fine and flow is normal	Cleaning does nothing for the cloudiness	If capture capacity was not the limitation, restoring it changes nothing.
Biological media is cleaned lightly in tank water	Little effect either way	Light rinsing removes loose debris while leaving most of the established biofilm intact.
Biological media is cleaned aggressively, or under running tap water	Cleaning makes things worse, with a delay	Removing established biofilm reduces the system's capacity to process organics and ammonia. The consequence appears over the following days, which obscures the connection.
The entire filter is stripped and everything replaced	Cleaning causes a new problem	Functionally equivalent to installing a new, uncycled filter.
A canister that has been off for a period is opened and cleaned before restart	Cleaning helps	Prevents ejecting accumulated and possibly anaerobic material into the tank.

[COMMON AQUARIUM PRACTICE] Independent testing reported by one hobby-science source found that thorough filter cleaning measurably impaired filter performance, supporting the widely held practice of cleaning mechanical media frequently and biological media rarely and gently, in tank water rather than under a tap (Aquarium Science, 6.8).

The practical rule

Separate the stages, separate the schedules

Mechanical stage (floss, pads, coarse sponge doing capture work): clean or replace often. This is the stage that gets dirty on purpose.

Biological stage (sponge, ceramic, sintered media hosting biofilm): disturb rarely, gently, and in tank water. Flow restriction is the signal to act, not appearance.

Never clean everything at once. If both stages need attention, stagger them by a week or more.

[REASONABLE INFERENCE] Because cleaning the biological stage produces its consequences over several days while cleaning the mechanical stage produces its benefit immediately, keepers systematically underattribute cloudiness to filter cleaning. The delay hides the causal link.

INTERVENTIONS

Part XLV — Should You Replace Filter Media?

"My water is cloudy, so I replaced the cartridge."

Filter cartridge replacement deserves specific treatment because it is the action most frequently recommended by product packaging and most frequently identified by experienced keepers as a cause of the problem it is supposed to solve.

What replacing media does

Effect	Direction	Notes
Restores mechanical capture capacity	Positive	A clogged or degraded medium captures poorly. New media captures well.
Removes clogged material restricting flow	Positive	Restores processing rate.
Removes established biological capacity	Negative and often substantial	The biofilm lives on the medium. Discarding the medium discards the biofilm.
Fails to address the actual cause	Neutral to negative	If the cloudiness is a bloom, dust, gas or coloration, none of these are affected by new media — and the biofilm loss may make the bloom worse.

[STRONG EVIDENCE] Replacing all filter media removes the established nitrifying and heterotrophic biofilm along with the medium. This is repeatedly identified across sources as a leading cause of bacterial blooms and of ammonia appearing in previously stable aquariums.

Better approaches

- Where a cartridge is degrading physically, replace it while retaining the old one in the filter alongside the new one for several weeks, so the biofilm can colonize the new medium before the old is removed.
- Where possible, convert cartridge-based filters to reusable foam or other durable media, so the mechanical component can be rinsed rather than discarded and the biological component is never thrown away.
- Replace only one component at a time when multiple media types are present.
- Treat "replace monthly" packaging instructions as a commercial recommendation rather than a husbandry requirement, and evaluate them against the observable state of the media and the flow.

[CONTESTED CLAIM] Manufacturer instructions for cartridge-based filters commonly specify monthly replacement. Experienced hobby sources characterize this as commercially motivated and biologically harmful. The mechanism supporting the criticism — that discarding the medium discards the biofilm — is sound. The strength of the language used varies considerably between sources.

Reinforcing the filtration principle without re-teaching it

A filter does three separable jobs. Mechanical media should be replaced or cleaned freely because its job is to get dirty. Biological media should be preserved because its job is to host a slow-growing community. Cartridge designs combine both into a single disposable unit, which is why they are the source of so many of the problems in this document.

INTERVENTIONS

Part XLVI — Should You Clean the Entire Aquarium?

The response most likely to turn a cosmetic problem into a biological one.

"My tank is cloudy so I cleaned everything" is a completely understandable instinct and, in most cases, the single most counterproductive action available.

What a comprehensive clean actually removes

- The established biofilm on filter media — the system's nitrogen-processing capacity.
- Biofilm on substrate, glass, hardscape and decorations — a smaller but nonzero contribution, and the food supply for grazing invertebrates.
- The microfauna community that grazes suspended bacteria and detritus.
- A large proportion of the water, and with it the chemistry the animals are acclimated to.

And what it usually does not remove: the actual cause. If the cause was a dead animal that has not been found, a source-water problem, dissolved gas, or an ongoing overfeeding habit, a comprehensive clean addresses none of it while removing the system's capacity to cope.

[REASONABLE INFERENCE] Comprehensive simultaneous cleaning predictably reduces biological capacity while rarely removing the cause of cloudiness. The commonly reported sequence — clean everything, water clears briefly, cloudiness returns worse within days, ammonia appears — is exactly what this mechanism predicts.

Targeted maintenance versus resetting

Targeted maintenance	Sterilizing / resetting
Identify the specific cause first	Address everything at once in the hope of catching the cause
Clean the mechanical stage; leave the biological stage	Replace or scrub all media
Siphon the specific area where debris accumulates	Deep-clean the entire substrate in one session
Remove the identified organic source	Remove and scrub decorations, hardscape and equipment
Partial water change appropriate to the situation	Near-total water replacement
Change one variable so the effect can be observed	Change every variable so nothing can be attributed
Preserves the microbial community	Requires the microbial community to re-establish
Diagnosis remains possible	Diagnosis becomes impossible

When a comprehensive intervention IS warranted

There are genuine exceptions, and stating them prevents this part from becoming its own oversimplification:

- A gross contamination event — a chemical spill into the tank, a large quantity of an inappropriate substance.
- A very large organic overload, such as a substantial quantity of food dumped in, where the priority is physically removing the material rapidly.
- A confirmed serious disease situation requiring a specific protocol.
- A situation where the substrate itself is the problem and is being replaced deliberately.

Note the common feature: in each case there is a specific identified reason, and the action targets that reason. That is different from cleaning everything because the water looks bad.

The principle

An aquarium is a biological system that took weeks to establish. The impulse to reset it in an afternoon because it looks wrong is the impulse to trade a visual problem for a biological one. Targeted action based on identification is slower, less satisfying, and considerably more effective.

WELFARE

Part XLVII — Cloudy Water and Dissolved Oxygen

The mechanism by which cloudiness can genuinely become dangerous.

If there is one physiological pathway connecting cloudy water to harm, this is it. Not the particles, not the appearance — the oxygen.

How each cause interacts with dissolved oxygen

Cause	Effect on dissolved oxygen	Magnitude	Timing of risk
Heterotrophic bacterial bloom	Consumes oxygen through aerobic respiration and through the decomposition driving it	Potentially significant in a dense bloom with a large organic load	Continuous; worst in warm, heavily stocked, poorly circulated tanks
Decomposition of a large organic mass	High biological oxygen demand	Potentially severe	Continuous until the material is removed
Green water	Produces oxygen in light, consumes it continuously	Can be substantial in either direction in a dense bloom	Minimum just before lights come on; possible supersaturation late in the photoperiod
Warm water	Reduces oxygen solubility	Physical and unavoidable	Aggravates every other factor
Substrate dust	Essentially none	Negligible	Not applicable
Microbubbles	Generally neutral or slightly positive	Minor	The exception is supersaturation, which is a different hazard
Tannins	Minimal direct effect	Indirect only, via the decomposition of the botanical material itself	Relevant when a large quantity of botanicals is added at once

[ESTABLISHED SCIENTIFIC PRINCIPLE] Oxygen solubility in water decreases as temperature increases. Aerobic microbial decomposition consumes dissolved oxygen. Photosynthetic organisms produce oxygen in light and consume it continuously, producing diel cycles in dense populations.

When adding aeration is a useful precaution

Increasing aeration is one of the few interventions in this entire document that is fast, cheap, low-risk and beneficial across a wide range of causes. It is close to a default safe action during any significant cloudy-water event in a stocked tank.

- Any dense bacterial bloom, particularly in warm or heavily stocked tanks.
- Any situation where decaying organic material is present or suspected.
- Dense green water, particularly for the overnight period.
- During and after a blackout, when photosynthetic oxygen production stops entirely.
- Whenever fish show any respiratory sign — before anything else is attempted.
- During elevated temperatures.
- After a filter failure or during any period of reduced circulation.

[REASONABLE INFERENCE] Supplemental aeration has minimal downside in a freshwater aquarium and addresses the principal mechanism by which cloudy-water conditions cause harm. It is defensible as an immediate precautionary step while diagnosis proceeds.

One qualification worth noting for planted tanks running injected CO₂: increased surface agitation drives CO₂ out of solution, which will reduce CO₂ availability to plants. That is a real trade-off, and in an oxygen emergency it is the correct trade to make — the plants recover, the fish may not.

Practical aeration measures

Increase surface agitation	Raise or redirect the filter return so it breaks the surface. Gas exchange happens at the surface, and turbulence increases the effective surface area.
Add an air stone	Rising bubbles both carry some gas exchange directly and, more importantly, drive circulation that renews the surface layer.
Lower the water level slightly	Increases the drop from a hang-on-back return and therefore the agitation.
Reduce temperature slightly	Cooler water holds more oxygen. Changes should be gradual.
Reduce feeding temporarily	Lowers both the organic input driving oxygen demand and the metabolic oxygen demand of the fish.
Increase circulation	Eliminates stagnant zones where local oxygen depletion is worst.

WELFARE

Part XLVIII — Cloudy Water and Fish Behavior

The observation that outranks every other observation in this document.

Everything else in this reference concerns identifying what is in the water. This part concerns whether the animals are currently in trouble, which is a separate question and a more urgent one. Behavior should be checked first, before appearance, before testing, and certainly before purchasing anything.

Warning behaviors

Behavior	What it may indicate	Urgency
Gasping at the surface	Low dissolved oxygen, or gill damage from ammonia or nitrite impairing oxygen uptake	Immediate. This is the highest-priority sign in the list.
Gathering at the filter outflow or an air stone	Seeking the most oxygenated water available	Immediate — functionally equivalent to gasping
Rapid or laboured gill movement	Respiratory distress: low oxygen, gill irritation, or nitrite affecting oxygen transport	High
Lethargy; resting on the substrate	Non-specific. Poor water quality, temperature problems, illness, or the normal behavior of some species	Moderate — interpret in context
Loss of balance or unusual swimming posture	Non-specific and potentially serious: neurological effects, swim bladder issues, severe water quality problems	High
Clamped fins	General stress indicator; common with poor water quality and with disease	Moderate to high
Hiding when normally visible	Stress; also normal after a disturbance or after adding new tankmates	Low to moderate — interpret in context
Refusing food	Non-specific stress or illness indicator; also normal for a day after introduction	Moderate
Flashing (rubbing against surfaces)	Irritation — from water chemistry or from external parasites	Moderate
Red or inflamed gills	Gill irritation, commonly associated with ammonia exposure	High

[COMMON AQUARIUM PRACTICE] Surface gasping and gathering at the outflow are widely recognized indicators of low dissolved oxygen or impaired oxygen uptake. They are among the most reliable behavioral signs available to a keeper because they are unambiguous and easy to recognize.

The essential point

These behaviors may indicate a problem requiring immediate attention even when the cloudiness itself is entirely harmless. A tank can have benign substrate dust and a simultaneous oxygen problem from a decaying animal behind the filter. The cloudiness and the distress are two observations, and only one of them is urgent.

What normal behavior tells you

Normal behavior is meaningful information and should be treated as such. Fish that are feeding readily, using the full water column, breathing at a normal rate and showing normal social behavior are not currently experiencing acute distress. That does not rule out a chronic problem, and it does not substitute for testing in a new or recently

disrupted tank. But it does substantially lower the urgency and creates room to diagnose carefully rather than react.

[REASONABLE INFERENCE] Normal livestock behavior is the single best justification for taking time to diagnose rather than intervening immediately. It converts an apparent emergency into a troubleshooting exercise, which is where better decisions get made.

WELFARE

Part XLIX — Livestock-Specific Considerations

Stated conservatively, because the evidence for species-specific sensitivity to cloudy water is thin.

The hobby generates a great deal of confident species-specific advice about cloudy water that does not survive scrutiny. This part states what can be defended and declines to state what cannot.

Group	Defensible considerations	Claims not supported by located evidence
Fish generally	Susceptible to low dissolved oxygen and to ammonia and nitrite toxicity, both of which can accompany cloudy-water conditions. Heavy suspended-solids loads affect gill function in fisheries contexts.	That cloudy water is inherently toxic to fish
Shrimp	Sensitive to sudden large changes in water chemistry, including copper and some medications. This makes aggressive intervention riskier than the cloudiness in many cases. Benefit from biofilm and microfauna, including the wood biofilm and microbial growth associated with new tanks.	That bacterial blooms are specifically dangerous to shrimp
Snails	No specific cloudy-water sensitivity identified. Graze biofilm and detritus.	That cloudy water harms snails
Fry	Small size and high metabolic rate make them relatively more vulnerable to oxygen depletion. Conversely, green water and the associated microfauna are a recognized and beneficial rearing environment.	That fry require crystal-clear water — the green-water rearing literature directly contradicts this
Labyrinth fish (bettas, gouramis)	Can breathe atmospheric air, providing partial insulation from dissolved-oxygen depression.	That they are immune to water-quality problems, or that their air-breathing makes low oxygen acceptable for the tank
Species from clear, fast-flowing habitats (e.g. hillstream loaches)	Higher oxygen requirements than average, making them more vulnerable to the oxygen-depletion pathway.	Specific sensitivity to turbidity itself
Blackwater species (many tetras, rasboras, apistogramma, licorice gouramis)	Adapted to tannin-stained, ion-poor, acidic water. Coloration is their natural condition rather than a problem.	That they require tannins to survive in an aquarium
Heavily stocked communities	Highest oxygen demand and therefore the least headroom during a bloom.	—

[ANECDOTAL OBSERVATION] Most species-specific claims about cloudy-water sensitivity in hobby discussion trace back to the general oxygen-depletion mechanism rather than to any species-specific finding. This document treats oxygen demand and nitrogenous toxicity as the operative pathways for all groups, with differences in vulnerability arising from oxygen requirements and body size rather than from anything specific to turbidity.

A note on stating uncertainty

For downstream content: it is more useful and more honest to say "the mechanism by which this could harm your fish is oxygen depletion, and here is how to check for it" than to assert that a particular species is especially vulnerable to cloudy water. The first statement is defensible and actionable. The second is neither.

CONTEXT

Part L — Cloudiness in Planted Aquariums

Where the largest number of causes overlap simultaneously.

A newly established planted aquarium is close to a worst case for diagnostic clarity, because a typical setup introduces five or six independent cloudiness mechanisms in the same afternoon.

The planted-tank interaction map

Element	Cloudiness contribution	Timing
Aquasoil or active substrate	Fine particulate; ammonium release; dissolved organics; humic coloration	Immediate for particulate; days to weeks for the rest
Fertilizers	Potential mineral precipitation if incompatible products are combined; nutrient availability for algae if plant uptake is low	Minutes for precipitation; days to weeks for algae
Root tabs	Localized nutrient release; particulate if the capsule is disturbed or breached	On insertion and if disturbed
Plant melt from emersed-grown stock	Decaying organic matter feeding heterotrophic bacteria	Days 3–14, which is exactly when keepers expect the setup bloom to be resolving
New planting and replanting	Substrate disturbance; released organics	Immediate, and again with every replanting session
CO2 injection	Microbubbles from fine diffusers; pH changes affecting carbonate solubility in hard water	During the injection photoperiod
Lighting	Drives green water when plant uptake cannot match available light	Days to weeks
Driftwood	Tannin coloration; surface biofilm; dissolved organics feeding blooms	Days to months
Decaying leaves and trimmings left in place	Ongoing organic input	Continuous if biomass is not exported

The biomass export point

Worth stating explicitly because it is frequently missing from planted-tank content: plants take up nutrients, but uptake is not removal. Nutrients taken into plant tissue return to the water when that tissue decays. A tank where plants grow vigorously and trimmings are left in place, or where old leaves melt and decompose in the substrate, is recycling nutrients rather than exporting them — and is adding organic material that can drive both bacterial blooms and algae. Pruning and physically removing plant material is the export step, and it is the step that completes the mechanism.

[ESTABLISHED SCIENTIFIC PRINCIPLE] Nutrient uptake into plant biomass removes nutrients from the water column only while that biomass remains intact. Decomposition returns them. Net export requires physical removal of the biomass.

Why new planted tanks are diagnostically difficult

Consider the realistic timeline of a new high-tech planted setup:

Day	Likely visible condition	Cause
0	Cloudy immediately on filling	Substrate fines
0–1	Additional milkiness after the fill	Dissolved gas
1–3	Clearing, then a yellow tint developing	Fines settling; humic release beginning
2–8	White haze developing	Heterotrophic bloom on released organics
3–14	Haze persisting or worsening	Plant melt adding organic matter
5–20	Brown film on glass and leaves	Diatoms
7–30	White film on the wood	Wood biofilm
14–60	Possible green tint	Green water if light exceeds plant uptake

A keeper experiencing this sequence reasonably concludes that nothing they try works, because each intervention addresses one phase while the next phase arrives. The genuinely useful message is that this sequence is largely normal, that most of it resolves as the system matures, and that the parts worth acting on are ammonium exposure to livestock, oxygen during heavy melt, and light management if green water appears.

[REASONABLE INFERENCE] The commonly reported experience that a new planted tank "stays cloudy for weeks despite everything" is best explained as a succession of different conditions rather than one persistent one. This reframing is likely to be more useful to keepers than any specific remedy.

WELFARE

Part LI — Fish-In versus Fishless Situations

The presence of animals changes the urgency, not the diagnosis.

The identification process is the same in both cases. What changes is how much time is available to complete it and what the cost of being wrong is.

Consideration	Fishless tank	Stocked tank
Urgency of diagnosis	Low — time is available	Proportional to livestock condition
Ammonia testing	Informative for tracking cycling progress	Essential and potentially urgent
Value of waiting	High — patience is genuinely the best tool	Conditional on the animals being unaffected
Cost of a wrong intervention	Low — the worst outcome is a delay	Potentially the loss of animals
Cost of no intervention	None	Potentially severe if the underlying condition is harmful
Aeration	Beneficial for the microbial community	Beneficial for the microbial community and potentially critical for the animals
Water changes	Optional; may slow cycling	A primary tool when nitrogenous compounds are present
Appropriate first action	Observe and identify	Check the animals, then identify

The framing that keeps welfare central

A bacterial bloom in an empty fishless setup is a microbiology observation. The identical bloom in a stocked tank where fish are gasping is an emergency. The water looks the same in both photographs. The difference is entirely in what is living in it and how those animals are doing — which is why livestock observation comes first in every framework in this document.

[REASONABLE INFERENCE] The presence of livestock should not change the diagnostic sequence, only its speed and the threshold for precautionary action such as aeration. Changing the diagnosis because animals are present leads to overreaction; ignoring their presence leads to underreaction.

DIAGNOSIS

Part LII — When More Than One Answer Is Correct

Multiple simultaneous causes are the normal case, not the exception.

Almost all cloudy-water advice implicitly assumes a single cause. Real aquariums frequently have two or three at once, often causally linked. Recognizing this prevents the most common troubleshooting failure: concluding that an intervention did not work when it worked on one of several problems.

Common combinations

Combination	How they link	What the keeper experiences
Aquasoil dust plus a bacterial bloom	The substrate releases both particulate and the dissolved organics that feed heterotrophs	Water clears partly after fine filtration, then goes white again a few days later. Feels like the problem is getting worse.
Driftwood tannins plus wood biofilm plus a bacterial bloom	One source — leaching organics — feeding three visible outcomes	Yellow water, white film on the wood, and haze in the column, all at once, all from the same piece of wood.
Disturbed substrate plus declining filter flow	Debris is released at the same time the system's capacity to capture it has quietly dropped	Post-maintenance cloudiness that never clears the way it used to.
Sunlight-driven green water plus overfeeding	Light provides the energy; feeding provides the nutrients	Reducing feeding helps a little; the bloom persists because light was the limiting factor.
Plant melt plus a bacterial bloom plus diatoms	Decaying plant tissue feeds bacteria; the immature system favors diatoms	The classic "my new planted tank has been a mess for three weeks."
Filter cartridge replacement plus overfeeding	Processing capacity removed at the same moment organic input rose	Rapid onset of a bloom and possible ammonia in an established tank.
Microbubbles plus substrate dust after a water change	Both are consequences of the same maintenance session	Milky water that partly clears within hours and then remains hazy.
Green water plus a bacterial bloom after an algaecide	The algaecide kills the algae; the dead algal biomass becomes dissolved organics feeding bacteria	Green water clears and is replaced by white water — which feels like a new problem and is in fact a consequence.
Crushed coral dust plus rising pH plus carbonate precipitation	The substrate contributes physical dust and dissolving carbonate simultaneously	Persistent chalky haze that returns after every disturbance.

[REASONABLE INFERENCE] Sequential and simultaneous causation is sufficiently common that any diagnostic framework should explicitly test for it. The practical marker is a partial response to a correctly matched intervention — which indicates the intervention worked and there is something else present.

How to work through a multi-cause situation

- 1 Identify the most likely primary cause and address it, changing one variable at a time.
- 2 Observe for two to three days before concluding anything, because different causes have different response timescales.
- 3 Interpret a partial improvement as evidence that you were partly right, not that the approach failed.
- 4 Re-run the identification steps on what remains. The residual condition may be a different category entirely from what you started with.

- 5 Look specifically for causal links: did addressing one thing create another? Algaecide creating organics, cleaning creating a bloom, a die-off creating oxygen demand.
- 6 Resist the urge to change several things at once out of frustration — that is precisely the point at which diagnosis becomes impossible.

REFERENCE MATRIX

Part LIII — Diagnosis by Timing

What becomes more or less plausible given when the cloudiness appeared.

This matrix is designed for retrieval. Given a timing signature, it lists what becomes more likely, what becomes less likely, and what to check.

Timing signature	More likely	Less likely	Check first
Cloudy immediately on filling a new tank	Substrate fines; hardscape residue; source-water turbidity; microbubbles	Bacterial bloom (no time has elapsed); green water; tannins at full intensity	Does it settle in a still glass? Does the tap water look the same?
Cloudy within an hour of a water change	Microbubbles; disturbed substrate	Bacterial bloom; algae	Does a glass sample clear from the top within the hour?
Cloudy 2–5 days after setup	Heterotrophic bacterial bloom; ongoing substrate release	Green water; detritus accumulation	Ammonia and nitrite if stocked; what organic material is in the tank
Cloudy after adding fish	Increased bioload exceeding current capacity; disturbed substrate from the addition; bag water introduced	Substrate dust; green water	Ammonia and nitrite; whether stocking was increased sharply
Cloudy after overfeeding or a suspected feeding event	Heterotrophic bloom; oxygen demand	Anything physical	Search for uneaten food; test ammonia; assess fish behavior
Cloudy after cleaning the filter	Released debris immediately; loss of biofilm if cleaning was aggressive	Green water; tannins	How thoroughly was it cleaned, and which media? Test ammonia.
Cloudy after replacing filter media or a cartridge	Loss of established biofilm capacity	Physical causes	Ammonia and nitrite — this is a genuine risk situation
Cloudy after adding driftwood	Tannin coloration; wood biofilm; bacterial response to leached organics	Substrate dust; precipitation	Is the water colored-but-clear, or genuinely hazy?
Cloudy after sunlight exposure changed	Green water	Bacterial bloom; particulate	White-cup test in daylight
Cloudy within minutes of fertilizing or dosing	Mineral precipitation	Bacterial; algal	Which products, in what order, into what water chemistry
Cloudy after moving rocks, plants or decorations	Released detritus and substrate fines	Bacterial (immediately); dissolved causes	Does it settle? Is fine media loading up?
Cloudy after a power outage	Debris ejected on filter restart; biofilm disruption; possible anaerobic material	Algae; precipitation	How long was flow stopped? Any odor on restart? Test ammonia.
Cloudy for weeks with no resolution	An ongoing organic source; inadequate filtration or flow; green water; sequential multiple causes	A simple one-off event	Re-run identification from the start — the current cause may not be the original one
Cloudy and worsening day over day	An ongoing and possibly escalating source — decaying material, continued overfeeding, a failing filter	A self-limiting setup phenomenon	Search thoroughly for a decaying source; check filter function; test

REFERENCE MATRIX

Part LIV — Diagnosis by Color

Eight appearance categories with follow-up questions and candidate interventions.

WHITE / MILKY

Likely possibilities	Heterotrophic bacterial bloom; microbubbles; fine substrate or mineral dust; mineral precipitation; filter media dust
Less-likely possibilities	Green water; tannins; large-particle detritus
Important follow-up questions	When exactly did it start, relative to any physical, chemical or biological event? Is the tank stocked, and how are the animals behaving? What are ammonia and nitrite? What was recently added, cleaned or replaced?
Useful observations	Standing-glass test to separate gas from particulate from microbial. White-cup test to exclude a color cast. Check whether fine media loads up within a day.
Potential interventions	Depends entirely on which of the four it is. Gas: nothing. Dust: fine filtration and siphoning. Precipitation: change the dosing practice. Bloom: address the organic source and maintain aeration.

GREEN

Likely possibilities	Suspended unicellular algae (green water)
Less-likely possibilities	Bacterial bloom; a green film on the glass being mistaken for green water
Important follow-up questions	Has light exposure changed — new fixture, longer photoperiod, sunlight, seasonal sun angle? Has plant mass changed? Has fertilizer dosing changed?
Useful observations	White-cup test in daylight is definitive. Wipe a patch of glass and look through it to exclude film algae.
Potential interventions	Eliminate direct sunlight; reduce photoperiod; increase plant competition; UV; blackout; diatom filtration for rapid clearing.

BROWN / TEA-COLORED, WATER STILL CLEAR

Likely possibilities	Dissolved tannins and humic substances from wood, leaves, botanicals or soil-based substrate
Less-likely possibilities	Suspended particulate; algae; bacterial
Important follow-up questions	What wood or botanicals are in the tank, and how recently were they added? Is the back glass still sharply visible? Are the animals behaving normally?
Useful observations	White-cup test shows amber but transparent. A light beam passes through with little scattering.
Potential interventions	None required. If unwanted: activated carbon, adsorbent resin, or ongoing water changes. Also consider whether it is desirable for the species kept.

BROWN AND HAZY, CLARITY REDUCED

Likely possibilities	Suspended detritus; fine soil or clay particulate; disturbed substrate
Less-likely possibilities	Dissolved tannins alone; microbubbles

Important follow-up questions	What was disturbed and when? Has filter flow changed? Is there a fine mechanical stage in the filter?
Useful observations	Standing-glass test shows settling. Fine media loads visibly. A light beam shows drifting particles.
Potential interventions	Fine mechanical filtration; siphoning during water changes; restore filter flow; redirect flow away from the substrate.

YELLOW

Likely possibilities	Dissolved organics at lower concentration than a full tea color — tannins, accumulated dissolved organic matter in an under-maintained tank, humic release from active substrate
Less-likely possibilities	Particulate causes; algae
Important follow-up questions	How long since the last water change? What is the wood and botanical inventory? Is the substrate an active soil-based product in its first weeks?
Useful observations	White-cup test shows a yellow cast with clarity preserved.
Potential interventions	Water changes; activated carbon; or acceptance. Persistent yellowing in an established tank with no wood suggests accumulated dissolved organics and a maintenance review.

GRAY

Likely possibilities	Fine substrate or mineral dust; a mild bacterial bloom; a low-density green-water bloom reading as grey; fine detritus
Less-likely possibilities	Tannins
Important follow-up questions	When did it start? Was anything disturbed? Has light exposure changed?
Useful observations	The white-cup test is particularly valuable here, because grey is the appearance most likely to be a color cast that the tank lighting is hiding.
Potential interventions	Identify the category first — grey is the least specific appearance and the most likely to be misdiagnosed.

CLEAR WITH VISIBLE PARTICLES

Likely possibilities	Detritus; plant fragments; uneaten food; sloughed biofilm; coarse substrate fines
Less-likely possibilities	Bacterial bloom; microbubbles; dissolved causes
Important follow-up questions	Is there a fine mechanical stage in the filter? Has flow declined? Are there dead zones with no circulation? Was anything disturbed?
Useful observations	The particles are individually resolvable and trackable, which distinguishes this from haze. Check whether they settle and where they accumulate.
Potential interventions	Add or improve fine mechanical filtration; restore flow; improve circulation; siphon accumulation zones.

SPARKLING / BUBBLY

Likely possibilities	Microbubbles from dissolved gas or from an equipment air leak
Less-likely possibilities	Everything else

Important follow-up questions	Was there a recent water change? Is any equipment new or recently reseated? Does the appearance change when the filter or air pump is switched off?
Useful observations	Bubbles rise and collect on glass and leaves. A glass sample clears from the top within the hour.
Potential interventions	Transient: nothing. Persistent: find the air leak, usually on the suction side of a canister filter.

CHALKY WHITE

Likely possibilities	Mineral precipitation — calcium carbonate, calcium phosphate, or similar
Less-likely possibilities	Bacterial bloom; algae; gas
Important follow-up questions	What was dosed, in what order, and how recently? What is the KH and GH? Was a pH-adjusting product used?
Useful observations	Onset is very fast and tied to a dosing event. Frequently reproducible — which is the strongest confirmation available.
Potential interventions	Change the dosing practice: premix, dose one product at a time, dose into high flow, avoid combining incompatible products.

REFERENCE MATRIX

Part LV — Diagnosis by Particle Behavior

What the suspended material does is more informative than what color it is.

Behavior over time separates causes that appearance cannot. These observations require nothing but patience and a glass.

Observation	If yes	If no	Diagnostic value
Does it settle overnight in a still glass?	Particulate with sufficient mass — substrate fines, detritus, precipitate	Very fine colloids, a microbial suspension, dissolved color, or gas	High — one of the best single tests available
Does the cloud clear from the top downward within an hour?	Microbubbles — essentially diagnostic	Not gas	Very high
Does it collect visibly in filter floss within a day?	Capturable particulate	Below the media capture threshold: bacteria, very fine colloids, or not particulate at all	High
Do bubbles stick to the glass and to plant leaves?	Gas is coming out of solution	Not gas	Very high
Can individual particles be resolved and tracked?	Discrete particulate — detritus, debris, biofilm fragments	The suspension is finer than the eye can resolve: bacteria, algae, clay, gas	Moderate to high
Does the appearance change when the filter or air pump is switched off?	Equipment is generating or sustaining the condition — bubbles or re-suspension	The condition is independent of equipment operation	High
Is the suspension visible as scattering in a flashlight beam?	Something is suspended	The problem may be dissolved coloration, glass films, or lighting	Moderate — confirms suspension but does not identify it
Does a sample in a white cup look green?	Suspended algae	Not algae	Very high
Is the water actually clear once removed from the tank?	The issue is on the glass, in the lighting, or in the viewing conditions — see Part LVII	The issue is genuinely in the water	Very high, and frequently overlooked
Do particles remain uniformly distributed rather than concentrating anywhere?	Neutrally buoyant or actively suspended: microbial or algal	Particulate that is being sorted by flow and gravity	Moderate
Does the cloudiness return within days of clearing it?	The cause is ongoing rather than a one-off event	The event was self-limiting	High for distinguishing an event from a condition

[COMMON AQUARIUM PRACTICE] These observations are used informally throughout the hobby and are mechanistically sound: they test settling velocity, buoyancy, capture size, gas behavior and optical properties, which are exactly the properties that differ between causes.

PRACTICAL METHOD

Part LVI — The White-Cup and Clear-Glass Tests

The highest-value zero-cost diagnostic in this document.

This part exists to give the technique a dedicated, retrievable home, because it resolves more diagnostic ambiguity per unit of effort than anything else available to a hobbyist.

Method

- 1 Use a clear, colorless glass — not tinted, not textured. A drinking glass or a jar is ideal.
- 2 Fill it with aquarium water from mid-column, away from the substrate and away from the filter outflow.
- 3 Fill a second identical glass with the source water the tank is filled from, dechlorinated the same way. This comparison is what makes the test conclusive rather than suggestive.
- 4 Carry both away from the aquarium and away from aquarium lighting.
- 5 View them side by side in natural daylight, against a sheet of white paper.
- 6 Then repeat the observation after an hour, and again the following morning, without moving or disturbing the glasses.

What each result indicates

Observation in the glass	Indicates
Green tint	Suspended algae
Amber or yellow tint, transparent	Dissolved tannins and humic substances
White or grey, uniform, no tint	Bacterial suspension, very fine dust, or gas — use the time observations to separate
Clears from the top within minutes to an hour	Microbubbles
Sediment collects at the bottom overnight; water above clears	Settleable particulate
Still uniformly cloudy the next morning	A microbial suspension or a very fine colloid
Visible discrete specks	Detritus or debris
Looks identical to the tap-water sample	The source water is the cause, or the perceived problem is not in the water at all
Looks completely clear while the tank looks cloudy	The issue is optical: glass films, lighting, viewing angle, or path length — see Part LVII

Limitations

- A very dilute suspension may not show a color cast in a small volume. Path length matters: a tall narrow glass viewed lengthwise shows more color than a shallow one.
- It does not identify which organisms are present, only which category.
- It does not distinguish bacteria from very fine mineral colloids without the settling observation, which requires patience.
- It says nothing about ammonia, nitrite, oxygen or any other chemical parameter.
- It is affected by the lighting used to view it — daylight is specified for a reason, and artificial indoor lighting can introduce its own color cast.

[USEFUL RULE OF THUMB] The white-cup test plus the overnight standing observation together separate the major categories — algal, dissolved-organic, particulate, gaseous and microbial — more reliably than any product, test kit or photograph. It should be the first step in almost every cloudy-water investigation.

PRACTICAL METHOD

Part LVII — Photography and Camera Misidentification

Why a tank can look far worse in a photograph than in the room — and occasionally far better.

A large share of cloudy-water troubleshooting now happens through photographs shared on social media. This introduces a category of error that did not exist when troubleshooting happened in front of the tank.

Factor	Effect on the image	How to compensate
Automatic white balance	Phone cameras aggressively correct color casts, which can neutralize a real green or amber tint — or, conversely, overcorrect neutral water into a color cast that is not there	Lock white balance manually, or include a known white reference such as a sheet of paper in the frame
Blue-heavy or full-spectrum aquarium LEDs	Shifts perceived color of both water and substrate; can make clear water look hazy and hazy water look worse	Photograph under room lighting with the aquarium light off, or photograph a water sample separately
Reflections on the glass	Room lights, windows and the photographer appear as haze layers over the image	Shoot at a slight angle, shield the glass, or shoot in a darkened room
Algae or hard-water film on the glass	Reads as cloudy water when the water is clear	Clean a patch of glass and photograph through it
Flash	Illuminates suspended particles dramatically, exaggerating turbidity; also creates a hotspot reflection	Avoid flash entirely for aquarium documentation
Path length and viewing angle	Shooting the long axis of a tank passes through far more water than shooting the short axis, exaggerating any tint	Photograph consistently from the same position when tracking change over time
Image compression and social-media processing	Reduces subtle tonal information; platforms apply their own processing	Share the least-processed version available
Background color	A black background makes a suspension far more visible; a white background makes color casts more visible	Note which background is in use when interpreting someone else's photograph
Screen calibration on the viewer's end	The person interpreting the photograph sees it on a different, uncalibrated display	Rely on described behavior and timing rather than on color judgment from an image

[REASONABLE INFERENCE] Photographic evaluation of aquarium water clarity is unreliable enough that behavioral and timing information should be weighted far more heavily than the appearance of an image when troubleshooting remotely.

The most useful question to ask about a photograph

Not "what does this look like" but "what did you do in the last three days, what does a glass of this water look like against white paper in daylight, and how are the fish behaving?" Those three answers outweigh any photograph.

APPLIED REASONING

Part LVIII — Scenario Library

Thirty situations analysed with the same structure, for retrieval and for reuse in answering real questions.

Each scenario below is analysed in the same seven-field format so that the reasoning pattern is visible and transferable, not just the conclusion. The fields separate what is actually known from what is being assumed, which is the discipline that keeps cloudy-water advice honest. Several scenarios deliberately end without a single answer, because a single answer would not be defensible from the information given.

How to read these

The most important field in most scenarios is **Important follow-up questions**. In a genuine troubleshooting exchange, those questions do more work than any list of possible causes, because they are what convert an unanswerable question into an answerable one.

Scenario A — “My brand-new tank went cloudy on day three.”

What we know	The tank is new. Onset is at day three, which is inside the classic heterotrophic bloom window. Something in the tank is releasing organic material and a bacterial population is responding to it.
What we do not know	Whether the tank is stocked. Whether it was filled with dechlorinated water. What the substrate is. Whether anything was added to seed the cycle. What the cloudiness actually looks like beyond the word “cloudy.”
Important follow-up questions	Is anything living in the tank right now? What does a glass of the water look like against white paper in daylight? What is the substrate? Have you added ammonia, food, or a bottled bacteria product?
Most plausible causes	Heterotrophic bacterial bloom is the leading candidate. Continued substrate release is plausible if the substrate is soil-based or was poorly rinsed. Microbubbles are unlikely at day three unless there is an ongoing air leak.
What tests matter	If unstocked: none are urgent. If stocked: ammonia and nitrite.
Immediate safety concerns	None if the tank is empty. If stocked, oxygen availability is the concern rather than the cloudiness itself.
What not to do prematurely	Do not replace filter media. Do not scrub the tank. Do not add a clarifier before identifying the cause. Do not conclude the cycle has failed.
Potential response	This is the most common and least alarming version of the question. In an unstocked tank, the accurate answer is that a bacterial bloom at day three is close to expected and that the correct action is aeration and patience. The information that changes the answer is whether animals are present.

Scenario B — “My established tank suddenly went cloudy after a water change.”

What we know	The tank was previously clear and stable. The onset is tied to a maintenance event, which makes the maintenance session the primary suspect rather than a coincidence.
What we do not know	How soon after the change it appeared. What the water change involved. Whether the filter was touched. Whether the substrate was vacuumed. What the appearance is.
Important follow-up questions	How long after the water change did it appear — minutes, hours, or a day? Did you clean the filter in the same session? Did you vacuum the substrate? How much water was changed? Did the refill go in with force?

Most plausible causes	Within minutes to an hour: microbubbles from cold pressurized tap water, or disturbed substrate. Within a day: released detritus, or a bloom following filter disturbance. Within two to four days: a bloom triggered by loss of biofilm if media was cleaned aggressively.
What tests matter	Ammonia and nitrite, specifically if the filter was cleaned or media replaced in the same session.
Immediate safety concerns	Low, unless the filter was disrupted. Then the concern is nitrification capacity rather than the appearance.
What not to do prematurely	Do not do another large water change immediately on the assumption that more is better. Do not clean the filter again to fix it.
Potential response	The order of operations question does most of the work here. Separating substrate vacuuming, filter maintenance and the water change into different sessions prevents this situation from being ambiguous in the future.

Scenario C — “My water is green and I can barely see the back of the tank.”

What we know	Green plus reduced visibility is close to diagnostic for suspended algae. Green water requires light and available nutrients, and it has been running long enough to reach high density.
What we do not know	What changed to trigger it. Whether the tank is planted. Whether there is direct sunlight. Whether the fish are affected.
Important follow-up questions	Does a glass of the water look green against white paper in daylight? Has the lighting or photoperiod changed recently? Does the tank get any direct or indirect sunlight, including seasonally? How heavily is it planted, and are the plants growing? Have the fish shown any change in behavior, especially early in the morning?
Most plausible causes	Suspended unicellular algae. The trigger is almost always a change in the light-to-uptake balance: a new fixture, a longer photoperiod, a window, a season, plant die-back, or a nutrient increase.
What tests matter	Nitrate and phosphate are informative for the nutrient side. Testing pH in the early morning and again in the evening can reveal the diel swing.
Immediate safety concerns	Dawn oxygen minimum in a dense bloom, and a potential crash if the bloom dies suddenly. Reduced visibility also means fish problems go unnoticed.
What not to do prematurely	Do not add an algaecide as a first move, because a mass die-off in a dense bloom converts an algae problem into an oxygen and organics problem. Do not do a huge water change expecting it to clear — the population regrows.
Potential response	Address light first, because light is usually the trigger and is the variable most under the keeper's control. Then nutrients and plant competition. UV is a legitimate tool here and works well on suspended algae, but it manages the symptom while the conditions remain.

Scenario D — “My water turned brown after I added driftwood.”

What we know	The onset is tied to a specific addition with a well-understood mechanism. Wood releases tannins and other humic substances into water.
What we do not know	Whether the water is colored-but-clear or genuinely turbid. Whether the keeper considers this a problem or is simply asking what it is.
Important follow-up questions	Can you still see the back glass sharply, or is the water hazy as well as brown? Was the wood boiled or soaked first? What species are you keeping? Is this bothering you aesthetically, or are you worried it is harmful?
Most plausible causes	Dissolved tannins and humic substances, which are colored but not turbid. If it is also hazy, add the possibility of a bacterial response to the leached organics and surface biofilm on the wood itself.
What tests matter	pH and KH, mainly to understand whether the buffering reserve is being consumed in a soft-water tank. This is informative rather than urgent.

Immediate safety concerns	Minimal in most cases. In very soft, poorly buffered water, heavy botanical loading can reduce pH more than expected.
What not to do prematurely	Do not treat it as an emergency. Do not add pH-raising products to counteract the color.
Potential response	The first thing to establish is whether this needs treating at all. For many species this water is closer to their natural condition than clear water is. If the keeper simply dislikes the look, carbon and water changes are effective and there is no reason to talk them out of it.

Scenario E — “My tank has been cloudy for three weeks and nothing works.”

What we know	The duration rules out a self-limiting one-off event. Either something is continuously sourcing the problem, or the cause has changed at least once during the three weeks, or the interventions being tried are mismatched to the actual cause.
What we do not know	Nearly everything. This scenario is characterised by the keeper having tried multiple things, which makes the history hard to reconstruct.
Important follow-up questions	What have you tried, in what order, and what happened after each? Has the appearance changed at all over the three weeks — color, texture, density? How old is the tank? What does a glass of it look like in daylight? What are your ammonia and nitrite readings? How much and how often do you feed?
Most plausible causes	Sequential multiple causes is the single most likely explanation for a new tank. For an established tank: an ongoing organic source such as a dead animal or persistent overfeeding, declining filter flow, inadequate mechanical filtration for the particle size present, or green water that has not been recognised as green water.
What tests matter	Ammonia, nitrite and nitrate. Nitrate is informative for maintenance history.
Immediate safety concerns	Depends on livestock condition, which should be established first.
What not to do prematurely	Do not add another product. Do not change several variables at once. The accumulated interventions are part of why this is hard to diagnose.
Potential response	The most useful response is to reset the diagnosis rather than suggest another remedy. Stop intervening, re-run identification from the start, and treat the current condition as a new question rather than a continuation of the original one.

Scenario F — “My water goes cloudy every time I clean the substrate.”

What we know	The cause and the trigger are both known. Vacuuming lifts fine material into suspension. The real question is why it does not clear promptly afterwards.
What we do not know	Whether it clears within hours or persists. What the substrate is. What mechanical filtration is in use.
Important follow-up questions	How long does it take to clear? Is there a fine mechanical stage in the filter, such as floss or a fine pad? Has the filter flow declined recently? What is the substrate — sand, gravel, soil?
Most plausible causes	Normal re-suspension of fines during vacuuming. Persistence beyond a few hours suggests either inadequate fine mechanical filtration or reduced flow.
What tests matter	None specifically.
Immediate safety concerns	None.
What not to do prematurely	Do not stop vacuuming the substrate to avoid the cloudiness — that trades a cosmetic issue for an organic accumulation issue.
Potential response	This is a filtration and expectation question rather than a water quality problem. Adding a fine polishing stage and vacuuming in sections rather than all at once addresses it. Some short-term cloudiness after vacuuming is normal and does not need eliminating.

Scenario G — “My fish are gasping and the water is cloudy.”

What we know	There are two observations here, and the gasping is the one that matters. Gasping indicates a respiratory problem, which may or may not share a cause with the cloudiness.
What we do not know	Everything about the cause. But the immediate response does not depend on knowing it.
Important follow-up questions	These come after the first actions, not before: What are ammonia and nitrite? What is the temperature? Is there surface movement? What changed in the last three days? Is anything possibly dead in the tank?
Most plausible causes	For the gasping: low dissolved oxygen, ammonia or nitrite toxicity, high temperature, or gill damage. A dense bacterial bloom or a green-water crash can drive oxygen down. Ammonia can accompany a bloom that followed filter disruption.
What tests matter	Ammonia, nitrite and temperature, urgently.
Immediate safety concerns	This is an emergency. Gasping fish are the highest-priority observation in this document.
What not to do prematurely	Do not spend time identifying the cloudiness before acting on the gasping. Do not add any product that promises to fix it.
Potential response	Act first, diagnose second. Increase surface agitation and aeration immediately. Check temperature. Perform a conservative, temperature-matched, dechlorinated water change. Search for anything decaying. Test. Only then consider what the cloudiness is.

Scenario H — “Is my tank cycled if the water is cloudy?”

What we know	The keeper is treating clarity as a proxy for cycling status. It is not one.
What we do not know	The actual cycling status, which only testing can establish.
Important follow-up questions	What are your ammonia, nitrite and nitrate readings? How long has the tank been running? Is it stocked? How did you cycle it?
Most plausible causes	The two are largely independent. A fully cycled tank can be cloudy from dust, and an uncycled tank can be perfectly clear.
What tests matter	Ammonia, nitrite and nitrate. This question cannot be answered without them.
Immediate safety concerns	Depends on the readings and on whether animals are present.
What not to do prematurely	Do not add fish based on the water looking clear. Do not conclude the cycle failed because the water is cloudy.
Potential response	The direct answer is that cloudiness tells you almost nothing about cycling status, and that clear water is one of the most misleading signals in the hobby. The nitrifying bacteria that constitute a cycle live on surfaces and are invisible.

Scenario I — “I added a bacteria starter and the tank went cloudy.”

What we know	A product was added and cloudiness followed. Two distinct mechanisms could explain it.
What we do not know	Which product, and whether ammonia was also being dosed.
Important follow-up questions	Which product, and how much relative to the label dose? Are you also dosing ammonia or feeding the tank? How soon after dosing did the cloudiness appear? Is the tank stocked?
Most plausible causes	The product itself is a dense suspension and can cloud the water directly for a short period. Separately, the organic carrier and any ammonia source can trigger a heterotrophic bloom over the following days.
What tests matter	Ammonia and nitrite if stocked.
Immediate safety concerns	Low if unstocked.

What not to do prematurely	Do not repeatedly re-dose in the hope of clearing it. Do not conclude the product failed.
Potential response	Distinguish immediate turbidity from the product itself, which clears within a day or two, from a bloom developing over several days, which is a separate and normal response. Neither means the product failed, and neither means it worked.

Scenario J — “Will a clarifier hurt my fish?”

What we know	The keeper is considering a product and asking a safety question, which is the right instinct.
What we do not know	What the cloudiness is, which determines whether the product can work at all. The specific product's formulation.
Important follow-up questions	What is causing the cloudiness? What is your filtration? Have you identified the cause yet, or is this the first thing you are trying?
What not to do prematurely	Do not use a clarifier before identifying the cause. Do not use one without adequate mechanical filtration to capture what it clumps.
Potential response	The honest answer has three parts. First, clarifiers work by clumping suspended particles so filtration can capture them, which means they address particulate cloudiness and do nothing about dissolved coloration and little that lasts about a bacterial bloom driven by an ongoing organic source. Second, no specific evidence of gill harm in aquarium use was located during this research, and the products are widely used without reported problems; that is an absence of located evidence rather than a demonstration of safety. Third, the meaningful risk is not toxicity but misapplication — clearing the appearance while the cause continues, and doing so without the mechanical filtration needed to remove the clumped material.

Scenario K — “My shrimp tank is cloudy — should I be worried?”

What we know	The tank contains animals that are sensitive to sudden chemistry changes and to certain additives, which raises the cost of aggressive intervention above the cost of the cloudiness in most cases.
What we do not know	What the cloudiness is. How old the tank is. What the shrimp are doing.
Important follow-up questions	How are the shrimp behaving — grazing normally, moving normally? How old is the tank? What substrate? Have you added or dosed anything recently? What are ammonia and nitrite?
Most plausible causes	In a new shrimp tank with active substrate: bacterial bloom plus substrate fines, both expected. In an established tank: an organic source, or a response to something recently added.
What tests matter	Ammonia, nitrite, and if the tank is new, an awareness that active substrates leach ammonium.
Immediate safety concerns	Copper-containing products and some medications are the genuine hazards here, not the cloudiness.
What not to do prematurely	Do not add algaecides, medications or clarifiers reflexively. Do not make large sudden changes to water chemistry.
Potential response	For shrimp specifically, the risk calculus favours patience more strongly than it does for most fish. A bacterial bloom in a new shrimp tank is normal, and the associated biofilm is food. The failure mode to warn against is treating the cloudiness with something that harms the shrimp.

Scenario L — “My water is cloudy but all my parameters are perfect.”

What we know	Ammonia, nitrite and nitrate are presumably acceptable. That rules out some causes and leaves most of them untouched.
What we do not know	What “perfect” means and which tests were run. What the cloudiness looks like.

Important follow-up questions	Which parameters did you test, and what were the actual numbers? What does a glass of the water look like in daylight? When did it start and what changed beforehand? How are the fish behaving?
Most plausible causes	Physical particulate, microbubbles, mineral precipitation, dissolved coloration, and green water are all fully compatible with good nitrogen readings. A mature bacterial bloom in a cycled tank can also coexist with zero ammonia, because the nitrifying population is handling it.
What tests matter	The tests already run are informative. Dissolved oxygen matters and is rarely tested at home; surface agitation is the practical proxy.
Immediate safety concerns	Low, given normal fish behavior.
What not to do prematurely	Do not assume the test kit is faulty. Do not escalate treatment because the numbers do not explain the appearance.
Potential response	This is an important teaching scenario. Standard test kits measure nitrogen compounds and pH. They do not measure turbidity, dissolved organics, suspended algae, gas saturation or particle load. Good readings and cloudy water is a normal and common combination, not a contradiction.

Scenario M — “Should I do a water change if the water is cloudy?”

What we know	The keeper is asking about the most-recommended and most-argued-about intervention in the hobby.
What we do not know	The cause, which is what determines the answer.
Important follow-up questions	Is the tank stocked? What is the cause of the cloudiness, if known? What are ammonia and nitrite? Are the fish behaving normally?
What tests matter	Ammonia and nitrite.
What not to do prematurely	Do not apply a blanket rule in either direction. Both “always change water” and “never change water during a bloom” are oversimplifications.
Potential response	The answer separates cleanly by situation. Stocked tank with any ammonia or nitrite, or with fish showing distress: yes, and promptly. Fishless cycling tank with a bloom: water changes are optional and may modestly slow the process, and doing nothing is reasonable. Particulate cloudiness: water changes help mechanically, especially combined with siphoning. Dissolved coloration: water changes dilute it and are one of the effective options. Green water: water changes do not resolve it, because the population regrows. The consistent principle is that a properly conducted water change — temperature matched, dechlorinated, not disturbing the biofilter — is very rarely harmful, and the argument against it applies to a narrower set of circumstances than it is usually given.

Scenario N — “My tank cleared up on its own — why?”

What we know	The problem resolved without intervention, which is genuinely informative about what it was.
What we do not know	—
Important follow-up questions	How long did it take? What did it look like? Did anything change in the meantime, including things you might not count as interventions?
Most plausible causes	Self-resolution over days to a few weeks is consistent with a heterotrophic bloom exhausting its substrate, with fines settling and being captured, with microbubbles outgassing, or with the system maturing past a setup phase.
What tests matter	None.
Immediate safety concerns	None.
Potential response	This question deserves a real answer rather than a shrug, because understanding why it cleared tells the keeper what it was and what to expect next time. Self-limitation is the signature of a bloom or a settling event, and it is a useful diagnostic in retrospect.

Scenario O — “My filter is running but the water stays cloudy.”

What we know	There is a filter, and it is not resolving the cloudiness. That is compatible with the filter working correctly on something it cannot capture.
What we do not know	What media is in the filter. Whether the flow has changed. What the cloudiness is.
Important follow-up questions	What media is in the filter, in what order? When was it last cleaned or replaced? Has the flow rate visibly declined? Does fine floss load up with material within a day?
Most plausible causes	The particles may be smaller than the media can capture, which is the case for bacteria and for suspended algae. There may be no fine mechanical stage at all. Flow may have declined below what is needed. Or the cause may not be particulate.
What tests matter	None specifically.
Immediate safety concerns	Low.
What not to do prematurely	Do not replace all the media in response. Do not assume the filter is broken.
Potential response	A filter running normally will not clear a bacterial bloom or green water, because the organisms pass through standard media. That is not a filter failure. If the cloudiness is particulate and the filter is not clearing it, the questions are media grade and flow.

Scenario P — “I can smell something and the water is cloudy.”

What we know	There is an odor and there is cloudiness, and an odor is a genuinely useful signal that hobbyists underuse.
What we do not know	What the odor is like, which is difficult to communicate in text but worth attempting.
Important follow-up questions	How would you describe the smell — rotten, sulphurous, musty, sweet, ammonia-like? Is anything possibly dead in the tank? When did you last feed, and how much? Have you disturbed the substrate recently? How are the fish?
Most plausible causes	A strongly foul or rotten odor with cloudiness points toward decaying organic matter, which is both a cloudiness source and an oxygen sink. A sulphurous smell after disturbing deep substrate suggests an anaerobic pocket. A musty or earthy smell is often associated with cyanobacteria.
What tests matter	Ammonia and nitrite.
Immediate safety concerns	Moderate to high. A significant decaying source is one of the few genuinely urgent cloudy-water situations, because of oxygen demand and ammonia production.
What not to do prematurely	Do not mask the smell with carbon before finding the source. Do not stir deep substrate further before improving aeration.
Potential response	This is one of the scenarios where the accompanying symptom outranks the cloudiness. The priority is finding and removing the source, and ensuring aeration while doing it.

Scenario Q — “The water is cloudy after I moved my tank.”

What we know	A major physical disruption occurred. Substrate was disturbed, the filter was stopped, and biofilm was exposed to air.
What we do not know	How long the filter was off. Whether the substrate was kept wet. Whether media was kept wet.
Important follow-up questions	How long was the filter off, and was the media kept submerged in tank water? How much of the original water was retained? Was the substrate kept wet or allowed to dry? Are the fish showing any signs of distress?

Most plausible causes	Massive re-suspension of fines and detritus, which is expected and clears. Separately and more importantly: potential loss of nitrifying capacity if the media dried out or was off for a long period, and release of anaerobic material from deep substrate.
What tests matter	Ammonia and nitrite, over the following two weeks. This situation carries a real risk of a partial recycle.
Immediate safety concerns	Moderate. Not from the cloudiness, but from the possible loss of biofiltration.
What not to do prematurely	Do not clean the filter. Do not replace media. Do not add livestock back faster than testing supports.
Potential response	Treat the cloudiness as expected and largely irrelevant, and treat the biofilter as the real question. Test daily for the first week or two and be ready to respond to ammonia.

Scenario R — “My water is cloudy and my nitrites are high.”

What we know	Nitrite is present, which means nitrification is incomplete. Both observations likely share a cause: the system's processing capacity is not matching its load.
What we do not know	Why capacity is short — new tank, disrupted filter, or increased bioload.
Important follow-up questions	How old is the tank? Was the filter recently cleaned or had media replaced? Was stocking recently increased? What is the ammonia reading? How are the fish behaving?
Most plausible causes	A tank in the middle of cycling, or an established tank whose nitrification was disrupted or overloaded. The bacterial bloom and the nitrite are both downstream of the same situation.
What tests matter	Ammonia, nitrite, and if available, chloride relative to nitrite for context on the salt-based mitigation.
Immediate safety concerns	Yes. Nitrite is directly toxic to fish, interfering with oxygen transport. This is a genuine welfare situation independent of the cloudiness.
What not to do prematurely	Do not clean the filter. Do not stop feeding entirely for extended periods if the tank is stocked, but do reduce input. Do not focus on the cloudiness.
Potential response	The nitrite is the story. Water changes to dilute it, increased aeration because nitrite impairs oxygen transport, reduced feeding, and no disruption of the biofilter. The cloudiness will resolve as the system catches up.

Scenario S — “Is cloudy water dangerous for my fish?”

What we know	A general question with no specific tank attached.
What we do not know	Everything about the specific situation.
Important follow-up questions	What does the cloudiness look like? What are the fish doing? How old is the tank? What are ammonia and nitrite?
Potential response	The accurate general answer is that cloudy water is usually a symptom rather than a hazard in itself, and that the danger, when there is one, comes through specific pathways: oxygen depletion from high biological demand, ammonia or nitrite from insufficient nitrification, and the underlying condition that caused both. The practical translation is that the fish's behavior tells you more about danger than the water's appearance does.

Scenario T — “My planted tank went cloudy after I added fertilizer.”

What we know	Onset is tied to a dosing event, and the timing is the strongest clue available.
What we do not know	Which fertilizers, in what order, and how quickly the cloudiness appeared.

Important follow-up questions	How soon after dosing — within minutes, or over days? Which products, and were any dosed together? What is your KH and GH? Did you dose into still water or into flow?
Most plausible causes	Within minutes: mineral precipitation, particularly if calcium- or magnesium-containing products were combined with phosphate-containing ones in hard water. Over days: increased nutrient availability supporting a bloom or an algal response, especially if plant uptake is low.
What tests matter	KH and GH for the precipitation pathway. Nitrate and phosphate for the nutrient pathway.
Immediate safety concerns	Low.
What not to do prematurely	Do not increase dosing to compensate. Do not add a clarifier for a precipitate that will recur with the next dose.
Potential response	The immediate-onset version has a clean and satisfying answer: change the dosing practice rather than treat the water. Dose products separately, premix into a larger volume, and dose into high flow. The reproducibility of the effect is the confirmation.

Scenario U — “My tank is cloudy and there is white film on the driftwood.”

What we know	Two observations from one source. Wood is releasing organics, which is feeding both a surface biofilm and, plausibly, a suspended bacterial population.
What we do not know	How new the wood is. Whether the fish are affected.
Important follow-up questions	How recently was the wood added? Is the water hazy, colored, or both? Do you have shrimp, snails, otocinclus or other grazers? Are the fish behaving normally?
Most plausible causes	A biofilm of bacteria and fungi on the wood surface, which is a normal and temporary stage for new wood, plus a bacterial response in the water column to the same leached organics.
What tests matter	Ammonia if the tank is newly stocked.
Immediate safety concerns	Low.
What not to do prematurely	Do not remove the wood. Do not scrub it repeatedly, which removes the visible film without addressing the leaching.
Potential response	This is one of the more reassuring scenarios. The white film is a well-known new-wood stage that resolves over weeks as the readily available organics are consumed. Many invertebrates and some fish graze it. Physical removal is optional and cosmetic.

Scenario V — “Why did my water go cloudy after I cleaned the filter?”

What we know	A maintenance action preceded the cloudiness, and there are two very different possible mechanisms with different timescales.
What we do not know	How the filter was cleaned and with what.
Important follow-up questions	How soon after cleaning did it appear? What exactly did you do — rinse the media in tank water, rinse it under the tap, or replace it? Which media? Is the tank stocked, and what are ammonia and nitrite?
Most plausible causes	Within minutes to hours: physically dislodged debris returned to the tank, which is harmless and clears. Within two to five days: a bacterial bloom following a reduction in biofilm, particularly if media was rinsed under chlorinated tap water or replaced.
What tests matter	Ammonia and nitrite. This scenario has one of the highest justifications for testing.
Immediate safety concerns	Moderate if media was replaced or rinsed in tap water, because nitrification capacity may have dropped.
What not to do prematurely	Do not clean the filter again. Do not replace the remaining media.

Potential response	The timing separates a cosmetic event from a functional one. Immediate cloudiness is debris. Delayed cloudiness with any ammonia is a capacity problem, and the response is reduced feeding, aeration, water changes as needed, and leaving the filter alone.
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Scenario W — “My tank has been cloudy since day one and it is now week three.”

What we know	Cloudiness present from filling and continuing for three weeks. Something is ongoing rather than resolving.
What we do not know	The substrate, the appearance, and whether it has changed character at any point.
Important follow-up questions	What is the substrate, and how was it prepared? Has the appearance changed at all — color, density, texture? Is the tank stocked? What filtration is running, and is there a fine mechanical stage? What does a glass of it look like in daylight?
Most plausible causes	Very fine substrate fines that remain suspended because nothing is capturing them, particularly with certain sands and clay-based products. A soil-based substrate still in its leaching phase. Or a sequence of causes handing off to one another, which is the usual answer for a new planted tank.
What tests matter	Ammonia and nitrite.
Immediate safety concerns	Depends on stocking.
What not to do prematurely	Do not tear the tank down. Do not add a clarifier without fine mechanical filtration.
Potential response	The productive move is to establish whether this is one persistent cause or a sequence. The white-cup and standing-glass tests do that quickly. If it is very fine mineral particulate, fine polishing media and patience resolve it; if it is a sequence, the answer is that the tank is behaving normally and is not yet finished maturing.

Scenario X — “Can I add fish while the water is cloudy?”

What we know	The keeper is planning an addition and correctly checking first.
What we do not know	Cycling status, which is the actual determinant.
Important follow-up questions	What are your ammonia, nitrite and nitrate readings right now? How long has the tank been running and how was it cycled? What is causing the cloudiness?
What tests matter	Ammonia, nitrite and nitrate, without exception.
Immediate safety concerns	Adding fish into an uncycled or under-capacity system is the concern, and it is a welfare concern rather than a water-clarity one.
What not to do prematurely	Do not add fish based on appearance in either direction — not because it looks clear, and not delaying indefinitely because it looks cloudy.
Potential response	The answer is determined by test results rather than by clarity. If ammonia and nitrite read zero with nitrate present and the tank has been processing a load, cloudiness from a physical cause is not a reason to wait. If the readings are not there, the cloudiness is irrelevant to the decision, because the answer is no regardless.

Scenario Y — “My water looks fine but my fish are acting strange.”

What we know	The visual signal and the behavioral signal disagree, and the behavioral signal is the reliable one.
What we do not know	Everything about the actual cause, which is not visible.
Important follow-up questions	What specifically are they doing? What are ammonia, nitrite and pH? What is the temperature? What changed in the last week, including anything added to the room? Is there surface movement?

Most plausible causes	Clear water is compatible with ammonia, nitrite, low oxygen, temperature problems, chlorine or chloramine from an untreated water change, copper from plumbing, disease, aerosol contamination from household products, and pH problems. None of these are visible.
What tests matter	Ammonia, nitrite, pH and temperature at minimum.
Immediate safety concerns	Potentially high, depending on the behavior.
What not to do prematurely	Do not conclude the water is fine because it looks fine.
Potential response	This scenario is the mirror image of most of this document and deserves emphasis for exactly that reason. Clear water is not evidence of good water quality. It is evidence of low turbidity and nothing else. Behavior is the better instrument.

Scenario Z — “I have tried everything and nothing works.”

What we know	The keeper is frustrated, has intervened repeatedly, and the accumulated interventions have made the situation harder to read.
What we do not know	The original cause, the sequence of interventions, and what the current condition actually is.
Important follow-up questions	What have you tried, in what order, and what happened after each one? What does the water look like right now, as opposed to when it started? Has anything improved even slightly? What are your test readings? How are the animals?
Most plausible causes	Serial interventions have often introduced new causes: an algacide creating organics, media replacement reducing capacity, repeated cleaning re-suspending material, clarifiers used without adequate filtration.
What tests matter	Ammonia, nitrite, nitrate.
Immediate safety concerns	Determined by livestock condition.
What not to do prematurely	Do not suggest another product. That is what has produced this situation.
Potential response	The most valuable response here is to stop rather than to add. Change nothing for a week except aeration and normal feeding, observe what the condition does on its own, and then diagnose the current state rather than the original complaint. A partial improvement after any intervention is information, and it has usually been discarded in this scenario because the improvement was not complete.

Scenario AA — “My water went cloudy overnight with no warning.”

What we know	Rapid onset with no recognised trigger. “No warning” usually means the trigger was not recognised rather than that there was none.
What we do not know	The trigger, which is what the questioning has to surface.
Important follow-up questions	Think back three days rather than one: did anything get added, moved, cleaned, dosed, or fed differently? Did anyone else in the household feed the tank? Was there a power interruption? Is every animal accounted for?
Most plausible causes	An unaccounted-for dead animal is a leading candidate for genuinely sudden onset in an established tank. Others: someone else feeding, a power outage restarting the filter and ejecting debris, a heater failure changing temperature, or a chemical introduction.
What tests matter	Ammonia and nitrite.
Immediate safety concerns	Moderate, because sudden onset in a previously stable established tank is more likely to have a live cause than a gradual change is.
What not to do prematurely	Do not accept “nothing changed” at face value, including from yourself.

Potential response	The counting-the-animals question is the highest-yield one in this scenario and is frequently the answer.
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Scenario BB — “My water turns cloudy every afternoon and clears by morning.”

What we know	The condition is cyclical and tied to a daily cycle, which narrows the possibilities sharply.
What we do not know	What is running on a timer.
Important follow-up questions	What is on a timer — lights, CO2, a dosing pump, an air pump? Does the cloudiness coincide with the photoperiod? Is it a haze or fine bubbles? Do bubbles collect on the glass?
Most plausible causes	Microbubbles from CO2 injection during the photoperiod is the most common explanation. A dosing pump producing a daily precipitate is a second. A daily oxygen and pH cycle in a green-water tank can also change appearance across the day.
What tests matter	pH morning and evening, which will reveal a CO2 or photosynthetic cycle.
Immediate safety concerns	Low, unless CO2 injection is high enough to affect the fish, which is a separate concern worth checking.
What not to do prematurely	Do not treat this as a water quality problem.
Potential response	The daily rhythm is the diagnosis. Matching the timing to what is on a timer resolves it almost every time.

Scenario CC — “The water is clear but there is a white haze on the glass and decorations.”

What we know	The problem is on surfaces, not in the water column, which puts it outside most of this document.
What we do not know	Whether the deposit wipes off.
Important follow-up questions	Does it wipe off easily, or is it hard and gritty? Is it above or below the waterline? What is your KH and GH? Do you top up with tap water or with RO?
Most plausible causes	Below the waterline and slimy: bacterial or algal biofilm. At and above the waterline and hard: mineral scale from evaporation and topping up with mineralised water. Hard deposit generally: calcium carbonate.
What tests matter	KH and GH.
Immediate safety concerns	None.
What not to do prematurely	Do not treat the water for a surface deposit.
Potential response	This is included specifically because it is frequently reported as “cloudy water” and is not water at all. Establishing whether the water in a glass is clear separates the two immediately.

Scenario DD — “The cloudiness came back after I fixed it.”

What we know	An intervention worked and the condition returned, which means the cause is ongoing rather than resolved.
What we do not know	What was done, and how long the clear period lasted.
Important follow-up questions	What did you do, and how long did it stay clear? Was the clearing gradual or immediate? Does it return on a schedule or after a particular activity?
Most plausible causes	The intervention addressed the symptom rather than the source — mechanical removal of particulate that is still being generated, a clarifier used against a bloom with an ongoing food supply, or UV switched off after clearing green water while the light and nutrient conditions remain.

What tests matter	Depends on the underlying cause.
Immediate safety concerns	Low.
What not to do prematurely	Do not simply repeat the same intervention indefinitely, though doing so is reasonable as a stopgap while addressing the source.
Potential response	Recurrence is diagnostically valuable: it establishes that the cause is continuous and shifts the investigation from “what is in the water” to “what keeps putting it there.”

[REASONABLE INFERENCE] Across these thirty scenarios, three questions recur as the highest-value openers regardless of the specific complaint: what changed in the last three days, what does a glass of the water look like against white paper in daylight, and how are the animals behaving. Any troubleshooting framework built for this topic should lead with those three.

TRIAGE

Part LIX — What Should I Do Right Now?

An ordered framework, because order matters more than the individual steps.

The most common failure in cloudy-water troubleshooting is not choosing the wrong remedy. It is choosing a remedy before establishing what the situation is. This framework is ordered so that the cheapest, most informative and most reversible actions come first.

The ordered sequence

- 1 **Look at the animals.** Are any fish gasping at the surface, crowding the filter outflow, breathing rapidly, or lying on the bottom? If yes, stop reading and go to Part LX. Nothing else in this list matters until that is addressed.
- 2 **Establish whether anything is dead or decaying.** Count the livestock. Look behind and under hardscape and equipment. Check for uneaten food. A decaying source is one of the few genuinely urgent causes and it is easy to miss.
- 3 **Recall what changed in the last 72 hours.** Additions, removals, cleaning, dosing, feeding, equipment, lighting, power interruptions, household activity, and anyone else who may have fed the tank. Timing narrows the possibilities faster than appearance does.
- 4 **Run the white-cup test.** A glass of tank water and a glass of dechlorinated source water, side by side, on white paper, in daylight. This separates algal, dissolved-organic, particulate and gaseous causes in under a minute.
- 5 **Let the glass stand and observe it again.** After an hour, and again the next morning. Clearing from the top means gas. Sediment on the bottom means settleable particulate. Still uniformly cloudy means a microbial suspension or a very fine colloid.
- 6 **Test ammonia and nitrite if the tank is stocked.** These determine whether the situation is a welfare matter or an appearance matter. Nitrate is worth adding for maintenance context.
- 7 **Ensure adequate surface agitation and aeration.** This is the one intervention that is appropriate before diagnosis in almost every case, because it addresses the primary harm pathway and carries essentially no risk. The exception worth noting is that in a CO₂-injected planted tank it will drive off injected CO₂, which is a trade-off the keeper should make knowingly.
- 8 **Only now decide whether to intervene, and choose the intervention that matches the identified cause.** Change one thing at a time and give it two to three days before judging it.

Sorting the situation into one of three categories

Category	Indicators	Response
Emergency	Fish gasping or in visible distress; measurable ammonia or nitrite with livestock present; strong foul odor; a known dead animal; sudden onset with behavioral change	Act immediately on oxygen and on dilution. Diagnose afterwards.
Cosmetic	Animals normal; ammonia and nitrite at zero; a known physical trigger such as vacuuming, a water change or a recent planting session	Optional intervention. Mechanical filtration and time. No urgency.
Normal and temporary	New tank in its first weeks; unstocked cycling tank with a bloom; new wood leaching; setup-phase substrate release; recently disturbed established tank	Aeration and patience. Actively resist intervening.

[REASONABLE INFERENCE] Ordering the response so that observation precedes intervention costs very little — in most cases a few minutes and one glass of water — and prevents the most common and most damaging errors in this topic, which are filter disruption and product stacking applied to a cause they cannot address.

TRIAGE

Part LX — When Cloudy Water Is an Emergency

A short list, deliberately.

Most cloudy water is not an emergency. Treating it as one is a significant source of harm, because panic interventions damage biofilters and stack incompatible products. But a small set of situations genuinely are urgent, and they should be recognisable without ambiguity.

Red flags

- **Fish gasping at the surface or crowded at the filter outflow.** The single clearest emergency indicator.
- **Rapid gill movement or visible respiratory distress** in one or more fish.
- **Multiple fish behaving abnormally at once** — simultaneous onset points to a water-column cause rather than an individual illness.
- **Any measurable ammonia or nitrite with livestock present.**
- **A strong foul, rotten or sulphurous odor**, which indicates significant decay and associated oxygen demand.
- **A known dead animal that has not been located.**
- **Sudden dense green water collapsing to grey or brown**, which indicates an algal die-off and a corresponding oxygen crash and organic load.
- **Cloudiness following a chemical addition**, particularly a medication, algacide, or any product not intended for aquarium use.
- **Cloudiness following a filter media replacement in a stocked tank**, where the risk is loss of nitrification rather than the appearance.
- **Fish deaths accompanying the cloudiness.**

Emergency response, in order

1. Increase surface agitation and aeration immediately. Raise the filter outflow to break the surface, add an air stone, or both.
2. Perform a conservative water change with temperature-matched, dechlorinated water. Twenty-five to fifty percent is a reasonable range; matching temperature and treating for chlorine matters more than the exact percentage.
3. Locate and remove any decaying material.
4. Stop feeding for a day or two.
5. Do not clean or replace filter media.
6. Test ammonia and nitrite and repeat the water change if either is present.
7. Only after the animals are stable, work out what the cloudiness was.

[COMMON AQUARIUM PRACTICE] Aeration plus a conservative, properly prepared water change is the standard emergency response across the hobby and across aquaculture, and it is standard because it addresses the two mechanisms — oxygen availability and dissolved toxicant concentration — that account for nearly all acute harm in these situations.

TRIAGE

Part LXI — When Doing Nothing Is the Right Answer

Restraint is an intervention, and it is frequently the correct one.

This part exists because the hobby has a strong bias toward action, and because nearly every commercial and social incentive pushes in the same direction. A keeper who does nothing has no product to buy, no video to watch and no update to post. That bias produces real harm.

Situation	Why patience is correct
Bacterial bloom in an unstocked cycling tank	The bloom is self-limiting and reflects the microbial community establishing itself. There is nothing at risk and intervention can slow the process.
Bacterial bloom in a stocked tank with zero ammonia and normal fish behavior	The nitrifying population is keeping pace. The bloom will exhaust its substrate. The only useful addition is aeration.
Substrate fines within the first days of a new setup	Mechanical filtration and settling will resolve it. Repeated water changes stir it back up.
Tannin coloration in a tank with wood or botanicals	It is a cosmetic preference, not a problem, and it is closer to natural conditions for many commonly kept species.
White biofilm on new driftwood	A recognised temporary stage. It is grazed by many invertebrates and resolves as the readily available organics are consumed.
Microbubbles after a water change	Dissolved gas equilibrates on its own within hours.
Brown diatom film in a tank under three months old	A standard early-succession stage that typically gives way as the system matures.
Brief cloudiness after substrate vacuuming	Expected re-suspension. Trying to eliminate it entirely usually means vacuuming less, which is the wrong trade.
Cloudiness during the first two weeks of a new planted tank with active substrate	Multiple overlapping normal processes. Intervening usually adds a variable rather than removing one.

The cost of unnecessary intervention

Doing nothing is not neutral in these situations — it is actively better than the alternative. Unnecessary intervention carries specific costs: filter cleaning and media replacement reduce nitrification capacity; stacked products interact unpredictably; repeated large water changes disturb settling material and can stress fish through chemistry swings; and every additional variable makes the actual cause harder to identify.

[REASONABLE INFERENCE] The conditions under which patience is correct share a common signature: the animals are behaving normally, nitrogenous compounds are absent or being processed, and the cause is a known transient process. Where all three hold, the expected value of intervention is negative.

CLAIM ASSESSMENT

Part LXII — Major Myth and Rule Evaluation

Thirty widely circulated statements, classified and explained.

The goal here is not to demonstrate that hobby advice is wrong. A good deal of it is sound. The goal is to identify the conditions under which each statement holds, because most of these claims are true somewhere and are repeated everywhere. A claim labelled **CONTESTED** is not a claim being dismissed — it is a claim whose validity depends on circumstances that the claim itself does not state.

“Cloudy water means the tank is not cycled.”

[UNSUPPORTED AQUARIUM FOLKLORE] These are largely independent. Cloudiness has physical, chemical, algal and microbial causes, most of which are unrelated to nitrification status. A fully cycled tank goes cloudy when its substrate is vacuumed. An uncycled tank can be perfectly clear. The kernel of truth is that new tanks are both more likely to be uncycled and more likely to be cloudy, which creates a correlation without a causal link.

“Cloudy water is always caused by a bacterial bloom.”

[UNSUPPORTED AQUARIUM FOLKLORE] Bacterial bloom is one cause among many, and it is over-diagnosed precisely because it is the most-discussed. Substrate fines, microbubbles, mineral precipitation, suspended algae, detritus and dissolved organics all produce cloudiness through entirely different mechanisms requiring entirely different responses.

“Never do a water change during a bacterial bloom.”

[CONTESTED CLAIM] This has a coherent rationale in a specific context and is misapplied outside it. The argument is that removing water removes some of the suspended population and some dissolved substrate, potentially extending the process, and this argument has some force for a fishless cycling tank where nothing is at risk. It does not hold where livestock are present with measurable ammonia or nitrite, where a decaying source is present, or where oxygen is a concern. A properly conducted water change — temperature matched, dechlorinated, not disturbing the biofilter — is very rarely harmful. Treating this as a universal prohibition has almost certainly caused more harm than the practice it warns against.

“Bacterial blooms are harmful to fish.”

[CONTESTED CLAIM] The bacteria themselves are overwhelmingly not pathogenic and the suspension is not directly toxic. The harm pathway is indirect: a dense heterotrophic population exerts substantial biochemical oxygen demand, which can depress dissolved oxygen, particularly in warm, heavily stocked or poorly circulated tanks. So the accurate statement is that blooms can be harmful through oxygen depletion, not that the bacteria attack fish.

“Clear water means healthy water.”

[UNSUPPORTED AQUARIUM FOLKLORE] Clarity is a measure of light scattering. Ammonia, nitrite, nitrate, chlorine, chloramine, copper, low dissolved oxygen, wrong temperature and unsuitable pH are all invisible. This is arguably the single most consequential misconception in the hobby, because it produces false confidence in exactly the situations where testing matters most.

“You should replace filter media regularly to keep water clear.”

[UNSUPPORTED AQUARIUM FOLKLORE] This is a marketing artifact of the disposable-cartridge business model rather than a husbandry principle. Biological media hosts the nitrifying population and should be preserved. Mechanical media should be rinsed — in tank water, not tap water — and replaced only when physically degraded. Chemical media genuinely does exhaust and genuinely does need replacing, which is the grain of truth the general claim grew from.

“Activated carbon fixes cloudy water.”

[UNSUPPORTED AQUARIUM FOLKLORE] Carbon adsorbs dissolved organic compounds. It is effective against dissolved coloration, including tannins, and against many dissolved organics. It does essentially nothing for suspended particles, bacteria, algae or gas bubbles, which is most cloudiness. It is a correct tool applied to the wrong problem most of the time it is recommended.

“UV sterilizers fix all cloudy water.”

[UNSUPPORTED AQUARIUM FOLKLORE] UV is genuinely effective against suspended organisms that pass through it, which makes it strong against green water and useful against suspended bacteria. It does nothing about particulate, dissolved coloration, gas or anything attached to a surface. It also depends heavily on dose — lamp output, contact time, flow rate, lamp age, sleeve cleanliness and water clarity — rather than on wattage alone, and turbidity reduces its effectiveness.

“Clarifiers are a good first response to cloudy water.”

[CONTESTED CLAIM] Clarifiers work by coagulation and flocculation, clumping fine suspended particles so that mechanical filtration can capture them. Used deliberately against particulate cloudiness with adequate fine filtration in place, they work as described. Used as a first response before identification, they frequently address the appearance of a problem while the cause continues, and they do little that lasts against a bloom with an ongoing food supply. No specific evidence of gill harm in aquarium use was located during this research; that is an absence of located evidence, not a demonstration of safety.

“Add more bacteria to fix a bacterial bloom.”

[UNSUPPORTED AQUARIUM FOLKLORE] A bacterial bloom is already a large bacterial population. Adding more bacteria does not reduce it. Bottled products may help establish nitrification, which is a different and legitimate purpose, but they are not a bloom remedy. One published comparison found that only one of five tested quick-start products significantly reduced ammonia relative to a control, which is worth knowing about the category generally.

“Cloudy water always clears in 24 hours.”

[UNSUPPORTED AQUARIUM FOLKLORE] Timescales differ by cause across three orders of magnitude. Microbubbles clear in hours. Substrate fines clear in hours to days. Bacterial blooms typically run several days to a few weeks. Green water persists indefinitely while its conditions persist. Tannins continue for months. Any specific duration promise is wrong for most causes.

“Green water is dangerous.”

[CONTESTED CLAIM] Green water is deliberately cultivated as a live-food and rearing medium, and there is published work associating live microalgae with improved larval survival in aquaculture. The genuine concerns are the daily oxygen cycle in a dense bloom — with a minimum before dawn — associated pH swings, loss of visibility for monitoring animals, and the oxygen crash that follows a sudden die-off. Those are real, and they are different from the claim that the algae are harmful.

“Blackout treatment always kills green water.”

[CONTESTED CLAIM] Extended darkness deprives photosynthetic organisms of energy and often works against green water. It does not address the conditions that produced the bloom, so recurrence is common. It also stresses plants, and a large simultaneous die-off adds organic load and oxygen demand. It is a legitimate tool with real caveats rather than a reliable fix.

“Cloudy water is caused by overfeeding.”

[CONTESTED CLAIM] Overfeeding is a genuine and common trigger, because uneaten food and increased waste both supply the dissolved organics that heterotrophic bacteria consume. It is one trigger among many, and diagnosing every cloudy tank as an overfed tank is both inaccurate and, for keepers who are not overfeeding, actively unhelpful.

“A bigger filter will prevent cloudy water.”

[CONTESTED CLAIM] Higher flow and greater media volume help with particulate cloudiness and improve circulation and gas exchange, both of which are useful. Neither removes bacteria or suspended algae, which pass through standard media, and neither addresses dissolved coloration or gas. Filtration is a genuine part of the picture and not a preventive against the category as a whole.

“You must rinse substrate until the water runs completely clear.”

[USEFUL RULE OF THUMB] Directionally correct and worth doing, with two qualifications. Thorough rinsing substantially reduces initial cloudiness and is the single most effective preventive for new-tank dust. But some products should not be rinsed at all — most aquasoils will break down — and even well-rinsed sand can cloud again when disturbed, because the fines are generated by abrasion between grains rather than only present as manufacturing residue.

“Cloudy water will clear if you just wait.”

[CONTESTED CLAIM] True for self-limiting causes: blooms, fines, microbubbles, setup-phase phenomena. False where a cause is ongoing: continuous overfeeding, a decaying animal, a failing filter, green water with a persistent light source, or continued substrate release. The useful diagnostic question is whether the cause is an event or a condition.

“Water changes always help cloudy water.”

[CONTESTED CLAIM] They help substantially with particulate and dissolved causes and with any situation involving nitrogenous compounds. They have limited durable effect on green water, because the population regrows, and their effect on a bacterial bloom depends on whether they remove the organic substrate driving it. A properly conducted water change is very rarely harmful, which is why the practice survives even where it does not resolve the specific cause.

“Tannins are bad for fish.”

[UNSUPPORTED AQUARIUM FOLKLORE] Humic substances are a normal component of many natural freshwater habitats. Published work shows dissolved organic carbon protecting zebrafish against ionoregulatory disturbance at low pH, and moderating metal toxicity. The picture is not uniformly positive: one study describes a double-edged effect in rainbowfish, with reduced respiratory stress at moderate concentrations alongside morbidity through ion loss, and a medaka proteomics study found suppression of innate immunity and coagulation proteins. The accurate summary is that humic substances have real physiological effects, both beneficial and adverse depending on concentration and context, and that they are not inherently harmful.

“Tannins boost fish immune systems.”

[CONTESTED CLAIM] This is widely repeated in marketing for botanicals and the evidence does not support it as stated. The medaka proteomics work found humic acid exposure suppressing innate immunity and coagulation-related proteins, which is the opposite direction. There are documented protective effects — against low-pH ionoregulatory stress and against metal toxicity — but those are chemical and physiological protection, not immune stimulation. The claim conflates the two.

“Tannins lower pH reliably.”

[CONTESTED CLAIM] The effect depends almost entirely on carbonate hardness. In soft, poorly buffered water, organic acids from wood and botanicals can lower pH noticeably. In water with meaningful KH, the buffering capacity resists the change and the pH may barely move. The important nuance is that KH is a consumable reserve rather than a permanent lock — it is depleted over time by acid production and by uptake, and a tank that resisted pH change for months can begin to shift once that reserve runs low.

“Green water means poor water quality.”

[CONTESTED CLAIM] Green water indicates that light and nutrients together exceeded what the plants and competing organisms in the system were taking up. That can happen in a tank with excellent nitrogen readings and a window nearby. It correlates with high nutrients often enough for the belief to persist, but light is frequently the actual limiting variable that changed.

“Cloudy water will kill your fish.”

[CONTESTED CLAIM] Cloudiness itself very rarely kills fish. The conditions that sometimes accompany it — oxygen depletion, ammonia, nitrite, a large decaying mass — absolutely can. Framing the appearance as the killer misdirects attention away from the mechanisms that actually matter, and it produces panic responses that cause their own harm.

“You need to tear down and restart a persistently cloudy tank.”

[UNSUPPORTED AQUARIUM FOLKLORE] A full teardown destroys the established biofilter and returns the tank to the condition most associated with cloudiness in the first place. It is close to the worst available response to persistent cloudiness. The narrow exceptions involve contamination that cannot be removed any other way.

“Cloudy water is caused by dirty gravel.”

[CONTESTED CLAIM] Accumulated detritus in substrate is a genuine reservoir of organic material and is released when disturbed, so there is a real mechanism here. But it is one cause among many, and the advice frequently leads keepers to vacuum aggressively when the actual cause is elsewhere — which produces more cloudiness in the short term.

“More filtration always equals clearer water.”

[CONTESTED CLAIM] True for particulate within the capture range of the media. Not true for anything smaller than the media can capture, for dissolved substances, or for gas. Adding filtration also adds flow, which can re-suspend settled material and, in some configurations, increases surface agitation in ways that matter for CO₂-injected tanks.

“Cloudiness in a new tank always resolves within a week.”

[UNSUPPORTED AQUARIUM FOLKLORE] New-tank cloudiness commonly runs longer than a week and frequently involves a sequence of different causes handing off to one another over several weeks. The one-week figure sets an expectation that makes normal progress feel like failure.

“Cloudy water in an established tank means something is seriously wrong.”

[CONTESTED CLAIM] It does mean something changed, and identifying the change is worthwhile. But the change is often benign — a maintenance session, a new decoration, a slightly heavier feeding, seasonal light. Sudden cloudiness in a long-stable tank deserves investigation, not alarm.

“You should stop feeding entirely until the water clears.”

[CONTESTED CLAIM] Reducing or briefly pausing feeding is a reasonable and low-risk measure during a bloom or an ammonia event, and healthy adult fish tolerate short fasts without difficulty. Extended complete fasting is a welfare cost imposed to address an appearance, and it is inappropriate for fry, for small-bodied species, and for any tank where the cause is not organic loading in the first place.

“Aquarium salt clears cloudy water.”

[UNSUPPORTED AQUARIUM FOLKLORE] No mechanism connects chloride addition to clearing suspended particles, bacteria or algae. Salt has a documented and specific use in mitigating nitrite toxicity, where chloride competes with nitrite uptake at the gill, which is a genuinely different claim and a legitimate one. It is not a clarity treatment, and it is not appropriate for many plants and soft-water species.

REFERENCE MATRIX

Part LXIII — Treatment-by-Cause Matrix

Which interventions address which causes, and which do not.

Read across a row to see the full set of options for a given cause. Read down a column to see the range of situations a given tool actually addresses — which is usually narrower than its marketing implies.

Cause	Water change	Fine filtration	Carbon	UV	Light change	Aeration	Patience	Other
Heterotrophic bacterial bloom	Situational	Minimal	Indirect	Partial	No	Supportive	Usually sufficient	Reduce organic input; find decaying material
Substrate fines / dust	Helps	Effective	No	No	No	No	Yes	Siphoning; time; better rinsing next time
Detritus in suspension	Helps	Effective	No	No	No	No	Partial	Restore flow; improve circulation; vacuum
Green water	Temporary	No	No	Effective	Primary tool	Supportive	No	Plant competition; blackout; diatom filtration
Dissolved tannins	Effective	No	Effective	No	No	No	Partial	Adsorbent resin; remove or pre-soak the source
Accumulated dissolved organics	Effective	No	Effective	No	No	No	No	Review maintenance and feeding
Microbubbles	No — may cause it	No	No	No	No	Speeds it	Effective	Find air leaks; pre-age refill water
Mineral precipitation	Partial	Partial	No	No	No	No	Partial	Change dosing practice; separate incompatible products
Active substrate leaching	Situational	Partial	Partial	No	No	No	Effective	Time; avoid stocking during the leaching phase
Decaying organic matter	Effective	Partial	Partial	No	No	Important	No	Locate and remove the source — the only real fix
Loss of biofilter capacity	Effective	No	No	No	No	Important	Partial	Reduce feeding; stop disturbing the filter; test daily
Source water turbidity	No	Effective	No	No	No	No	No	Pre-filter or pre-settle fill water
Cyanobacteria (benthic)	Partial	No	No	Minimal	Helpful	Helpful	No	Physical removal; improve flow; address nutrients
Diatoms	Minimal	No	No	No	Minimal	No	Effective	Grazers; time as the system matures

[REASONABLE INFERENCE] The pattern visible down the columns is the useful takeaway: no single intervention addresses more than a few causes, and the two entries that appear most widely useful — aeration and patience — are the two that nobody sells.

REFERENCE MATRIX

Part LXIV — Wrong-Treatment Matrix

Common mismatches between problem and response.

Each row below pairs a real situation with the treatment most commonly applied to it and explains why the pairing fails. These are not hypothetical; they are the recurring patterns visible in hobby discussion.

Problem	Common wrong treatment	Why it does not address the cause
Bacterial bloom	Cleaning or replacing filter media	Removes nitrifying capacity at the moment the system most needs it, and can turn a cosmetic problem into an ammonia problem
Bacterial bloom	Adding more bottled bacteria	The tank already contains a very large bacterial population. Adding bacteria does not reduce a bacterial population.
Bacterial bloom	Activated carbon	Carbon adsorbs dissolved compounds, not suspended cells. Any effect is indirect, through removing some of the dissolved organics feeding the bloom.
Green water	Water changes alone	The algae reproduce faster than dilution removes them, and a water change may add nutrients if the source water contains them
Green water	Activated carbon	Carbon does not remove suspended cells
Green water	Algaecide as a first response	A mass die-off in a dense bloom converts an algae problem into an oxygen and dissolved-organics problem, and frequently produces a bacterial bloom immediately afterwards
Tannin coloration	Clarifier	Clarifiers clump suspended particles. Tannins are dissolved and have nothing to clump.
Tannin coloration	UV sterilizer	UV inactivates organisms. It does not remove dissolved color.
Substrate fines	Bottled bacteria	The cloudiness is mineral particulate. Bacteria are irrelevant to it.
Substrate fines	Repeated large water changes	Refilling disturbs the substrate again, re-suspending what had begun to settle
Microbubbles	Any product at all	It is dissolved gas coming out of solution and it resolves on its own within hours
Mineral precipitation	Clarifier or water change	Treats the current instance while the dosing practice that caused it remains unchanged, so it recurs on the next dose
Decaying animal or food	Clarifier, carbon, or any clarity product	Masks or partly removes the visible consequence while the source continues to consume oxygen and produce ammonia
Loss of biofilter capacity after media replacement	Cleaning the tank thoroughly	Removes further surface-attached biofilm, deepening the exact deficit causing the problem
Persistent unidentified cloudiness	Full teardown and restart	Destroys the biofilter and returns the tank to the immature state most associated with cloudiness
Cloudiness in a new tank	Adding fish to “get the cycle going”	Exposes animals to ammonia and nitrite. Fishless cycling achieves the same outcome without the welfare cost.
Cloudiness with fish gasping	Identifying the cloudiness before acting	The gasping is the emergency and it requires immediate aeration regardless of what the cloudiness turns out to be
Diatom film in a young tank	Chemical treatment or nutrient manipulation	A normal early-succession stage. Intervention adds variables to a system that is still stabilising.
Any cloudiness	Stacking several products at once	Makes the cause unidentifiable, introduces unpredictable interactions, and removes the ability to learn anything from the outcome

The pattern underneath all of these

Every row is the same error in a different costume: a treatment chosen from the appearance rather than from the cause. The appearance is the output of the system. The treatment has to attach to the input.

COMMUNITY LANDSCAPE

Part LXV — Current Hobbyist Pain Points

What people are actually asking, and why the answers they receive conflict.

How to treat this section

What follows is a characterisation of recurring themes in hobbyist discussion, drawn from public forum and social discussion during research. It is **not evidence** about aquarium biology and should not be cited as such. It is evidence about what confuses people, which is a different and separately useful thing. Every claim in this part is about the discourse, not about the water.

“Why is my brand new tank cloudy and how long will it last?”

Why it is confusing	The keeper has no baseline for what normal looks like, and every source gives a different duration.
Conflicting advice encountered	Answers range from “it will clear in 24 hours” to “this means your cycle crashed” to “add this product.” All three are given confidently.
What the evidence suggests	A bacterial bloom in a new tank is a normal and largely self-limiting phase, typically running days to a few weeks depending on the organic supply, temperature, and how mature the surface-attached community is.
What remains uncertain	The duration in any individual tank, because it depends on variables nobody measures.

“Should I do a water change or not?”

Why it is confusing	The keeper has encountered two confident and opposite rules and has no way to evaluate them.
Conflicting advice encountered	One camp says water changes prolong the bloom by removing substrate and disrupting the process. The other says water changes are always the answer.
What the evidence suggests	The correct answer is situational, turning on whether livestock are present, whether nitrogenous compounds are measurable, and whether the cause is particulate, dissolved, or microbial. Both universal rules are wrong as universal rules.
What remains uncertain	Whether water changes meaningfully extend bloom duration in a fishless tank — this is asserted more often than it is demonstrated.

“Is cloudy water hurting my fish?”

Why it is confusing	The visible symptom feels alarming and the actual risk is invisible.
Conflicting advice encountered	Answers split between “totally harmless, just wait” and “this is an emergency.”
What the evidence suggests	The cloudiness is usually not directly harmful. The harm pathways are oxygen depletion, ammonia and nitrite, which are separately checkable.
What remains uncertain	How often a bloom actually depresses oxygen to a harmful degree in a typical home aquarium — the mechanism is well established, the frequency is not well characterised.

“I have tried everything and it will not clear.”

Why it is confusing	Multiple sequential causes are being treated as one persistent cause, and every intervention adds a variable.
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Conflicting advice encountered	Each new responder suggests a different product, none of which are wrong for some cause, and none of which match this keeper's actual cause.
What the evidence suggests	Sequential and simultaneous causes are common, especially in new planted tanks. Partial improvement is evidence of a partly correct diagnosis, not of failure.
What remains uncertain	Nothing specific — this is a diagnostic and communication problem rather than a scientific one.

“My parameters are perfect but the water is cloudy.”

Why it is confusing	The keeper believes test kits measure water quality comprehensively.
Conflicting advice encountered	Responses often assume the test kit is faulty or that the keeper is testing wrong.
What the evidence suggests	Standard hobby test kits measure nitrogen compounds and pH. They do not measure turbidity, dissolved organics, suspended algae, gas saturation, oxygen or particle load. Good readings with cloudy water is a normal combination.
What remains uncertain	—

“Is this a bacterial bloom or something else?”

Why it is confusing	“Bacterial bloom” has become the default diagnosis for any white cloudiness, which crowds out the alternatives.
Conflicting advice encountered	Confident identification from photographs, which is unreliable.
What the evidence suggests	White cloudiness has at least four common causes with different responses: microbial suspension, microbubbles, fine particulate, and mineral precipitate. Simple physical tests separate them.
What remains uncertain	—

“My planted tank has been cloudy for three weeks.”

Why it is confusing	A new planted setup introduces five or six independent cloudiness mechanisms simultaneously, on different timescales.
Conflicting advice encountered	Advice targets one mechanism at a time while the others continue.
What the evidence suggests	The typical new planted-tank sequence — fines, gas, humic release, bloom, plant melt, diatoms, wood biofilm — is largely normal and resolves as the system matures.
What remains uncertain	How much of the variation between tanks is substrate-driven versus planting-density-driven.

“My water is brown — is that bad?”

Why it is confusing	Discoloration and turbidity are being conflated, and aesthetic preference is being framed as a health question.
Conflicting advice encountered	Answers range from “that is beautiful blackwater, leave it” to “get carbon in there immediately.” Both are preferences dressed as advice.
What the evidence suggests	Dissolved humic substances are a normal feature of many natural habitats with documented physiological effects that are context-dependent rather than uniformly good or bad.
What remains uncertain	Optimal concentrations for aquarium species, which are not established.

“Can I add fish yet?”

Why it is confusing	Clarity is being used as a cycling indicator.
Conflicting advice encountered	Some responders answer from the appearance, which is unanswerable from the appearance.
What the evidence suggests	Cycling status is determined by test results. Clarity is essentially uncorrelated with it.
What remains uncertain	—

“Why does it go cloudy every time I clean?”

Why it is confusing	The keeper is being told to clean more and to clean less by different sources.
Conflicting advice encountered	One camp treats any post-cleaning cloudiness as a sign of over-cleaning; another treats it as under-cleaning.
What the evidence suggests	Brief re-suspension after vacuuming is expected and benign. Persistence afterwards points to inadequate fine mechanical filtration or declining flow. Aggressive filter cleaning is a genuine and separate risk.
What remains uncertain	—

[ANECDOTAL OBSERVATION] The recurring structure across these pain points is that hobbyists are given rules without conditions. Nearly every conflict listed above dissolves once the conditions under which each rule applies are stated. That is the primary editorial opportunity this topic offers.

CONTENT RESEARCH

Part LXVI — Search Intent Landscape

Query patterns and the intent behind them.

A note on numbers

No search-volume figures are given anywhere in this part. Reliable volume data was not available during this research, and inventing plausible-looking numbers would make the document less trustworthy rather than more useful. What follows is a structural characterisation of query types and the intent behind them.

Query clusters

Cluster	Representative queries	Underlying intent
Diagnostic — general	why is my aquarium cloudy; why is my fish tank cloudy; cloudy aquarium water causes	Identify an unknown condition. Broadest and least specific.
Diagnostic — by color	white cloudy aquarium water; green aquarium water; brown aquarium water; yellow aquarium water; milky fish tank water	The keeper has observed a specific appearance and wants it matched to a cause.
Diagnostic — by timing	cloudy water after water change; new tank cloudy water; cloudy water after cleaning filter; cloudy water day 3; tank cloudy after adding fish	The keeper has a trigger in mind and wants confirmation. The highest-value cluster, because timing is genuinely diagnostic.
Duration and reassurance	how long does bacterial bloom last; how long until aquarium water clears; aquarium still cloudy after a week	Seeking reassurance and a timeline. High emotional weight, low technical specificity.
Safety	is cloudy water bad for fish; will cloudy water kill my fish; is bacterial bloom dangerous; cloudy water shrimp safe	Risk assessment. Should lead with the behavioral check rather than the cause.
Solution-seeking — general	how to clear cloudy aquarium water; how to fix cloudy fish tank water; clear aquarium water fast	Wants a remedy, often before identification. The cluster where mismatched advice does the most damage.
Solution-seeking — specific product	does activated carbon clear cloudy water; UV sterilizer green water; aquarium water clarifier safe; accu-clear review	Evaluating a specific tool. Best served by explaining what the tool addresses and what it does not.
Terminology	what is a bacterial bloom; what are tannins in aquarium; what is green water; diatoms vs cyanobacteria	Definitional. Well served by clear category boundaries.
Comparison and differentiation	bacterial bloom vs green water; is my water cloudy or is it algae; tannins vs cloudy water; bacterial bloom or ammonia	The keeper knows there are multiple causes and wants to distinguish them. Directly served by the identification tests in this document.
Cycle-related	is my tank cycled if water is cloudy; cloudy water during cycling; bacterial bloom cycling normal	Conflating clarity with cycling status. A correction opportunity.
Context-specific	cloudy water planted tank; cloudy water shrimp tank; cloudy water betta; cloudy water goldfish; aquasoil cloudy water	Wants advice tailored to a setup. Genuinely different answers by context, especially for active substrates.
Troubleshooting persistent cases	aquarium cloudy for weeks; cloudy water will not clear; tried everything cloudy water	Frustration-stage queries. Best served by resetting the diagnosis rather than offering another remedy.

What the query landscape implies for content

- The by-color and by-timing clusters are where accurate differentiation adds the most value, because both are genuinely diagnostic and both are poorly served by generic answers.
- The solution-seeking clusters arrive before identification. Content that leads with a remedy serves the query and fails the keeper; content that leads with a thirty-second identification step serves both.
- The safety cluster carries the most emotional weight and should lead with the behavioral check, because that is the answer to the question actually being asked.
- The comparison cluster indicates a meaningful segment of keepers who already know there are multiple causes. That audience is underserved by beginner-level content.
- Reassurance is a legitimate content goal. A keeper asking how long a bloom lasts is asking whether they have ruined something.

RETRIEVAL REFERENCE

Part LXVII — Question Library

Thirty-five common questions with self-contained answers.

Each answer below is written to stand alone, without requiring the surrounding document for context. This is deliberate: the section is built for retrieval, and a retrieved answer that depends on unretrieved context is not useful.

Why is my aquarium water cloudy?

Because something is suspended in it, dissolved in it, or coming out of solution in it. The main categories are bacterial suspensions, suspended algae, fine mineral or organic particulate, dissolved organic coloration, gas bubbles, and mineral precipitates. Which one it is determines what, if anything, to do — and a glass of tank water viewed against white paper in daylight separates most of them in under a minute.

How long does a bacterial bloom last?

Commonly several days to a few weeks. Duration depends on how much organic material is available, temperature, how established the surface-attached bacterial community is, oxygen availability, and whether the organic supply is continuing. There is no reliable fixed number, and any source giving one is guessing.

Is cloudy water harmful to fish?

Usually not directly. The realistic harm pathways are oxygen depletion from high biological demand, and ammonia or nitrite from insufficient nitrification. Both are checkable: watch for gasping or crowding at the filter outflow, and test. The fish tell you more about danger than the water does.

Should I do a water change if my water is cloudy?

If the tank is stocked and ammonia or nitrite is measurable, or if fish are showing distress: yes, promptly, with temperature-matched dechlorinated water. If the tank is unstocked and cycling: it is optional and may slightly slow the process. For particulate or dissolved causes: it helps. For green water: it will not resolve it, because the population regrows.

Will my cloudy water clear on its own?

If the cause is an event, yes — blooms, fines, microbubbles and setup-phase processes are self-limiting. If the cause is ongoing — a decaying animal, continued overfeeding, a light source driving green water, a failing filter — it will not, and waiting just delays addressing it. The useful question is whether the cause is an event or a condition.

Does cloudy water mean my tank is not cycled?

No. These are largely independent. New tanks are more likely to be both, which creates a correlation, but cycling status can only be established by testing ammonia, nitrite and nitrate.

What is the difference between a bacterial bloom and green water?

A bacterial bloom is a suspension of heterotrophic bacteria and appears white, grey or milky. Green water is a suspension of unicellular algae and appears green. A sample in a white cup viewed in daylight distinguishes them immediately. They also respond to entirely different interventions: blooms to organic-source management, green water to light management.

Can I add fish while the water is cloudy?

That depends on test results, not on clarity. If ammonia and nitrite read zero with nitrate present, cloudiness from a physical cause is not a reason to wait. If those readings are not there, the answer is no regardless of how the water looks.

Why did my tank go cloudy after a water change?

Within minutes to an hour, it is usually microbubbles from dissolved gas in cold pressurised tap water, or substrate disturbed by the refill. Within days, it may be a bacterial response if the filter was cleaned in the same session. The timing separates the two.

Why did my tank go cloudy after cleaning the filter?

Immediately: dislodged debris returning to the tank, which is harmless. After two to five days: a bacterial bloom following reduced biofilm, especially if media was rinsed in tap water or replaced. The second case warrants ammonia testing.

Does activated carbon clear cloudy water?

It removes dissolved organic compounds, so it works on dissolved coloration such as tannins. It does essentially nothing for suspended particles, bacteria, algae or gas bubbles, which account for most cloudiness.

Do UV sterilizers clear cloudy water?

They work well on suspended organisms that pass through them, which makes them effective against green water and useful against suspended bacteria. They do nothing for particulate, dissolved coloration or gas. Their effectiveness depends on delivered dose — lamp output, contact time, flow rate, lamp age, sleeve cleanliness and water clarity — not on wattage alone.

Are water clarifiers safe?

They work by clumping fine particles so filtration can capture them, and no specific evidence of gill harm in aquarium use was located during this research — which is an absence of located evidence rather than a demonstration of safety. The practical risk is misapplication: using one without adequate mechanical filtration, or using one to clear an appearance while the cause continues.

Why is my aquarium water brown?

Almost always dissolved tannins and other humic substances leaching from driftwood, botanicals, leaf litter or soil-based substrate. If the water is colored but still sharply transparent, that is the answer. If it is also hazy, something particulate is present as well.

Are tannins bad for fish?

No, as a general statement. Humic substances occur naturally in many freshwater habitats and have documented protective effects against low-pH ionoregulatory stress and against metal toxicity. There are also documented adverse effects at some concentrations, including ion loss and suppression of certain immune-related proteins. The honest summary is that the effects are real, context-dependent, and not uniformly good or bad.

How do I get rid of green water?

Address light first, since it is usually what changed and it is the variable most under your control: eliminate direct sunlight, shorten the photoperiod, reduce intensity. Then increase competition with healthy fast-growing plants. UV is effective and fast. Blackout works but does not address the cause and stresses plants. Water changes alone do not resolve it.

Why is my new tank cloudy?

Most likely one or several of: substrate fines from filling, dissolved gas from pressurised tap water, humic release from an active substrate, and a heterotrophic bacterial bloom developing over the first days. In a new planted tank these often occur in sequence, which is why the cloudiness seems to change character over weeks.

Is a bacterial bloom dangerous?

The bacteria themselves are overwhelmingly harmless and not pathogenic. The risk is indirect: a dense population exerts substantial oxygen demand, which matters most in warm, heavily stocked or poorly circulated tanks. Maintaining strong surface agitation during a bloom addresses the actual risk.

What causes white cloudy water specifically?

Four common causes: a heterotrophic bacterial suspension, microbubbles from dissolved gas, fine mineral or substrate particulate, and mineral precipitation from dosing. A standing glass separates them — gas clears from the top within the hour, particulate settles overnight, a microbial suspension stays uniformly cloudy.

Why does my water go cloudy every time I vacuum the gravel?

Vacuumping lifts fine material into suspension, which is expected. The question worth asking is why it does not clear within a few hours, and the usual answers are no fine mechanical stage in the filter, or declining filter flow.

Can overfeeding cause cloudy water?

Yes. Uneaten food and increased waste supply dissolved organics that heterotrophic bacteria consume, which is a well-understood pathway. It is one trigger among many, though, and not every cloudy tank is an overfed tank.

Why is my water cloudy but my parameters are fine?

Because standard test kits measure nitrogen compounds and pH, and cloudiness is caused by turbidity, dissolved organics, suspended algae or gas — none of which those kits measure. Good readings with cloudy water is a normal and common combination, not a contradiction.

Should I replace my filter media to fix cloudy water?

Almost never, and it frequently makes things worse. Biological media hosts the nitrifying population. Removing it at the moment the system is under load can convert a cosmetic problem into an ammonia problem. Rinse mechanical media in tank water instead.

Is it normal for a new planted tank to stay cloudy for weeks?

It is common, and it is usually a sequence of different normal processes rather than one persistent problem: fines, then humic release, then a bacterial bloom, then plant melt feeding it further, then diatoms. Recognising it as a sequence tends to be more useful than any single remedy.

Does a bigger filter prevent cloudy water?

It helps with particulate and improves circulation and gas exchange, both of which are useful. It does not remove bacteria or suspended algae, which pass through standard media, and it does nothing about dissolved coloration or gas.

Why does my water look cloudy only in photos?

Automatic white balance, flash, blue-heavy aquarium LEDs, glass reflections and film on the glass all distort perceived clarity. Photographing a glass of the water in daylight against white paper is far more reliable than photographing the tank.

What is the white film on my driftwood?

A biofilm of bacteria and fungi feeding on organic compounds leaching from the wood. It is a normal, temporary stage for new wood, it is grazed by many invertebrates and some fish, and it resolves over weeks. It is often accompanied by haze in the water column from the same source.

Should I use a blackout to clear green water?

It often works, because the algae are photosynthetic. It does not address the conditions that produced the bloom, so recurrence is common; it stresses plants; and a large simultaneous die-off adds organic load and oxygen demand. It is a legitimate tool with real caveats.

Can I use tap water directly if it looks cloudy?

Cloudy tap water is usually dissolved air, which clears within minutes in a glass. If a standing glass of treated tap water is still cloudy the next morning, the source water has genuine turbidity and pre-filtering or pre-settling the fill water is worth considering. Always dechlorinate regardless.

How do I tell if it is algae on the glass or cloudy water?

Wipe a patch of glass and look through it. If the view is clear through the wiped patch, the problem was on the glass. Also check a glass of the water itself — if that is clear, the water is clear.

Why did my water go cloudy after I added fertilizer?

If it happened within minutes, mineral precipitation is the likely answer, particularly if calcium- or magnesium-containing products were combined with phosphate-containing ones in hard water. Dosing products separately, premixing, and dosing into flow usually resolves it.

Is green water good for fry?

It is deliberately cultivated for exactly that purpose. It supports the microfauna that small fry eat and there is published aquaculture work associating live microalgae with improved larval survival. It is a management decision rather than a problem in that context.

My fish are gasping and the water is cloudy — what do I do?

Act on the gasping first and diagnose the cloudiness later. Increase surface agitation and aeration immediately, check temperature, perform a conservative temperature-matched dechlorinated water change, look for anything decaying, and test ammonia and nitrite. Do not clean the filter.

Does cloudy water affect plants?

Heavy turbidity reduces light reaching the plants, which can slow growth if it persists. Dense green water competes for both light and nutrients. Short-term cloudiness has little effect. Tannin coloration reduces light somewhat but is generally tolerated well by low-light species.

How can I prevent cloudy water in a new tank?

Rinse substrate thoroughly unless it is an aquasoil that should not be rinsed. Fill slowly onto a plate or bag to avoid blasting the substrate. Include a fine mechanical stage in the filter. Avoid overfeeding during the first weeks. Expect some cloudiness anyway and plan to leave it alone.

EXTENDED CAUSES

Part LXVIII — Additional Causes Identified During Research

Causes that appear in real reports but rarely in the standard lists.

The standard hobby list of cloudy-water causes runs to about four items. Research surfaced a considerably longer tail. Several of these are individually uncommon but collectively account for a meaningful share of the cases that resist standard diagnosis.

Cause	Mechanism	Recognition clue
Spawning events	Released milt and eggs add organic material and can visibly cloud the water directly	Sudden onset with breeding behavior beforehand; often clears within a day or two
Medications	Many are dyes or suspensions; some kill bacterial populations, releasing organics and reducing nitrification	Onset tied to dosing; frequently a color cast as well as haze
Water conditioners	Some products are themselves cloudy or react with tank water; ammonia-binding products can interact with test results	Immediate onset on dosing; clears within hours
Phosphate precipitation	Phosphate combining with calcium in hard water forms an insoluble white precipitate	Chalky white haze minutes after phosphate dosing in hard water
Aragonite and crushed coral dust	Manufacturing dust plus ongoing dissolution and abrasion	Persistent chalky haze in tanks using calcareous substrate; returns after every disturbance
Exhausted or degrading resins	Some ion-exchange and adsorbent media shed fines as they degrade, and exhausted media may release what it previously held	Onset after a long period without media replacement; fine particulate downstream of the media
New filter media dust	Ceramic, sintered glass and some carbon products carry manufacturing dust	Onset within hours of installing new media; clears within a day or two if the media was not rinsed
Construction and renovation residue	Airborne dust settling on the water surface; in extreme cases, chemical contamination	Household context; often affects multiple tanks in the same room simultaneously
Plumbing and source-water changes	Municipal main flushing, hydrant use, or pipe work introduces sediment, iron or higher chlorine	Tap water looks different; other household fixtures affected; often a utility notice
Source water turbidity	Well water with sediment, or seasonal changes in municipal supply	A standing glass of treated tap water is cloudy the next morning
Euglena blooms	A flagellated organism producing a green bloom that behaves differently from typical green algae and is notably resistant to some treatments	Green water that responds poorly to standard measures; sometimes a distinctly different shade
Protozoan and infusoria blooms	Protozoa grazing on a bacterial bloom can themselves become numerous enough to cloud the water, as a successional stage after the bacterial peak	Cloudiness that persists or changes character after a bacterial bloom appears to be resolving
Fungal growth on organic matter	Fungi colonising uneaten food, dead plant material or new wood	Visible fuzzy growth on a specific object rather than uniform haze
Air pump and airline leaks	A leak on the suction side of a canister filter draws air into the system, producing a continuous fine bubble stream	Persistent bubbles rather than a one-off event; changes when equipment is switched off
Overheating and temperature spikes	Higher temperature reduces oxygen solubility and increases bacterial metabolic rate, intensifying blooms and their oxygen demand	Seasonal or heater-failure context; blooms that are worse than the organic load would suggest
Power outages	Flow stops, biofilm suffers, deposited material is ejected on restart, and anaerobic conditions can develop in filter media	Onset immediately on restoration of power; sometimes an odor

Cause	Mechanism	Recognition clue
Substrate compaction and anaerobic pockets	Deep fine substrate can develop anaerobic zones; disturbing them releases gas and reduced compounds	Sulphurous smell and dark substrate when disturbed
New silicone or sealant curing	Uncured sealant can leach into the water	New or resealed tank; often accompanied by an odor
Untreated new decorations and hardscape	Manufacturing residue, dust, or unsuitable materials leaching	Onset within hours of adding a specific item
Excess CO2 diffuser output	Very fine diffusers produce a persistent mist that reads as haze	Present only during the CO2 photoperiod; bubbles visible on close inspection

[ANECDOTAL OBSERVATION] Several entries above are characterised from hobbyist reports rather than from published study, and are labelled as such. They are included because they have plausible mechanisms and because they account for cases that otherwise get diagnosed incorrectly as bacterial blooms by default.

EXTENDED CAUSES

Part LXIX — Source Water as a Cause

An underexamined variable, with a simple test.

Nearly all cloudy-water troubleshooting assumes the problem originates inside the tank. Sometimes it arrives with the fill water, and this possibility is rarely checked because it is rarely suggested.

The tap-glass versus tank-glass comparison

- 1 Fill one clear glass with water from the tank, mid-column.
- 2 Fill a second identical glass with water from the tap, treated with dechlorinator the same way you would for a water change.
- 3 Set both aside, away from the aquarium, on white paper in a room with daylight.
- 4 Observe both immediately, after one hour, and the next morning.

Result	Interpretation
Tap glass cloudy at first, clear within minutes	Dissolved air. Completely normal and the reason water changes often produce brief milkiness.
Tap glass still cloudy the next morning	The source water has genuine turbidity. Worth pre-filtering or pre-settling fill water, and worth contacting the utility or checking a well filter.
Tap glass has a color cast	Dissolved iron, manganese, tannins from a surface source, or organic matter in a well supply.
Both glasses look the same and both are cloudy	The source water is the likely cause, or at least a contributor.
Tank glass cloudy, tap glass clear	The cause is in the tank. Proceed with the rest of this document.
Both glasses clear while the tank looks cloudy	The issue is optical — glass films, lighting, viewing angle or path length.

Source-water factors worth knowing

Seasonal variation	Municipal supplies change with source conditions. Spring runoff, drought, and switching between reservoirs and wells all alter turbidity, hardness and organic content.
Main flushing and hydrant use	Routine utility work disturbs sediment in distribution pipes. This produces short-lived but real turbidity and sometimes discoloration.
Chloramine versus chlorine	Many utilities have switched to chloramine, which does not off-gas the way chlorine does and requires a conditioner rated for it. This is a livestock issue rather than a clarity issue but belongs in the same conversation.
Well water	Can carry sediment, iron, manganese and dissolved gases at concentrations that municipal water rarely does. Iron in particular can produce a visible cast on oxidation.
Home filtration and softeners	Water softeners exchange calcium and magnesium for sodium, which changes GH without changing TDS much. RO and RO/DI systems remove nearly everything, including the buffering that keeps pH stable.
Aeration before use	Letting fill water stand with an air stone for a few hours equilibrates dissolved gas and eliminates the microbubble effect entirely. It is a simple habit for keepers who find the post-water-change milkiness bothersome.

BACKGROUND SCIENCE

Part LXX — How Turbidity Is Actually Measured

Included for accuracy and vocabulary, not as a suggestion that hobbyists need instruments.

Hobbyists do not need turbidity meters, and nothing in this part should be read as implying otherwise. It is included because understanding what turbidity is as a measured quantity clarifies why appearance is such an unreliable guide, and because the vocabulary appears in aquaculture and water-treatment sources that hobbyists sometimes encounter.

Core concepts

Turbidity	An optical property: the degree to which suspended material causes light to be scattered and absorbed rather than transmitted in straight lines. It is a measure of clarity, not a measure of what is present. Regulatory guidance describes the responsible particles as spanning roughly the size range from fine clay to visible sediment.
Nephelometry	The standard measurement method: a light source illuminates the sample and a detector positioned at 90 degrees measures scattered light. The instrument reports the intensity of that scattered light.
NTU — Nephelometric Turbidity Units	The most commonly cited unit, based on a white-light source. Because white light is absorbed by colored water, a NTU reading cannot compensate for dissolved coloration and will read a tannin-stained but particle-free sample as turbid.
FNU — Formazin Nephelometric Units	Uses a near-infrared source, which is much less affected by dissolved color. This is why FNU is preferred for naturally colored waters — and it is a precise instrumental parallel to the distinction this document draws between coloration and turbidity.
Secchi depth	A field method: a patterned disc is lowered until it disappears from view. Simple, robust, and used widely in pond and lake management. The aquarium equivalent is noticing at what depth the back glass stops being sharp.
Total suspended solids	A gravimetric measure: filter a known volume, dry the filter, weigh what was captured. Measures mass rather than optical effect, and the two do not correlate simply because small particles scatter far more light per unit mass than large ones.

Why this matters for interpretation

Two facts from this vocabulary carry real practical weight. First, the NTU-versus-FNU distinction demonstrates that the confusion between colored water and turbid water is not a hobbyist failing — it is a genuine measurement problem that instrument designers had to solve deliberately. Second, turbidity has documented downstream consequences beyond appearance: experimental work in a food-science context found that reducing turbidity from roughly 23 to 11 NTU improved ultraviolet inactivation of *E. coli* by more than two orders of magnitude. That is the mechanism behind the advice that a UV unit will underperform in cloudy water, stated quantitatively.

[STRONG EVIDENCE] Turbidity substantially reduces the effectiveness of ultraviolet disinfection by scattering and absorbing UV and by shielding organisms. This is well established in water treatment and directly relevant to aquarium UV performance.

BACKGROUND SCIENCE

Part LXXI — Why Particle Size Matters

The single variable that explains most of what filtration can and cannot do.

Particle size determines three things simultaneously: how strongly a particle scatters light, how fast it settles, and whether any given filter medium can capture it. Almost every confusing observation about cloudy water resolves once size is taken into account.

Approximate size	What is in this range	Behavior
Dissolved — not particles	Tannins, humic substances, ammonia, nitrate, most fertilizers	Never settles, never filtered mechanically, may color the water, requires adsorption or dilution
Roughly 0.5–2 micrometres	Individual bacteria	Passes through virtually all aquarium mechanical media; scatters light strongly; effectively does not settle
Roughly 2–15 micrometres	Common green-water algae such as <i>Chlorella</i> and <i>Scenedesmus</i>	Passes through standard media; scatters light strongly; motile species actively maintain position
Roughly 1–10 micrometres	Clay and very fine mineral colloids	Settles extremely slowly; may stay suspended for weeks; capturable only by very fine media or after flocculation
Roughly 10–100 micrometres	Fine sand fines, fine detritus, bacterial flocs	Settles over hours; capturable by fine floss and polishing pads
Above roughly 100 micrometres	Visible detritus, plant fragments, uneaten food	Settles readily; captured by standard mechanical media; individually visible

Three consequences worth stating plainly

- **A filter running perfectly will not clear a bacterial bloom or green water.** The organisms are smaller than the media can capture. This is not a filter fault and no amount of additional filtration of the same grade changes it.
- **Small particles scatter light far more per unit mass than large ones.** A tiny mass of very fine material can make water look far worse than a much larger mass of coarse debris. This is why a tank can look alarming and contain very little.
- **Settling time scales steeply with size.** Coarse material settles in minutes, fine sand fines in hours, clay colloids over days to weeks, and bacteria effectively not at all. The overnight standing-glass test is a direct measurement of this, which is why it works so well.

[ESTABLISHED SCIENTIFIC PRINCIPLE] Settling velocity increases with particle size and with the density difference from water, while light scattering per unit mass increases as particle size decreases toward the wavelength of visible light. These two relationships together explain the bulk of observed aquarium clarity behavior.

FRAMEWORK

Part LXXII — Four Kinds of Water Quality

Chemical, biological, physical, and visual — and why conflating them causes most of the confusion in this topic.

Much of the difficulty in this subject comes from a single vocabulary problem: the phrase “water quality” is used for four different things that can vary independently of one another.

Dimension	What it covers	How it is assessed	Relationship to clarity
Chemical	Ammonia, nitrite, nitrate, pH, KH, GH, dissolved oxygen, chlorine, metals, dissolved organics	Test kits, meters	Almost entirely independent. Most of these are invisible at any concentration.
Biological	The nitrifying community, heterotrophic populations, algae, pathogens, biofilm, microfauna	Indirectly, through nitrogen readings and observation over time	Partially related. Suspended populations affect clarity; the important surface-attached ones do not.
Physical	Suspended solids, particle size distribution, temperature, flow, gas saturation	Observation, thermometer, the standing-glass test	Directly related. This is what turbidity actually measures.
Visual	What the water looks like to a person standing in front of the tank	The eye, influenced by lighting, glass condition, viewing angle and path length	This is clarity, and it is the least informative of the four about the animals' welfare.

The claim this framework supports

Visual clarity is a physical-dimension observation being used as a proxy for the chemical and biological dimensions, which it does not reliably track. That single substitution generates the two mirror-image errors this document keeps returning to: assuming clear water is healthy, and assuming cloudy water is dangerous. Both are the same mistake.

[REASONABLE INFERENCE] Separating these four dimensions explicitly is likely to be the single most useful conceptual move available to anyone explaining this topic, because it gives keepers a reason why clear water can be dangerous and cloudy water can be fine — rather than asking them to accept both as isolated facts.

MASTER FRAMEWORK

Part LXXIII — Full Decision Tree

The complete diagnostic path in branch form.

This is the consolidated version of the reasoning distributed through the rest of the document. It is written as a sequence of branches so it can be followed directly or adapted into a flowchart.

Branch 0 — Livestock check (always first)

Question	Are any animals gasping at the surface, crowded at the filter outflow, breathing rapidly, lying on the bottom, or otherwise behaving abnormally?
If yes	EMERGENCY. Increase aeration immediately, check temperature, perform a conservative temperature-matched dechlorinated water change, search for decaying material, test ammonia and nitrite, do not touch the filter. Diagnose the cloudiness afterwards.
If no	Proceed to Branch 1. Normal behavior means there is time to diagnose properly.

Branch 1 — Is it actually in the water?

Test	Fill a clear glass from mid-column and look at it against white paper in daylight. Also wipe a patch of the tank glass and look through it.
If the glass is clear	The problem is on the glass, in the lighting, or in the viewing conditions. See Part LVII. Stop here.
If the glass is cloudy or colored	Proceed to Branch 2.

Branch 2 — Colored, or turbid, or both?

Amber or brown, and transparent — the far side of the glass is sharp	Dissolved tannins and humic substances. Likely possibilities: driftwood, botanicals, leaf litter, soil-based substrate. Additional checks: what wood or botanicals are present and how recently added; pH and KH if the water is soft. Emergency red flags: none. Next step: decide whether this is a problem or a preference. If unwanted, use carbon, adsorbent resin or water changes.
Green	Suspended algae. Likely possibilities: increased light, sunlight exposure, longer photoperiod, nutrient increase, reduced plant uptake. Additional checks: light history; plant condition; nitrate and phosphate. Emergency red flags: dawn gasping, or a sudden collapse of a dense bloom. Next step: address light first, then competition, then consider UV.
White, grey, milky, or hazy without a color cast	Proceed to Branch 3.
Clear water with individually visible particles	Proceed to Branch 5.

Branch 3 — The standing-glass test

Clears from the top downward within minutes to an hour	Microbubbles. Likely possibilities: recent water change with cold pressurised tap water, an air leak on the suction side of a canister filter, a fine CO2 diffuser. Additional checks: does it change when equipment is switched off; do bubbles collect on glass and leaves. Emergency red flags: none in normal aquarium conditions. Next step: nothing if transient; find the leak if persistent.
Sediment collects on the bottom overnight, water above clears	Settleable particulate. Proceed to Branch 4.

Still uniformly cloudy the next morning A microbial suspension or a very fine colloid. Proceed to Branch 6.

Branch 4 — Settleable particulate

Likely possibilities	Substrate fines, disturbed detritus, mineral precipitate, filter media dust, source water sediment.
Additional checks	What was disturbed and when. Whether the tank is new. Whether a dosing event preceded it by minutes. Whether there is a fine mechanical stage in the filter and whether flow has declined. Whether a standing glass of treated tap water is also cloudy.
Emergency red flags	None. This category is essentially always cosmetic.
Next step	Fine mechanical filtration, siphoning, restored flow, and time. If it followed dosing within minutes, change the dosing practice rather than treating the water.

Branch 5 — Clear water with visible particles

Likely possibilities	Detritus, plant fragments, uneaten food, sloughed biofilm, coarse fines.
Additional checks	Is there a fine mechanical stage? Has flow declined? Are there dead zones with no circulation? When was the filter last serviced?
Emergency red flags	None, unless accompanied by odor, which changes the branch.
Next step	Improve mechanical filtration and circulation; siphon accumulation zones; service the filter without replacing biological media.

Branch 6 — Persistent uniform haze

Sub-question	How old is the tank, and what happened in the last five days?
New tank, days 2–14	Heterotrophic bacterial bloom, very likely normal. Additional checks: ammonia and nitrite if stocked. Emergency red flags: any ammonia or nitrite with livestock present; gasping. Next step: aeration and patience. Do not touch the filter.
Established tank, following filter cleaning or media replacement	Bloom following reduced biofilm. Additional checks: ammonia and nitrite, urgently. Emergency red flags: measurable ammonia or nitrite. Next step: reduce feeding, maintain aeration, water changes as needed, leave the filter alone.
Established tank, following overfeeding or a suspected decaying source	Bloom on an organic source. Additional checks: count the livestock; search behind hardscape and equipment; check for uneaten food; smell the water. Emergency red flags: strong foul odor; gasping; ammonia. Next step: find and remove the source. This is the highest-priority action in this branch.
Established tank, no identifiable trigger	Investigate more broadly. Additional checks: has anyone else fed the tank; was there a power interruption; has flow declined; has the temperature changed; is every animal accounted for. Next step: work through Part XXXII.
Persisting for weeks despite intervention	Assume multiple or sequential causes. Next step: stop intervening for a week except for aeration and normal feeding, then re-run this tree from Branch 0 on the current condition rather than the original one.

The two rules that govern the whole tree

Livestock behavior outranks water appearance at every branch. If the animals are in distress, the branch you are on stops mattering.

Change one variable at a time and wait two to three days. Every branch above assumes you can attribute a change to a cause, and stacking interventions destroys that ability.

MASTER FRAMEWORK

Part LXXIV — Quick Identification Table

For rapid reference and for adaptation into a graphic.

If it looks like this	Think about this first	Check this next
Milky white, uniform, developed over days in a new tank	Heterotrophic bacterial bloom	Ammonia and nitrite if stocked; how much organic material is in the tank
Milky white, appeared within an hour of a water change	Microbubbles from dissolved gas	Does a glass clear from the top within the hour?
Milky white, appeared within minutes of dosing	Mineral precipitation	Which products, in what order, and what is the KH and GH
Cloudy immediately on filling a new tank	Substrate fines	Does it settle overnight? Is there a fine mechanical stage in the filter?
Green, reduced visibility	Suspended algae	Light history — new fixture, longer photoperiod, sunlight, season
Brown or amber, still sharply transparent	Dissolved tannins	What wood or botanicals are present, and how recently added
Brown and hazy, reduced clarity	Suspended detritus or fine soil particulate	What was disturbed; filter flow; fine media loading
Yellow, clarity mostly preserved, no wood in the tank	Accumulated dissolved organics	Time since the last water change; feeding and maintenance history
Grey, no obvious tint	Fine dust, a mild bloom, or a low-density green bloom	White-cup test in daylight — grey is the least specific appearance
Sparkling with fine bubbles on glass and leaves	Dissolved gas or an equipment air leak	Does it change when the filter or air pump is switched off?
Chalky white, appeared very fast, recurs on a schedule	Mineral precipitation from dosing	Dosing order and water hardness
Clear water with drifting visible specks	Detritus and debris	Mechanical filtration grade; flow; circulation dead zones
White film on driftwood plus haze in the water	Wood biofilm and a bacterial response to the same leached organics	How recently the wood was added
Brown film on glass and leaves in a young tank	Diatoms	Tank age — under three months is the classic window
Dark green or blue-green sheets peeling off surfaces	Cyanobacteria	Does it peel off in sheets and smell musty? Check flow and dead zones.
Cloudy only in photographs	Camera white balance, flash, or glass film	Photograph a glass of the water in daylight instead
Cloudy, and the fish are gasping	The gasping, not the cloudiness	Aeration immediately, then temperature, then ammonia and nitrite

MASTER FRAMEWORK

Part LXXV — What Not To Do

A consolidated list, because the avoidable errors are more consistent than the correct responses.

- **Do not replace biological filter media** in response to cloudy water. This is the single most damaging common response, because it removes nitrification capacity at the moment the system is under load.
- **Do not rinse filter media under chlorinated tap water.** Rinse in removed tank water instead.
- **Do not tear down and restart the tank.** This returns the system to the immature state most associated with cloudiness.
- **Do not add multiple products at once.** It makes the cause unidentifiable, introduces unpredictable interactions, and removes the ability to learn anything from the outcome.
- **Do not use an algaecide as a first response to dense green water.** A mass die-off converts an algae problem into an oxygen and dissolved-organics problem.
- **Do not use a clarifier before identifying the cause** or without adequate mechanical filtration to capture what it clumps.
- **Do not add more bottled bacteria to fix a bacterial bloom.** The tank already contains a very large bacterial population.
- **Do not stop feeding entirely for an extended period.** Reducing or briefly pausing is reasonable; prolonged fasting is a welfare cost imposed to address an appearance.
- **Do not perform repeated very large water changes in quick succession** in an attempt to force clarity. This disturbs settling material and can stress fish through chemistry swings.
- **Do not add fish because the water looks clear.** Clarity says nothing about cycling status.
- **Do not delay adding fish solely because the water looks cloudy** if testing confirms the tank is cycled and the cause is physical.
- **Do not clean the substrate, clean the filter and perform a large water change in the same session.** Separate them so that cause and effect stay legible.
- **Do not diagnose from a photograph** when behavior and timing information is available instead.
- **Do not treat cloudiness before checking the animals.**
- **Do not assume nothing changed.** In an established tank that suddenly went cloudy, something did, and the most common answers are a dead animal, someone else feeding, or declining filter flow.
- **Do not conclude an intervention failed because the improvement was partial.** Partial improvement usually means the diagnosis was partly right and more than one cause is present.

[REASONABLE INFERENCE] Nearly every item above shares a structure: an action taken from the appearance rather than from the cause, or an action that reduces the system's own capacity to resolve the situation. Framing the advice that way is likely to transfer better than the list itself.

MASTER FRAMEWORK

Part LXXVI — When to Escalate or Seek Help

And what information to bring.

Situations warranting outside help

- Fish are dying and the cause is not identifiable after working through the diagnostic steps.
- Ammonia or nitrite remains measurable despite repeated water changes over several days.
- Multiple animals show the same abnormal behavior simultaneously and water testing shows nothing.
- Cloudiness follows a suspected chemical contamination — aerosols, cleaning products, paint, pesticide, or anything not intended for aquarium use.
- Source water has changed and testing suggests a supply problem, in which case the water utility is the right contact.
- A disease outbreak is suspected alongside the water-quality issue, where an aquatic veterinarian is the appropriate escalation.
- A high-value or difficult-to-replace collection is at risk and the keeper is not confident in the diagnosis.

What information to bring

- 1 Tank volume, age, and how it was cycled.
- 2 Complete stocking list with approximate sizes and when each was added.
- 3 Filtration: type, media in order, when each was last cleaned or replaced.
- 4 Substrate type and whether it is an active or inert product.
- 5 Full test results with actual numbers, the test method used, and how recently the kit was purchased.
- 6 A complete list of everything added to the tank in the last month, including conditioners, fertilizers, medications and foods.
- 7 Maintenance schedule: water change frequency and volume, feeding frequency and amount.
- 8 A precise timeline of when the cloudiness started and what preceded it.
- 9 A description of livestock behavior, which matters more than any photograph.
- 10 A photograph of a glass of the water against white paper in daylight, alongside a photograph of the tank.
- 11 Everything already tried, in order, with what happened after each.

The most commonly missing piece

In practice, the item most often absent from a request for help is the list of what has already been tried. Without it, responders repeat suggestions that have already failed, which wastes everyone's time and reinforces the keeper's sense that nothing works.

BACK MATTER

Glossary

Terms used in this document, defined for standalone retrieval.

Adsorption	The adhesion of dissolved molecules onto a solid surface. The mechanism by which activated carbon removes dissolved organics and coloration. Distinct from absorption, which implies uptake into a volume.
Aerobic	Occurring in the presence of oxygen.
Anaerobic	Occurring in the absence of oxygen. Anaerobic pockets in deep fine substrate can produce hydrogen sulphide and other reduced compounds.
Autotroph	An organism that obtains carbon from inorganic sources such as carbon dioxide. Nitrifying bacteria are autotrophs; they build biomass from inorganic carbon using energy from oxidising ammonia or nitrite.
Bacterial bloom	A rapid increase in suspended, typically heterotrophic bacteria, producing white or grey cloudiness. Distinct from the nitrifying population, which grows on surfaces and does not cloud water.
Benthic	Living on or attached to a surface at the bottom of a body of water. Benthic cyanobacteria form mats on substrate and hardscape.
Biochemical oxygen demand (BOD)	The amount of dissolved oxygen consumed by microorganisms breaking down organic matter. The mechanism by which a dense bacterial bloom can endanger fish.
Biofilm	A community of microorganisms attached to a surface within a matrix they secrete. The functional home of the nitrifying population, and the white film seen on new driftwood.
Coagulation	Neutralisation of the charges that keep fine particles dispersed, allowing them to come together. The first stage of clarifier action.
Colloid	A suspension of particles small enough that they remain dispersed almost indefinitely rather than settling.
Comammox	Complete ammonia oxidation: bacteria capable of oxidising ammonia all the way to nitrate within a single organism. Certain Nitrospira are comammox organisms and have been found in home aquarium biofilters.
Cyanobacteria	Photosynthetic bacteria, often called blue-green algae. Commonly form benthic mats in aquariums; mats can detach and float, which is a different appearance from planktonic bloom.
Diatom	Single-celled algae with silica cell walls. Produce the brown film typical of young aquariums.
Dissolved organic carbon (DOC)	Organic carbon present in dissolved form. Includes humic substances and is the substrate that heterotrophic bacteria consume.
Flocculation	The aggregation of destabilised particles into larger clumps that can be captured by filtration or settle out. The second stage of clarifier action.
FNU	Formazin Nephelometric Units. A turbidity unit based on a near-infrared light source, less affected by dissolved coloration than NTU.
Gas bubble disease	A condition caused by total dissolved gas supersaturation, in which gas comes out of solution within the tissues and bloodstream of fish.
Green water	A suspension of unicellular algae dense enough to color the water green. Common genera include Chlorella and Scenedesmus, typically a few to fifteen micrometres across.
Heterotroph	An organism that obtains carbon from organic compounds. The bacteria responsible for bacterial blooms are heterotrophs, which is why they respond so quickly to organic input.
Humic substances	A broad class of complex organic compounds from decomposed plant and animal material, including humic and fulvic acids. Responsible for the amber coloration of blackwater habitats.

KH — carbonate hardness	The concentration of carbonate and bicarbonate, which buffers against pH change. Best understood as a consumable reserve that is depleted over time rather than a permanent lock on pH.
Nephelometry	Measurement of turbidity by detecting light scattered at ninety degrees to an illuminating beam.
Nitrification	The biological oxidation of ammonia to nitrite and nitrite to nitrate, carried out by autotrophic bacteria and archaea living on surfaces.
NTU	Nephelometric Turbidity Units. The most commonly cited turbidity unit, based on a white-light source and therefore affected by dissolved coloration.
Photoreactivation	Repair of ultraviolet-induced DNA damage by organisms exposed to visible light afterwards. Minimal above a sufficient delivered UV dose.
Planktonic	Suspended and drifting in the water column, as opposed to attached to a surface.
Total dissolved gas (TDG)	The combined pressure of all gases dissolved in water, expressed relative to atmospheric pressure. Values above one hundred percent indicate supersaturation.
Turbidity	An optical property describing the degree to which suspended material scatters and absorbs light. A measure of clarity, not of composition.
Total suspended solids (TSS)	The mass of suspended particulate per unit volume, measured by filtration and weighing. Does not correlate simply with turbidity, because fine particles scatter far more light per unit mass.

BACK MATTER

Key Evidence Summary

The findings that carry the most weight, and the gaps that remain.

Findings supported by published research

- Nitrifying bacteria and the heterotrophic bacteria responsible for blooms are metabolically distinct, with heterotrophs growing far faster on organic carbon while autotrophic nitrifiers grow slowly on inorganic energy sources. Reported doubling times for *Nitrosomonas europaea* span roughly 7.4 to 24 hours.
- Home aquarium biofilters host a successional microbial community including comammox *Nitrospira*, and function as fixed-film bioreactors — which is why the nitrifying population is attached rather than suspended.
- In one published comparison of bottled quick-start products, only one of five significantly reduced ammonia relative to control.
- Ultraviolet disinfection efficacy depends on delivered dose rather than lamp wattage, with published aquaculture figures indicating higher doses are required for waterborne algae than for bacteria.
- Turbidity substantially reduces ultraviolet disinfection efficacy. Experimental work in a food-science context found that reducing turbidity from roughly 23 to 11 NTU improved *E. coli* log reduction by more than two orders of magnitude.
- Dissolved organic carbon from a blackwater source protected zebrafish against ionoregulatory disturbance at low pH.
- Humic substances show a dual effect in rainbowfish — reduced respiratory stress at moderate concentrations alongside morbidity associated with ion loss.
- Humic acid exposure suppressed innate immunity and coagulation-related proteins in a medaka proteomic study, which contradicts the common marketing claim that tannins boost fish immunity.
- Dissolved organic carbon moderates metal toxicity to fish.
- Granular activated carbon removes a large majority of humic substances and substantially reduces color.
- Coagulants including chitosan and conventional metal salts achieve high turbidity removal in water treatment, with reported figures varying considerably between studies and conditions.
- Live microalgae in larval rearing has been associated with substantially improved survival in at least one marine aquaculture species.
- Total dissolved gas supersaturation causes gas bubble disease in fish, and management guidance keeps saturation well below the levels at which signs appear.
- Turbidity measured with white light cannot compensate for dissolved coloration, which is why near-infrared FNU measurement exists.

Manufacturer positions relied on

- An aquasoil manufacturer's own documentation describes three distinct visual effects from the product — particulate cloudiness from crumbling granules, white turbidity from bacteria feeding on released organics, and harmless humic yellowing — and advises waiting two to three weeks before stocking, filling slowly, and running the filter only after the water clears.
- Clarifier manufacturers describe their products as clumping suspended particles for mechanical capture, and at least one advises a partial water change after twenty-four hours to remove settled material. Both descriptions are consistent with the coagulation and flocculation mechanism and both implicitly confirm that the products do not address dissolved causes.

Genuine gaps and disagreements documented rather than resolved

- Whether water changes meaningfully extend bacterial bloom duration in a fishless tank. This is asserted confidently in hobby sources and was not found demonstrated.
- Whether humic substances reduce ammonia and nitrite toxicity in recirculating systems remains an open question in the literature reviewed.
- No specific evidence of gill harm from aquarium water clarifiers in normal use was located. This is an absence of located evidence and should not be presented as a safety finding.
- Optimal humic substance concentrations for commonly kept aquarium species are not established.
- The frequency with which bacterial blooms actually depress dissolved oxygen to harmful levels in typical home aquariums is not characterised, although the mechanism is well established.
- Several extended causes listed in Part LXVIII are characterised from hobbyist reports with plausible mechanisms rather than from published study, and are labelled accordingly.
- Whether purifying resins meaningfully assist with bloom clearing is contested between sources.

On what could not be traced

Several widely repeated hobby claims could not be traced to any original source during this research, including specific bloom duration figures, the assertion that clarifiers damage gills, and the claim that aquarium salt clears cloudy water. Where that is the case, this document says so rather than supplying a citation that would look authoritative and would not be.

BACK MATTER

Bibliography and Source Notes

Sources consulted, with honest notes on what could not be verified.

On the completeness of these entries

Where full bibliographic details — volume, page numbers, DOI — were not captured during research, they are omitted rather than reconstructed. An incomplete citation that can be verified is more useful than a complete-looking one that cannot. Access dates fall within the research period preceding this document's compilation.

Aquarium Science. Cloudy Water (section 6.2.3), Purification Resins (7.4.6), and Filter Maintenance (6.8). aquariumscience.org. Independent hobbyist-scientist reference; used for the heterotrophic energetics argument, microbial succession to protozoa, microbubble mechanism, and several contested positions.

Bagchi, S., et al. (2025). Microbial community succession in home aquarium biofilters, including comammox *Nitrospira* and fixed-film bioreactor framing; also reports comparative testing of bottled quick-start products, of which one of five significantly reduced ammonia relative to control.

Hommes, N. G., et al. (2003). Growth characteristics of *Nitrosomonas europaea*, reporting doubling times in the range of approximately 7.4 to 24 hours. Accessed via Gupta et al. (2020).

Duarte, R. M., Smith, D. S., Val, A. L., & Wood, C. M. (2016). Dissolved organic carbon from the Rio Negro protects zebrafish against ionoregulatory disturbance caused by low pH. *Scientific Reports*.

Holland, A., et al. (2013). Effects of humic substances on *Melanotaenia splendida splendida*: reduced respiratory stress at 10 and 20 mg/L alongside morbidity associated with ion loss, described as a double-edged effect. *Environmental Science and Pollution Research*.

Morozov, A., et al. Proteomic analysis of medaka exposed to humic acid, reporting suppression of innate immunity and coagulation-related proteins. *International Journal of Molecular Sciences*.

McGeer, J. C., et al. (2002). Dissolved organic carbon moderates metal toxicity to fish.

Meinelt, T., et al. Discussion of whether humic substances reduce ammonia and nitrite toxicity in recirculating aquaculture systems; identifies this as an open question.

Chhetri, et al. Effect of turbidity on ultraviolet inactivation of *Escherichia coli*: reducing turbidity from 23.32 to 10.93 NTU improved log reduction by more than 2.15 log. *Food Science and Nutrition*.

Kwan, et al. Live microalgae in cobia larval rearing, with survival improving from 8.2 percent to 23.3 percent. *Fish and Shellfish Immunology Reports*.

Kuřerová, et al. Humic substance characterisation and removal by granular activated carbon, reporting over 70 percent humic removal and approximately 50 percent color reduction. *Engineering Proceedings*.

Chávez-Mejía, et al. Ionisation behavior of carboxyl and phenolic groups in humic substances. *Polymers*.

Ahmad, et al. Chitosan as a coagulant, reporting 87.67 percent turbidity removal at 20 mg/L with 30 minutes settling; comparative review figures of approximately 68.9 percent for chitosan against over 95 percent for alum and ferric chloride. *Polymers*.

Integrated Aqua Systems. Ultraviolet sterilizer dose guidance for aquaculture, citing approximately 15,000 microwatt-seconds per square centimetre for bacteria and approximately 22,000 for waterborne algae, and noting that wattage alone is insufficient to specify a unit.

Aquatic Equipment and Design. Ultraviolet dose ranges of approximately 30 to 180 millijoules per square centimetre for common targets, with 246 millijoules per square centimetre or more required for infectious pancreatic necrosis virus.

UltraAqua. Guidance that photoreactivation is minimal at delivered doses of 15 millijoules per square centimetre and above.

United States Environmental Protection Agency. Volunteer Stream Monitoring: A Methods Manual — turbidity definition and the approximate particle size range of 0.004 to 1.0 millimetres.

Fondriest Environmental. Turbidity, Total Suspended Solids and Water Clarity — explanation of why white-light NTU measurement cannot compensate for dissolved coloration while near-infrared FNU measurement can.

Aqua Design Amano (ADA). Aqua Soil Amazonia product documentation: guidance on ammonium leaching, waiting two to three weeks before stocking, filling slowly, running the filter only after the water clears, distinguishing granule breakdown from bacterial turbidity from humic yellowing, and the usefulness of ultraviolet treatment for setup-stage microbial cloudiness.

API. ACCU-CLEAR product information, describing the product as a colorless inorganic polymer that clumps suspended particles including dirt, silt and dust.

Aqueon. Water clarifier product information, advising a partial water change after twenty-four hours to remove settled particles.

Hydralife. Green water composition, identifying *Chlorella*, *Scenedesmus*, *Nannochloropsis* and *Euglena* at approximately 2 to 15 micrometres, typically as mixed community cultures.

American Killifish Association. Green water for rearing, including its role in supporting *Daphnia* and rotifers, and the note that *Chlorella* is too large for the smallest fry to digest directly.

WikiVet; PetMD; Toledo Zoo. Gas bubble disease and total dissolved gas supersaturation, including the practice of keeping pond oxygen saturation below 125 percent and reported signs in silver carp above 130 percent total dissolved gas.

Interstate Technology and Regulatory Council. Harmful Cyanobacterial Blooms guidance, including the distinction between benthic and planktonic forms and the detachment of benthic mats.

That Pet Place. Description of the typical new-tank visual sequence from white bacterial cloudiness to brown diatom film.

Public hobbyist forums and social discussion. Reviewed to characterise recurring questions, points of confusion and conflicting advice. Used only as evidence about the discourse, explicitly not as evidence about aquarium biology. See Part LXV.

Claims that could not be traced to any source

- Specific fixed durations for bacterial blooms, such as the frequently repeated "24 to 48 hours" or "one to two weeks." No original source was located for any specific figure.
- The claim that water clarifiers damage fish gills. Widely repeated in hobby discussion; no supporting source located.
- The claim that aquarium salt clears cloudy water. No mechanism and no source located. Salt does have a documented and separate use in mitigating nitrite toxicity.
- The claim that tannins boost fish immune systems. Marketing-adjacent; the located proteomic evidence points in the opposite direction.
- Numeric search-volume figures for any query discussed in Part LXVI. None were available and none are given.

[REASONABLE INFERENCE] The presence of this list is itself a finding. A substantial fraction of confidently stated cloudy-water advice has no traceable origin, which is the structural reason the topic generates so much conflicting guidance.